S7106

Version 1.0m

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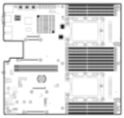
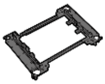
Contents

Before you begin…

Check the box contents!

The retail motherboard package should contain the following:

	S7106 Motherboard x 1
	SATA Single Cable x 2
	Rear IO shielding x 3
	CPU clip for Narrow Non-Fabric CPU Carrier x 2
	1 x S7106 Quick Installation Guide
	1 x TYAN® Driver CD
	M.2 Screw kit x1

IMPORTANT NOTE:
1. Sales samples may not come with any of the accessories listed above.
If you have ordered a sales sample and you are missing any of the above items, please contact your sales representative to help order accessories.
2. The standard rear I/O shielding listed in the above packing contain does not support RJ45 OCP Accessories.

Chapter 1: Instruction

1.1 Congratulations

You have purchased the powerful TYAN® S7106 motherboard, based on the Intel® C621/C622 Lewisburg chipset. The S7106 is designed to support Intel® Xeon® Scalable Processor Families, and Up to 512GB RDIMM/ 1024GB LRDIMM/ 768GB LRDIMM 3DS* DDR4 memory. Leveraging advanced technology from Intel®, the S7106 is capable of offering scalable 32 and 64-bit computing, high-bandwidth memory design, and lightning-fast PCI-E bus implementation.

The S7106 not only empowers you in today's demanding IT environment but also offers a smooth path for future application upgradeability. All of these rich feature sets provide the S7106 with the power and flexibility to meet demanding requirements for today's IT environments.

Remember to visit the TYAN® website at **http://www.tyan.com**. There you can find all the information on all TYAN® products as well as all the supporting documentation, FAQs, Drivers and BIOS upgrades.

1.2 Hardware Specifications

S7106GM2NR Specifications

Processor	**Socket Type / Q'ty**	LGA3647/ (2)
	Supported CPU Series	Intel Xeon Scalable Processor Families
	Thermal Design Power (TDP) wattage	Max up to 205W
Chipset	**PCH**	S7106GM2NR Intel C621
Memory	**Supported DIMM Qty**	(8)+(8) DIMM slots
	DIMM Type / Speed	DDR4 RDIMM/RDIMM 3DS/LRDIMM/LRDIMM 3DS 2933/2666 / Intel Optane DC Persistent Memory Module (DCPMM)
	Capacity	Up to 512GB RDIMM/ 1,024GB LRDIMM/ 768GB LRDIMM 3DS *Follow latest Intel DDR4 Memory POR
	Memory channel	6 Channels per CPU
	Memory voltage	1.2V

	Notification		DDR4-2933 speed only supported with 2nd Gen Intel Xeon Scalable Processor (codename: Cascade Lake) and in a 1 DIMM per channel configuration. / Intel Optane DC Persistent Memory Module (DCPMM) only supported with specific 2nd Gen Intel Xeon Scalable Processor (codename: Cascade Lake). Please contact Tyan Technical Support for more detail.
Expansion Slots	PCI-E		(2) PCI-E Gen3 x24 slots
	Others:		(2) PCI-E Gen3 x16 OCP 2.0 slots (con.A+con.C or con.F+con.G) (1) PCI-E Gen3 x2 for M.2 Port (2242/2260/2280)
LAN	Port Q'ty		(2) GbE ports, (1) PHY dedicated for IPMI
	Controller		Intel I210-AT
	PHY		Realtek RTL8211E
Storage	SATA	Connector	(2) Mini-SAS HD (8-ports)
		Controller	Intel C621
		Speed	6.0 Gb/s
		RAID	RAID 0/1/10/5 (Intel RSTe)
		M.2 connector	(1) M.2 connector (2242/2260/2280) by PCI-E interface
	sSATA	Connector	(1) Mini-SAS HD (4-ports) + (2) SATA-III
		Controller	Intel C621
		Speed	6.0 Gb/s
		RAID	RAID 0/1/10/5 (Intel RSTe) only for 4 SATA devices
Graphic	Connector type		D-Sub 15-pin
	Resolution		Up to 1920x1200
	Chipset		Aspeed AST2500
Input /Output	USB		(2) USB3.0 ports (via cable), (2) USB3.0 ports (at rear), (2) USB 2.0 ports (via cable), (1) Type-A USB3.0 port
	COM		(2) ports (1 at rear, 1 via cable)

	VGA	(2) D-Sub ports (1 at rear, 1 via cable)
	RJ-45	(2) GbE ports, (1) Dedicated for IPMI
	Front Panel	(1) 2x12-pin SSI front panel header
	SATA	(2) SATA-III connectors + (3) Mini-SAS HD (4-in-1) connectors
	Power	(1) ATX 24-pin + (2) 8-pin power connectors
	Others	(1) ID button
TPM (Optional)	**TPM Support**	Please refer to our TPM supported list.
System Monitoring	**Chipset**	Aspeed AST2500
	Fan	Total (5) 4-pin headers
	Temperature	Monitors temperature for CPU & memory & system environment
	Voltage	Monitors voltage for CPU, memory, chipset & power supply
	LED	Over temperature warning indicator, Fan & PSU fail LED indicator
	Others	Watchdog timer support
Server Management	**AST2500 iKVM Feature**	IPMI 2.0 compliant baseboard management controller (BMC), Supports storage over IP and remote platform-flash, USB 2.0 virtual hub
	AST2500 IPMI Feature	24-bit high quality video compression, 10/100/1000 Mb/s MAC interface
BIOS	**Brand / ROM size**	AMI, 32MB
	Feature	PXE boot support, ACPI 5.0, SMBIOS 3.0/PnP/Wake on LAN, ACPI sleeping states S4,S5, User-configurable H/W monitoring
Physical Dimension	**Form Factor**	EATX
	Board Dimension	12"x13" (305x330mm)
Regulation	**FCC (DoC)**	Class B
	CE (DoC)	Yes
Operating Environment	**Operating Temp.**	10° C ~ 35° C (50° F~ 95° F)
	Non-operating Temp.	- 40° C ~ 70° C (-40° F ~ 158° F)

	In/Non-operating Humidity	90%, non-condensing at 35° C
RoHS	**RoHS 6/6 Compliant**	Yes
Operating System	**OS supported list**	Please refer to our Intel OS supported list.
Package Contains	**Motherboard**	(1) S7106 Motherboard
	Manual	(1) Web User's manual, (1) Quick Installation Guide
	Installation CD	(1) TYAN installation CD

S7106GM2NR-L2 Specifications

Processor	**Socket Type / Q'ty**	LGA3647/ (2)
	Supported CPU Series	Intel Xeon Scalable Processor Families
	Thermal Design Power (TDP) wattage	Max up to 205W
Chipset	**PCH**	Intel C622
Memory	**Supported DIMM Qty**	(8)+(8) DIMM slots
	DIMM Type / Speed	DDR4 RDIMM/RDIMM 3DS/LRDIMM/LRDIMM 3DS 2933/2666/ Intel Optane DC Persistent Memory Module (DCPMM)
	Capacity	Up to 512GB RDIMM/ 1,024GB LRDIMM/ 768GB LRDIMM 3DS *Follow latest Intel DDR4 Memory POR
	Memory channel	6 Channels per CPU
	Memory voltage	1.2V
	Notification	DDR4-2933 speed only supported with 2nd Gen Intel Xeon Scalable Processor (codename: Cascade Lake) and in a 1 DIMM per channel configuration. / Intel Optane DC Persistent Memory Module (DCPMM) only supported with specific 2nd Gen Intel Xeon Scalable Processor (codename: Cascade Lake). Please contact Tyan Technical Support for more detail.
Expansion Slots	**PCI-E**	(2) PCI-E Gen3 x24 slots

	Others:		(2) PCI-E Gen3 x16 OCP 2.0 slots (con.A+con.C or con.F+con.G) (1) PCI-E Gen3 x2 for M.2 Port (2242/2260/2280)
LAN	Port Q'ty		(2) GbE ports, (1) PHY dedicated for IPMI
	Controller		Intel I210-AT
	PHY		Realtek RTL8211E
Storage	SATA	Connector	(2) Mini-SAS HD (8-ports)
		Controller	Intel C622
		Speed	6.0 Gb/s
		RAID	RAID 0/1/10/5 (Intel RSTe)
		M.2 connector	(1) M.2 connector (2242/2260/2280) by PCI-E interface
	sSATA	Connector	(2) SATA-III, (1) Mini-SAS HD (4-ports)
		Controller	Intel C622
		Speed	6.0 Gb/s
		RAID	RAID 0/1/10/5 (Intel RSTe) only for 4 SATA devices
Graphic	Connector type		D-Sub 15-pin
	Resolution		Up to 1920x1200
	Chipset		Aspeed AST2500
Input /Output	USB		(2) USB3.0 ports (via cable), (2) USB3.0 ports (at rear), (2) USB 2.0 ports (via cable), (1) Type-A USB3.0 port
	COM		(2) Ports (1 at rear, 1 via cable)
	VGA		(2) D-Sub ports (1 at rear, 1 via cable)
	RJ-45		(2) GbE ports, (1) Dedicated for IPMI
	Front Panel		(1) 2x12-pin SSI front panel header
	SATA		(2) SATA-III connectors + (3) Mini-SAS HD (4-in-1) connectors
	Power		(1) ATX 24-pin + (2) 8-pin power connectors
	Others		(1) ID button

TPM (Optional)	**TPM Support**	Please refer to our TPM supported list.
System Monitoring	**Chipset**	Aspeed AST2500
	Fan	Total (5) 4-pin headers
	Temperature	Monitors temperature for CPU & memory & system environment
	Voltage	Monitors voltage for CPU, memory, chipset & power supply
	LED	Over temperature warning indicator, Fan & PSU fail LED indicator
	Others	Watchdog timer support
Server Management	**AST2500 iKVM Feature**	IPMI 2.0 compliant baseboard management controller (BMC), Supports storage over IP and remote platform-flash, USB 2.0 virtual hub
	AST2500 IPMI Feature	24-bit high quality video compression, 10/100/1000 Mb/s MAC interface
BIOS	**Brand / ROM size**	AMI, 32MB
	Feature	Hardware Monitor, SMBIOS 3.0/PnP/Wake on LAN, Boot from USB device/PXE via LAN/Storage, User Configurable FAN PWM Duty Cycle, Console Redirection, ACPI sleeping states S4,S5, ACPI 6.1
Physical Dimension	**Form Factor**	EATX
	Board Dimension	12"x13" (305x330mm)
Regulation	**FCC (DoC)**	Class B
	CE (DoC)	Yes
Operating Environment	**Operating Temp.**	10° C ~ 35° C (50° F~ 95° F)
	Non-operating Temp.	- 40° C ~ 70° C (-40° F ~ 158° F)
	In/Non-operating Humidity	90%, non-condensing at 35° C
RoHS	**RoHS 6/6 Compliant**	Yes
Operating System	**OS supported list**	Please refer to our AVL support lists.
Package Contains	**Motherboard**	(1) S7106 Motherboard
	Manual	(1) Web User's manual, (1) Quick

		Installation Guide
	Installation CD	(1) TYAN installation CD

1.3 Software Specifications

For the latest AST2500 User's Guide and OS (operation system) support, please visit the Tyan's Web site at http://www.tyan.com for the latest information

Chapter 2: Board Installation

You are now ready to install your motherboard.

How to install our products right… the first time

The first thing you should do is read this user's manual. It contains important information that will make configuration and setup much easier. Here are some precautions you should take when installing your motherboard:

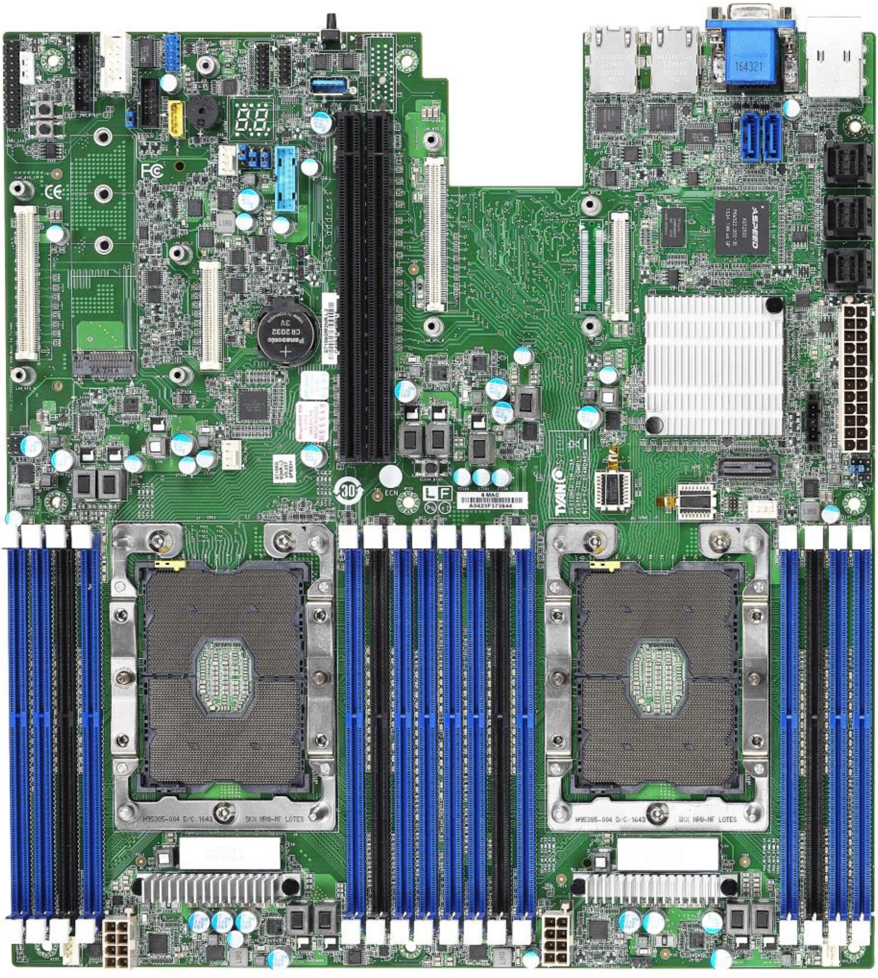
(1) Ground yourself properly before removing your motherboard from the antistatic bag. Unplug the power from your computer power supply and then touch a safely grounded object to release static charge (i.e. power supply case). For the safest conditions, MiTAC recommends wearing a static safety wrist strap.
(2) Hold the motherboard by its edges and do not touch the bottom of the board, or flex the board in any way.
(3) Avoid touching the motherboard components, IC chips, connectors, memory modules, and leads.
(4) Place the motherboard on a grounded antistatic surface or on the antistatic bag that the board was shipped in.
(5) Inspect the board for damage.

The following pages include details on how to install your motherboard into your chassis, as well as installing the processor, memory, disk drives and cables.

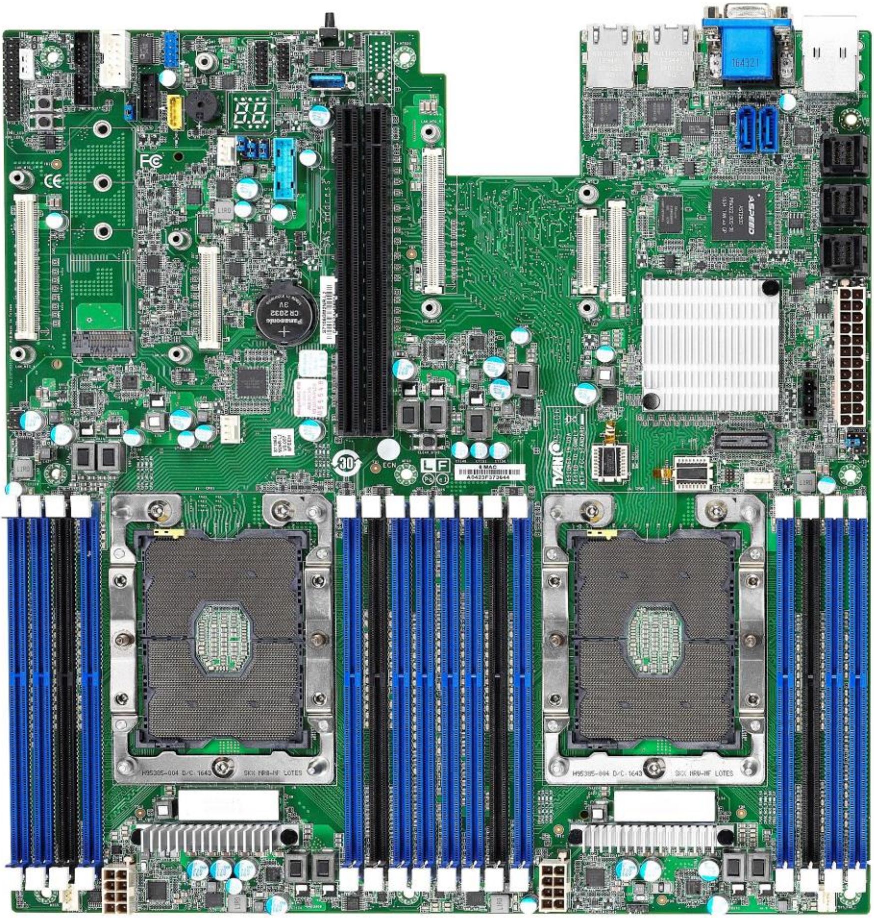
Caution!

1. To avoid damaging the motherboard and associated components, do not use torque force greater than 5~7kgf/cm (4.35~6.09 lb/in) on each mounting screw for motherboard installation.
2. Do not apply power to the board if it has been damaged.

2.1 Board Image



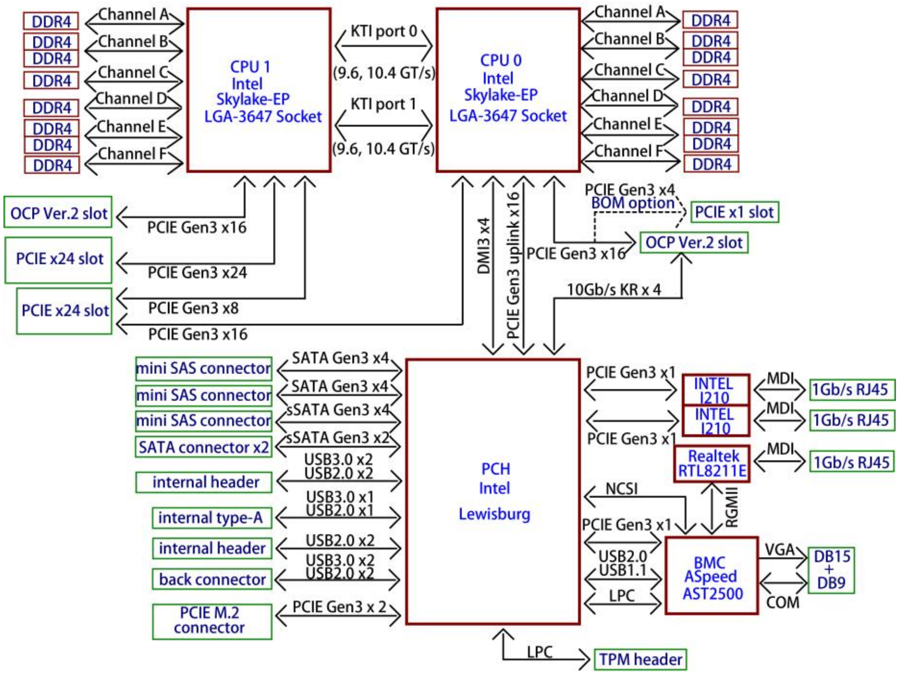
S7106GM2NR



S7106GM2NR-L2

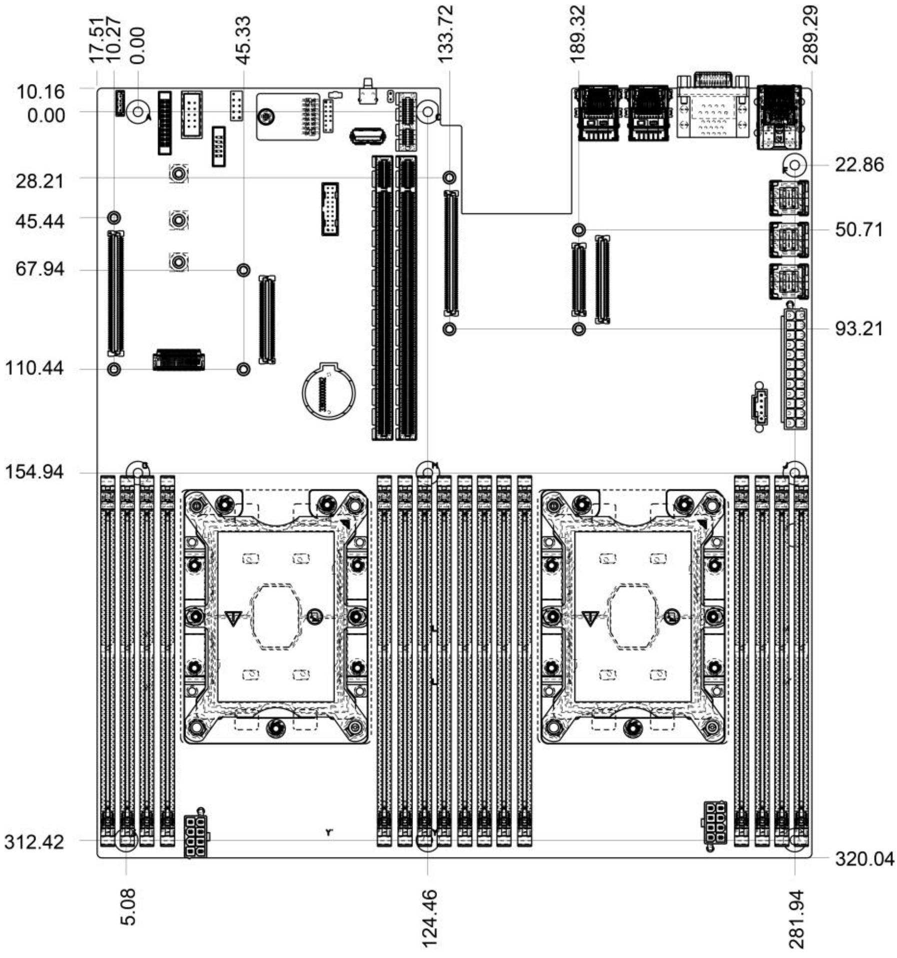
This picture is representative of the latest board revision available at the time of publishing. The board you receive may not look exactly like the above picture.

2.2 Block Diagram

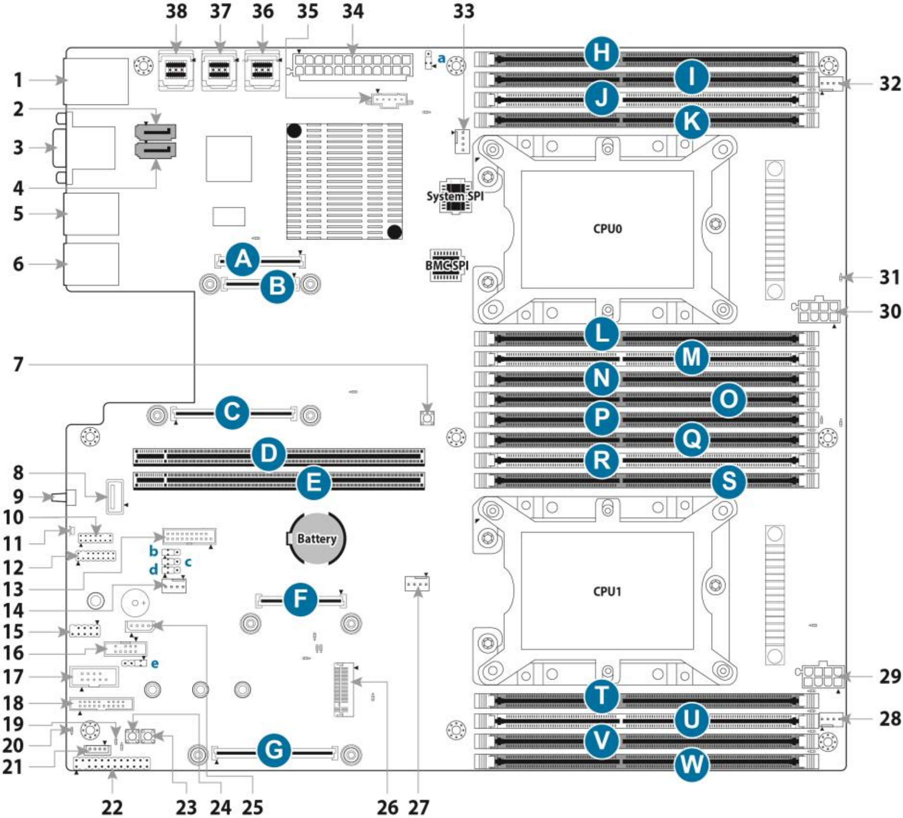


S7106 Block Diagram

2.3 Motherboard Mechanical Drawing



2.4 Board Parts, Jumpers and Connectors



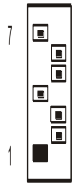
This diagram is representative of the latest board revision available at the time of publishing. The board you receive may not look exactly like the above diagram. The DIMM slot numbers shown above can be used as a reference when reviewing the DIMM population guidelines shown later in the manual. For the latest board revision, please visit our web site at **http://www.tyan.com**.

Motherboard Components

Connectors	
1. Dedicated IPMI port+ USB3.0x2 (J112)	20. BMC Heartbeat LED (BMC_LED4)
2. SSATA Connector 4 (SSATA4)	21. PCH_SATA_RAID_KEY (J51)
3. VGA+COM Connector (VGA_COM1)	22. Front Panel Header (FPIO_1)
4. SSATA Connector 5 (SSATA5)	23. Reset Button (RST_ BTN1)
5. LAN Connector (LAN2)	24. Power Button (PWR_BTN1)
6. LAN Connector (LAN1)	25.IPMB Pin Header (IPMB_HD1)
7. Clear CMOS Button (CLEAR_BTN1)	26. M.2 NGFF PCIe 22x42/60/80mm Socket (U58)
8. Vertical Type A USB Connector (TYPEA_USB3)	27. CPU1 FAN Header (CPU1_FAN)
9. Rear ID LED Button(ID LED_BTN1)	28. SYS_FAN2 Header(SYS_FAN_2)
10. VGA Header (FPIO_VGA3)	29. SSI 8-pin CPU1 Power Connector (PW2)
11. Rear ID LED (ID_LED1)	30. SSI 8-pin CPU0 Power Connector (PW1)
12. TYAN Module Header (DBG_HD1)	31. CPU0 PVCCP POWER OK LED (P0_PG_LED1)
13. Front Panel USB3.0 Header (USB3_FPIO1)	32. SYS_FAN 1 Header (SYS_FAN_1)
14. Rear IO_FAN 2 (SYS_FAN_3)	33. CPU0 FAN Header (CPU0_FAN)
15. Internal USB2.0 Header (J26)	34. ATX Power Connector (24 pin) (PWR1)
16. sSATA SGPIO Header for SSATA4/SSATA5 (SSATA_SGPIO1)	35. PSMI Pin Header (PSMI_HD1)
17. COM2 Header (HD_COM2)	36. 36-pin Vertical Mini SAS HD Connector (SATA 3.0 signals)(PCH_SSATA_0123)
18. FAN Tachometer Header (FAN_HD1)	37. 36-pin Vertical Mini SAS HD Connector (SATA 3.0 signals)(PCH_SATA_0123)
19. On board BMC IPMI ALERT LED (IPMI_LED1)	38. 36-pin Vertical Mini SAS HD Connector (SATA 3.0 signals) (PCH_SATA_4567)
Headers/Jumpers	
a. ME Recovery Mode Jumper (J113)	d. PSU Throttling Jumper (J125)
b. BMC COM PORT Jumper (J119)	e. PC BEEP Jumper (J120)
c. BMC COM PORT Jumper (J118)	
PCIE Slots	
A+C. OCP 2.0 PCI-E Gen3x16 MEZZ slot f/CPU0 for LAN Card	**F+G.** OCP 2.0 PCI-E Gen3x16 MEZZ slot f/CPU1 for Storage MEZZ Card
Standalone C. OCP 1.0 PCI-E Gen3x8 MEZZ slot f/CPU0 for LAN Card	**Standalone G.** OCP 1.0 PCI-E Gen3x8 MEZZ slot f/CPU1 for Storage MEZZ Card
B. OCP 2.0 KR interface(only available in S7106GM2NR-L2 SKU)	**D.** PCI-E Gen3x24 slot(x24 link) f/ CPU1
E. PCI-E Gen3x24 slot(x24 link) f/ CPU0	
Memory Slots	
H. CPU0 Memory Slot (P0_MC0_DIM_CH_C0)	**P** CPU1 Memory Slot (P1_MC0_DIM_CH_C0)
I. CPU0 Memory Slot (P0_MC0_DIM_CH_B0)	**Q** CPU1 Memory Slot (P1_MC0_DIM_CH_B0)

J CPU0 Memory Slot (P0_MC0_DIM_CH_B1)	**R** CPU1 Memory Slot (P1_MC0_DIM_CH_B1)
K CPU0 Memory Slot (P0_MC0_DIM_CH_A0)	**S** CPU1 Memory Slot (P1_MC0_DIM_CH_A0)
L CPU0 Memory Slot (P0_MC1_DIM_CH_D0)	**T** CPU1 Memory Slot (P1_MC1_DIM_CH_D0)
M CPU0 Memory Slot (P0_MC1_DIM_CH_E1)	**U** CPU1 Memory Slot (P1_MC1_DIM_CH_E1)
N CPU0 Memory Slot (P0_MC1_DIM_CH_E0)	**V** CPU1 Memory Slot (P1_MC1_DIM_CH_E0)
O CPU0 Memory Slot (P0_MC1_DIM_CH_F0)	**W** CPU1 Memory Slot (P1_MC1_DIM_CH_F0)

SSATA4/SSATA5: sSerial Advanced Technology Attachment Gen3(sSATA)



Name	TYPE
S1	GND
S2	SATA TX DP
S3	SATA TX DN
S4	GND
S5	SATA RX DN
S6	SATA RX DP
S7	GND
Connects to the Serial ATA ready drives via the Serial ATA cable.	

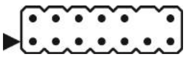
TYPEA_USB3: Vertical (Type_A) USB3.0 Connector

Signal	Pin	Pin	Signal
+5V	**1**	**2**	USB2.0_DATA_N
USB2.0_DATA_P	**3**	**4**	GND
USB3.0_RX_N	**5**	**6**	USB3.0_RX_P
GND	**7**	**8**	USB3.0_TX_N
USB3.0_TX_P	**9**		

ID LED_BTN1: Rear IO IDLED Button

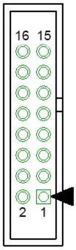
Signal	Pin	Pin	Signal
FP_IDLED_BTN_N	**1**	**2**	GND

FPIO_VGA3: VGA Header



Signal	Pin	Pin	Signal
GND	**1**	**2**	VGA2_5V
GND	**3**	**4**	HD_VGA_R
GND	**5**	**6**	HD_VGA_G
GND	**7**	**8**	HD_VGA_B
GND	**9**	**10**	HD_VGA_DAT
VGA_HD_HS	**11**	**12**	KEY
VGA_HD_CLK	**13**	**14**	HD_VGA_VS

DBG_HD1: TYAN Module Header

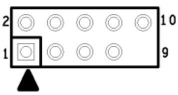


Signal	Pin	Pin	Signal
P3V3	1	2	FRAME_N
LAD0	3	4	KEY
LAD1	5	6	PLT_RST_N
LAD2	7	8	GND
LAD3	9	10	CLK_33M
DBG_SERIRQ	11	12	GND
DBG_PRES_N	13	14	VCC3_AUX
TPM_ADDR_MB	15	16	PCH_TPM_PP_EN

USB3_FPIO1: Front USB3.0 Header

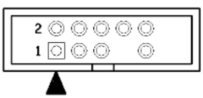
Signal	Pin	Pin	Signal
+5V	1	2	P0_RX_N
P0_RX_P	3	4	GND
P0_TX_N	5	6	P0_TX_P
GND	7	8	P0_N
P0_P	9	10	OC_N
P1_P	11	12	P1_N
GND	13	14	P1_TX_P
P1_TX_N	15	16	GND
P1_RX_P	17	18	P1_RX_N
+5V	19	20	Key

J26: Internal USB2.0 Header



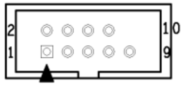
Signal	Pin	Pin	Signal
+5V	1	2	+5V
USB DATA1-	3	4	USB DATA2-
USBDATA1+	5	6	USB DATA2+
GND	7	8	GND
KEY	9	10	GND

SSATA_SGPIO1: SSATA SGPIO Header for SSATA4/SSATA5



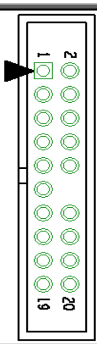
Signal	Pin	Pin	Signal
SMBCLK	1	2	SDATA IN
SMBDATA	3	4	SDATA OUT
GND	5	6	SLOAD
KEY	7	8	SCLOCK
VCC3_AUX	9	10	HD_ERROR_LED

HD_COM2: COM Port Header



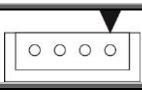
Signal	Pin	Pin	Signal
COM_DCD	1	2	COM_DSR
COM_RXD	3	4	COM_RTS
COM_TXD	5	6	COM_CTS
COM_DTR	7	8	COM_NRI
GND	9	10	KEY

FAN_HD1: FAN Tachometer Header Header



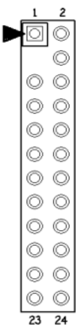
Signal	Pin	Pin	Signal
TACH1	1	2	TACH6
TACH2	3	4	TACH7
TACH3	5	6	TACH8
TACH4	7	8	TACH9
TACH5	9	10	TACH10
GND	11	12	KEY
PWM2	13	14	PWM1
TACH11	15	16	SMB_DATA
TACH12	17	18	SMB_CLK
NC	19	20	PWM3

J51: PCH_SATA_RAID_KEY (HW Key for Intel VROC – NVMe only)



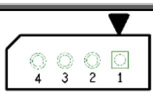
Pin	1	2
Signal	LAN1_LED1_FP+	LAN1_LINK_ACT#

FPIO1: Front Panel Connector

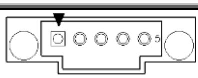


Signal	Pin	Pin	Signal
FP_PW_LED_PW	1	2	FP_PWER (VCC3_Aux)
KEY	3	4	FP_ID_LED_PW
FP_PW_LED_GND	5	6	FP_ID_LED_N
HDD_LED_PW	7	8	BMC_HW_FAULT_N
HDD_ACT_LED_N	9	10	BMC_SYS_FAULT_N
FP_PWR_BTN_N	11	12	LAN0_LED_P
GND	13	14	LAN0_LED_N
FP_RST_BTN_JP_N	15	16	FP_SMB_DAT
GND	17	18	FP_SMB_CLK
FP_IDLED_BTN_N	19	20	FP_INTRUSION_N
SYS_Air_Inlet	21	22	LAN1_LED_P
FP_NMI_BTN_N	23	24	LAN1_LED_N

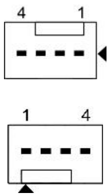
IPMB_HD1: IPMB Connector

	Signal	Pin	Pin	Signal
	IPMB_DAT	1	2	GND
	IPMB_CLK	3	4	VCC3_AUX

PSMI_HD1: PSMI Connector

	Pin	1	2	3	4	5
	Signal	SMB_CLK	SMB_DAT	SMB_ALERT#	GND	V3P3

CPU0/1 FAN/ SYS FAN1~3: 4-pin CPU Fan Connector

	Pin	1	2	3	4
	Signal	GND	VCC12	FAN_TACH	PWM

Use this header to connect the cooling fan to your motherboard to keep the system stable and reliable.

Note: A 4-pin fan is required for fan support 4pin Control

Clear_BTN1: RTC reset Button for clear CMOS

	Pin	1	2	3	4
Normal (Default)	Signal	GND	GND	RST_N	RST_N

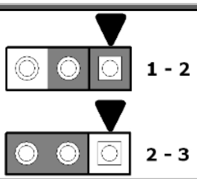
PWR_BTN1: POWER Button

	Pin	1	2	3	4
Normal (Default)	Signal	GND	GND	PWR_BTN1	PWR_BTN1

RST_BTN1: Reset Button

	Pin	1	2	3	4
Normal (Default)	Signal	GND	GND	FP_RST_BTN_N	FP_RST_BTN_N

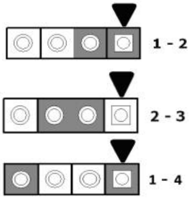
J113: ME Recovery Mode Jumper



Pin	1	2	3
Signal	NC	FM_ME_RCVR_N	1K-Pull down

Pin 1-2 Closed: Normal Mode (**Default**)
Pin 2-3 Closed: ME In Force Update Mode

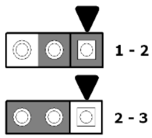
J120: BUZZER Disable Jumper



Pin	1	2	3	4
Signal	VCC5	NA	BUZ_1	BUZ_2

Pin 3-4 Closed: Normal Mode (**Default**)
Pin 2-3 Closed: Disable PC Beep
Pin 1-4 Closed: Use the External Speaker

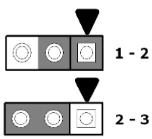
J118: BMC COM Port debug Jumper



Signal	Pin	Pin	Signal
BMC_COM2_RXD	1	2	RXD_2
BMC_COM5_RXD	3		

1-2: BMC COM Port2(Default)
2-3: BMC COMSOLE Port5

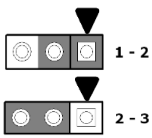
J119: BMC COM Port debug Jumper



Signal	Pin	Pin	Signal
BMC_COM2_TXD	1	2	TXD_2
BMC_COM5_TXD	3		

1-2: BMC COM Port2(Default)
2-3: BMC COMSOLE Port5

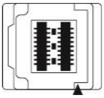
J125: PSU Throttling Jumper



Signal	Pin	Pin	Signal
NC	1	2	PSU_SMB_ALERT_BUFF_N_R
VCC3_Aux	3		

1-2: Normal Mode(Default)
2-3: Enable PSU Throttling Function

Mini SAS HD Connector: (PCH_SATA_0123/ PCH_SATA_4567/ PCH_SSATA_0123)



Signal	Pin	Pin	Signal
SM_DAT	PIN A1	PIN A2	SGPIO _CLK
GND	PIN A3	PIN A4	SATA6G_RX_P1
SATA6G_RX_N1	PIN A5	PIN A6	GND
SATA6G_RX_P3	PIN A7	PIN A8	SATA6G_RX_N3
GND	PIN A9	PIN B1	GND
SGPIO _LOAD	PIN B2	PIN B3	GND
SATA6G_RX_P0	PIN B4	PIN B5	SATA6G_RX_N0
GND	PIN B6	PIN B7	SATA6G_RX_P2
SATA6G_RX_N2	PIN B8	PIN B9	GND
SGPIO_SATA_DATAOUT0	PIN C1	PIN C2	GND
GND	PIN C3	PIN C4	SATA6G_TX_P1
SATA6G_TX_N1	PIN C5	PIN C6	GND
SATA6G_TX_P3	PIN C7	PIN C8	SATA6G_TX_N3
GND	PIN C9	PIN D1	NA
SM_CLK	PIN D2	PIN D3	GND
SATA6G_TX_P0	PIN D4	PIN D5	SATA6G_TX_N0
GND	PIN D6	PIN D7	SATA6G_TX_P2
SATA6G_TX_N2	PIN D8	PIN D9	GND

2.5 LED Definitions

<table>
<tr><td rowspan="7">BMC_LED4</td><td rowspan="7">BMC Heart Beat LED</td><td>Pin</td><td colspan="2">Signal</td></tr>
<tr><td>+</td><td colspan="2">+VCC3_SB</td></tr>
<tr><td>-</td><td colspan="2">GND</td></tr>
<tr><td>State</td><td colspan="2">Description</td></tr>
<tr><td>OFF</td><td>OFF</td><td>The LED shuts off when the BMC controller cannot be detected or properly initiated.</td></tr>
<tr><td rowspan="2">Blinking</td><td rowspan="2">Green</td><td rowspan="2">The LED blinks per second to indicate that the BMC controller is working normally</td></tr>
<tr></tr>
<tr><td rowspan="6">P0_PG_LED1</td><td rowspan="6">CPU1 PWOK LED</td><td>Pin</td><td colspan="2">Signal</td></tr>
<tr><td>+</td><td colspan="2">+ VCC3</td></tr>
<tr><td>-</td><td colspan="2">GND</td></tr>
<tr><td>State</td><td colspan="2">Description</td></tr>
<tr><td>OFF</td><td>OFF</td><td>The LED shuts off when the power of CPU1 is abnormal.</td></tr>
<tr><td>ON</td><td>Green</td><td>The LED lights up when the power of CPU1 is normally.</td></tr>
<tr><td rowspan="6">P1_PG_LED1</td><td rowspan="6">CPU1 PWOK LED</td><td>Pin</td><td colspan="2">Signal</td></tr>
<tr><td>+</td><td colspan="2">+ VCC3</td></tr>
<tr><td>-</td><td colspan="2">GND</td></tr>
<tr><td>State</td><td colspan="2">Description</td></tr>
<tr><td>OFF</td><td>OFF</td><td>The LED shuts off when the power of CPU1 is abnormal.</td></tr>
<tr><td>ON</td><td>Green</td><td>The LED lights up when the power of CPU1 is normally.</td></tr>
<tr><td rowspan="6">ID_LED1</td><td rowspan="6">Rear ID LED</td><td>Pin</td><td colspan="2">Signal</td></tr>
<tr><td>+</td><td colspan="2">+ VCC3_AUX</td></tr>
<tr><td>-</td><td colspan="2">GND</td></tr>
<tr><td>State</td><td colspan="2">Description</td></tr>
<tr><td>OFF</td><td>OFF</td><td>OFF</td></tr>
<tr><td>ON</td><td>Green</td><td>ON</td></tr>
<tr><td rowspan="6">IPMI_LED1</td><td rowspan="6">HW Sensor Alert LED</td><td>Pin</td><td colspan="2">Signal</td></tr>
<tr><td>+</td><td colspan="2">+ VCC3_AUX</td></tr>
<tr><td>-</td><td colspan="2">GND</td></tr>
<tr><td>State</td><td colspan="2">Description</td></tr>
<tr><td>OFF</td><td>OFF</td><td>HW Sensor status is normal</td></tr>
<tr><td>ON</td><td>Red</td><td>HW Sensor status is abnormal.</td></tr>
</table>

2.6 Installing the Processor and Heatsink

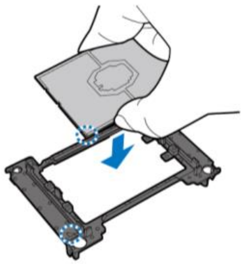
The types of processors supported by the S7106 are listed in the ***1.2 Hardware*** Specifications section on page 4. Check our website at **http://www.tyan.com** for the latest list of validated **Intel**® processors for this specific motherboard.

NOTE: MiTAC is not liable for damage as a result of operating an unsupported configuration.

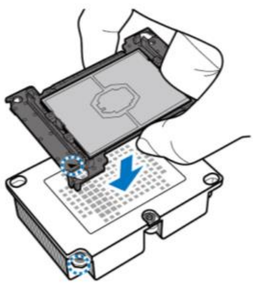
Processor Installation (Dual Socket P/ LGA3647 for Intel Skylake CPU)

Follow the steps below to install the processors and heat sinks.
Please note that the illustrations are based on LGA3647 socket which may not look exactly like the motherboard you purchased. Therefore, the illustrations should be held for your reference only.
NOTE: Please save and replace the flip CPU protection cap when returning for service.

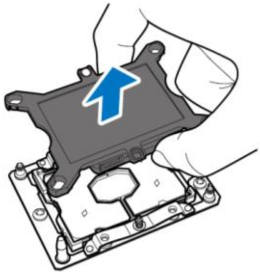
1. Align and install the processor on the carrier.



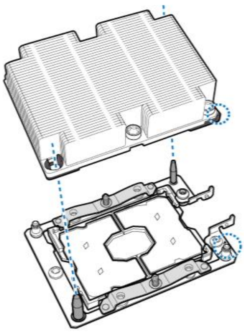
2. Carefully flip the heatsink. Then install the carrier assembly on the bottom of the heatsink and make sure the gold arrow is located in the correct direction.



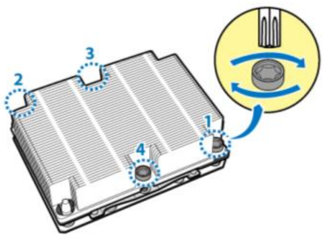
3. Remove the CPU cover. **NOTE: Save and replace the CPU cover if the processor is removed from its socket.**



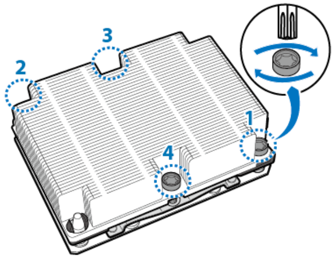
4. Align the heatsink with the CPU socket by the guide pins and make sure the gold arrow is located in the correct direction. Then place the heatsink onto the top of the CPU socket.



5. Align the heatsink by the guide pins and make sure the gold arrow is located in the correct direction. Then place the heatsink onto the top of the processor.

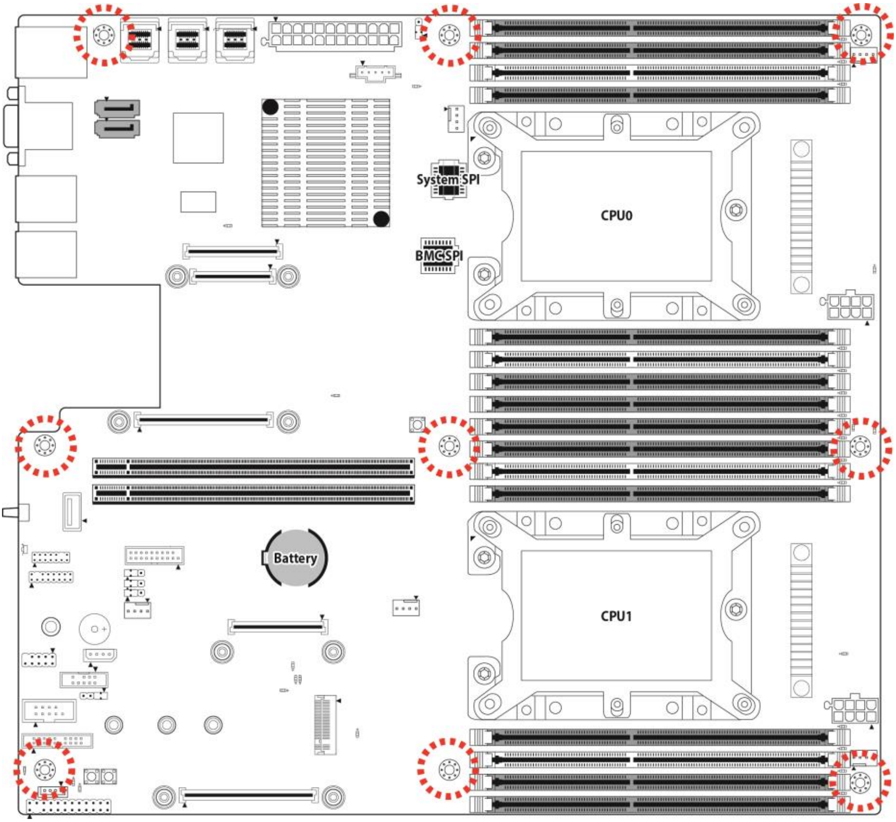


6. To secure the heatsink, use a Security T30 Security Torx to tighten the screws in a sequential order (1->2->3->4). When dissembling the heatsink, loosen the screws in reverse order (4->3->2->1).



2.7 Tips on Installing Motherboard in Chassis

Before installing your motherboard, make sure your chassis has the necessary motherboard support studs installed. These studs are usually metal and are gold in color. Usually, the chassis manufacturer will pre-install the support studs. If you are unsure of stud placement, simply lay the motherboard inside the chassis and align the screw holes of the motherboard to the studs inside the case. If there are any studs missing, you will know right away since the motherboard will not be able to be securely installed.

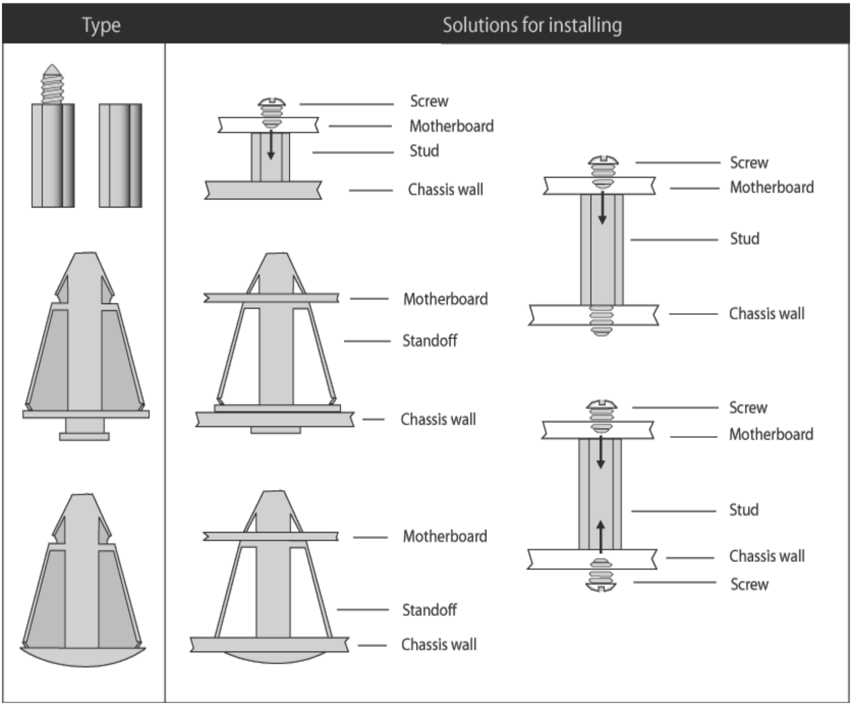


Note: Be especially careful to look for extra stand-offs. If there are any stand-offs present that are not aligned with a mounting hole on the motherboard, it will likely short components on the back of the motherboard when installed. This will cause malfunction and/or damage to your motherboard.

Some chassis include plastic studs instead of metal. Although the plastic studs are usable, MiTAC recommends using metal studs with screws that will fasten the motherboard more securely in place.

Below is a chart detailing what the most common motherboard studs look like and how they should be installed.

Mounting the Motherboard



2.8 Installing the Memory

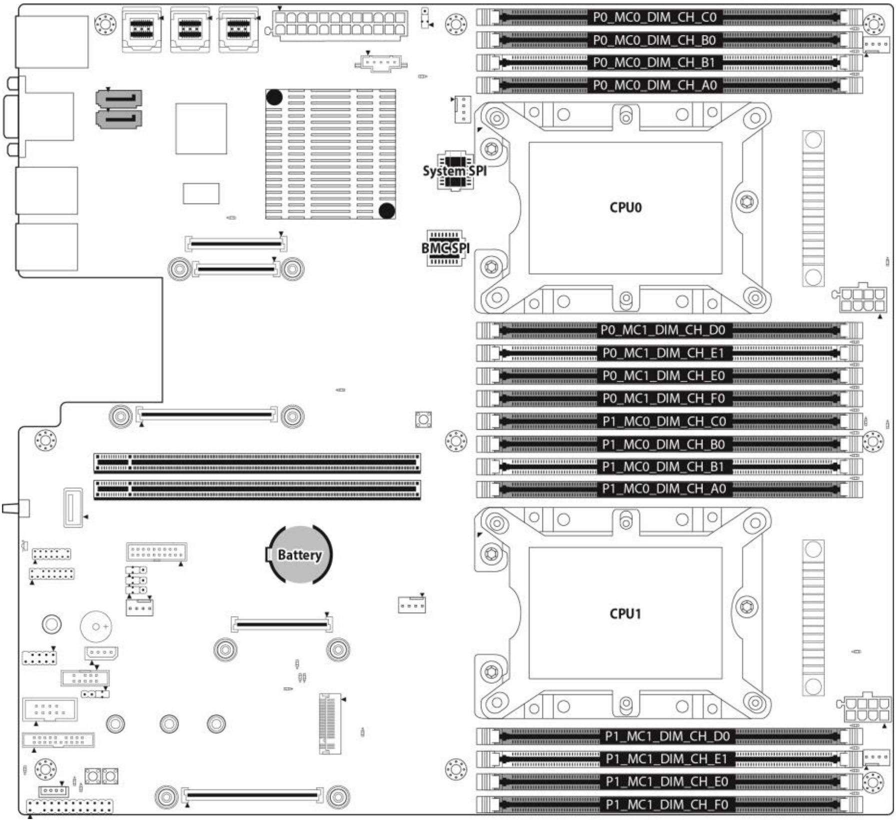
Before installing memory, ensure that the memory you have is compatible with the motherboard and processor. Check the TYAN Web site at http://**www.tyan.com** for details of the type of memory recommended for your motherboard.

- This platform supports (8)+(8) DDR4 RDIMM/RDIMM 3DS/LRDIMM/ LRDIMM 3DS 2933/2666
- Up to 512GB RDIMM/ 1,024GB LRDIMM/ 768GB LRDIMM 3DS are supported
- 1.2V DDR4 DIMMs are supported
- All installed memory will be automatically detected. No jumpers or settings need to be changed for memory detection.
- All memory must be of the same type and density. **Different memory types can NOT be mixed and matched on the same motherboard.**
- 2933 MHz memory is supported by 2nd Generation Intel Xeon Scalable Processor Platinum 8200 and Gold 6200 series processors only.
- 16Gb-based memory modules are supported by 2nd Gen Intel Xeon Scalable-SP processors only.
- Intel® Optane™ DC Persistent Memory Module (DCPMM) Implementation Requirements
  - DCPMM memory is supported by 2nd Generation Intel Xeon Scalable Processor Platinum 8200, Gold 6200, Gold 5200 series and Silver 4215 processors only.
  - All installed DCPMMs must be in the same size. Mixing DCPMMs of different capacities is not supported.
  - To maximize performance, balance all memory channels.
  - No DDR4 1Rx8 for either DCPMM Memory Mode or App-Direct Mode
  - Maximum 6 DCPMMs per processor (install 1 in each memory channel).
  - Minimum 2 DCPMMs per processor (1 per memory controller).

Key Parameters for DIMM Configuration

Parameter	Possible Values
# of Channels	1, 2, 3, 4, 5 or 6
# of DIMMs Populated per channel	1DPC or 2DPC
DIMM Type	RDIMM (w/ECC), 3DS-RDIMM, LRDIMM, 3DS-LRDIMM, DCPMM
DIMM construction	●non-3DS RDIMM Raw Cards: A/B (2Rx4), C (1Rx4), D (1Rx8), E (2Rx8) ●3DS RDIMM Raw Cards: A/B (4R/4) ●non-3DS LRDIMM Raw Cards: D/E (4Rx4) ●3DS LRDIMM Raw Cards: A/B (8Rx4) ●DPCMM

DIMM Location



NOTE: NOTE:

1. √ indicates a populated DIMM slot.
2. Use paired memory installation for max performance.
3. Populate the same DIMM type in each channel, specifically
   - Use the same DIMM size
   - Use the same # of ranks per DIMM
4. Dual-rank DIMMs are recommended over single-rank DIMMs.
5. Un-buffered DIMM can offer slightly better performance than registerd DIMM if populating only a single DIMM per channel.
6. Always install with CPU0 Socket and DIMM_0 Slot first, following the alphabetical order.

Recommended Memory Population Table

Single CPU Installed(CPU0 only)	Quantity of memory installed							
	1	2	3	4	5*	6	7*	8
CPU0_DIMM_A0	√	√	√	√	√	√	√	√
CPU0_DIMM_B1							√	√
CPU0_DIMM_B0			√	√	√	√	√	√
CPU0_DIMM_C0			√		√	√	√	√
CPU0_DIMM_D0		√		√	√	√	√	√
CPU0_DIMM_E1								√
CPU0_DIMM_E0				√	√	√	√	√
CPU0_DIMM_F0						√	√	√

Dual CPU Installed (CPU0 & CPU1)	Quantity of memory installed							
	2	4	6	8	10	12	14	16
CPU0_DIMM_A0	√	√	√	√		√		√
CPU0_DIMM_B1								√
CPU0_DIMM_B0		√	√	√		√		√
CPU0_DIMM_C0			√			√		√
CPU0_DIMM_D0				√		√		√
CPU0_DIMM_E1								√
CPU0_DIMM_E0				√		√		√
CPU0_DIMM_F0					Not Recommended	√	Not Recommended	√
CPU1_DIMM_A0	√	√	√	√		√		√
CPU1_DIMM_B1								√
CPU1_DIMM_B0		√	√	√		√		√
CPU1_DIMM_C0			√			√		√
CPU1_DIMM_D0				√		√		√
CPU1_DIMM_E1								√
CPU1_DIMM_E0				√		√		√
CPU1_DIMM_F0						√		√

DCPMM Support Location

Symmetric Population within the Socket													
	iMC1						iMC0						
	Channel F		Channel E		Channel D		Channel C		Channel B		Channel A		
Modes	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	
AD	DCPMM	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	2-2-2
MM	DCPMM	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	2-2-2
AD + MM	DCPMM	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	2-2-2
AD	-	DRAM1	-	DRAM1	DCPMM	DRAM1	-	DRAM1	-	DRAM1	DCPMM	DRAM1	2-1-1
MM	-	DRAM3	-	DRAM3	DCPMM	DRAM3	-	DRAM3	-	DRAM3	DCPMM	DRAM3	2-1-1
AD + MM	-	DRAM4	-	DRAM4	DCPMM	DRAM4	-	DRAM4	-	DRAM4	DCPMM	DRAM4	2-1-1
AD	-	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	-	DRAM1	DCPMM	DRAM1	DCPMM	DRAM1	2-2-1
MM	-	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	-	DRAM2	DCPMM	DRAM2	DCPMM	DRAM2	2-2-1
AD + MM	-	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	-	DRAM4	DCPMM	DRAM4	DCPMM	DRAM4	2-2-1
AD	-	DCPMM	-	DRAM1	-	DRAM1	-	DCPMM	-	DRAM1	-	DRAM1	1-1-1
MM	-	DCPMM	-	DRAM2	-	DRAM2	-	DCPMM	-	DRAM2	-	DRAM2	1-1-1
AD + MM	-	DCPMM	-	DRAM4	-	DRAM4	-	DCPMM	-	DRAM4	-	DRAM4	1-1-1
AD	-	DCPMM	DRAM1	DRAM1	DRAM1	DRAM1	-	DCPMM	DRAM1	DRAM1	DRAM1	DRAM1	2-2-1

Asymmetric Population within the Socket													
	iMC1						iMC0						
	Channel F		Channel E		Channel D		Channel C		Channel B		Channel A		
Modes	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	Slot 1	Slot 0	
AD	-	DRAM1	-	DRAM1	-	DRAM1	-	DRAM1	-	DRAM1	DCPMM	DRAM1	2-1-1
AD*	-	DRAM1	-	DRAM1	-	DRAM1	-	DRAM1	-	DRAM1	DCPMM	DRAM1	2-1-1

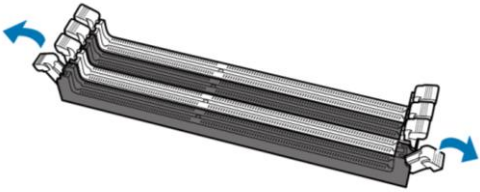
- * 2nd socket has no DCPMM DIMM
- Mode definitions: AD=App Direct Mode, MM=Memory Mode, AD+MM=Mixed Mode
- For MM, general DDR4+DCPMM ratio is between 1:4 and 1:16. Excessive capacity for DCPMM can be used for AD.
- For each individual population, rearrangements between channels are allowed as long as the resulting population is consistent to Purley PDG rules
- For each individual population, the same DDR4 DIMM has to used in all slots, as specified by the Purley PDG rules
- For each individual population, sockets are normally symmetric with exceptions for 1 DCPMM per socket and 1 DCPMM per node case.

	DDR4 Type				Capacity
DRAM1	RDIMM	3DS RDIMM	LRDIMM	3DS LRDIMM	Any Capcity
DRAM2	RDIMM	3DS RDIMM	LRDIMM	3DS LRDIMM	Any Capacity and Preferably, >=32GB
DRAM3	RDIMM	-	-	-	16GB or 32GB
DRAM4	RDIMM	3DS RDIMM	LRDIMM		Any Capacity

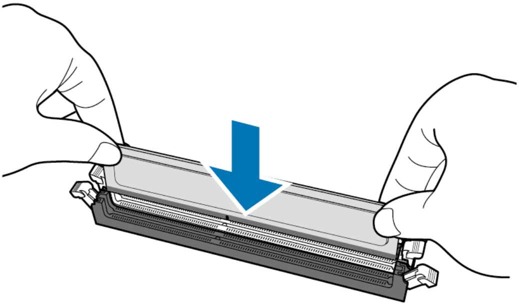
Memory Installation Procedure

Follow these instructions to install memory modules into the S7106.

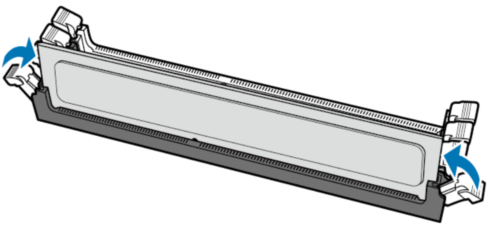
1. Unlock a DIMM socket by Press the retaining clip outwardly in the following illustration.



2. Align the memory module with the socket，such that the DIMM NOTCH match the KEY SLOT on the socket.



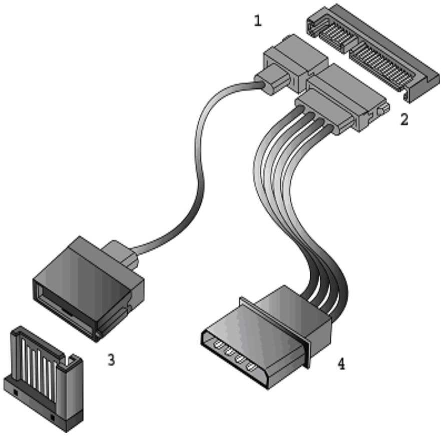
3. Seat the module firmly into the socket by gently pressing down until it sits flush with the socket. The locking levers pop up into place.



2.9 Attaching Drive Cables

Attaching SATA Cables

The following illustrates how to make a SATA Cable connection. If you are in need of SATA/SAS cables or power adapters please contact your local sales representative.



1. SATA drive cable connection

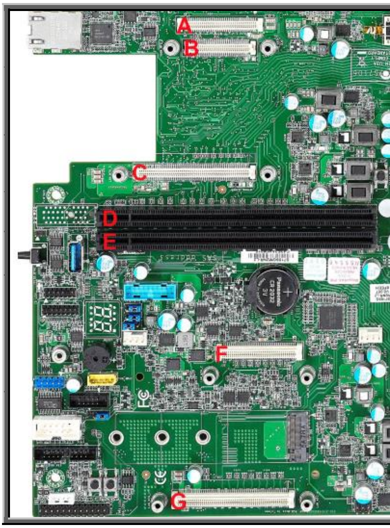
2. SATA drive power connection

3. SATA cable motherboard connector

4. SATA drive power adapter

2.10 Installing Add-In Cards

Before installing add-in cards, it’s helpful to know if they are fully compatible with your motherboard. For this reason, we’ve provided the diagrams below, showing the slots that may appear on your motherboard.

	A+C: OCP 2.0 PCI-E Gen3x16 MEZZ slot f/CPU0 for LAN card
	B: OCP 2.0 KR interface (only available in S7106GM2NR-L2 SKU)
	Standalone C: OCP 1.0 PCI-E Gen3x8 MEZZ slot f/CPU0 for LAN card
	D: PCI-E Gen3x24 slot(x24 link) f/ CPU1
	E: PCI-E Gen3x24 slot(x24 link) f/ CPU0
	F+G: OCP 2.0 PCI-E Gen3x16 MEZZ slot f/CPU1 for Storage Mezz Card
	Standalone G: OCP 1.0 PCI-E Gen3x8 MEZZ slot f/CPU1 for Storage Mezz Card

Simply find the appropriate slot for your add-in card and insert the card firmly. Do not force any add-in cards into any slots if they do not seat in place. It is better to try another slot or return the faulty card rather than damaging both the motherboard and the add-in card.

TIP: It’s a good practice to install add-in cards in a staggered manner rather than making them directly adjacent to each other. Doing so allows air to circulate within the chassis more easily, thus improving cooling for all installed devices.

NOTE: You must always unplug the power connector from the motherboard before performing system hardware changes to avoid damaging the board or expansion device.

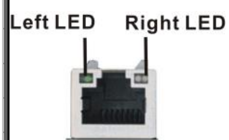
2.11 Connecting External Devices

Connecting external devices to the motherboard is an easy task. The motherboard supports a number of different interfaces through connecting peripherals. See the following diagrams for the details.

NOTE:
An extended function of system ID button is added to the latest BMC firmware release. When hold the front (or rear) ID button for 5 seconds, it will perform the same behavior as pressing the system power button to power on (or off) the system. The implementation is convenient to onsite service technicians that need to press the power button when he/she is at the rear side of the system. Please visit https://www.tyan.com/Motherboards=S7106=S7106GM2NR-L2=downloads=EN to download the latest BMC firmware release and flash it to your systems.

Onboard LAN LED Color Definition

The **Three (3)** onboard Ethernet ports have green and Amber LEDs to indicate LAN status. The chart below illustrates the different LED states.

	1Gbps LAN Link/Activity LED Scheme			
	**Description**		**Left LED (Link/Activity)**	**Right LED (Speed)**
	No Link		OFF	OFF
	10 Mbps	**Link**	Green	OFF
		Active	Blinking Green	OFF
	100 Mbps	**Link**	Green	Solid Green
		Active	Blinking Green	Solid Green
	1Gbps	**Link**	Green	Solid Amber
		Active	Blinking Green	Solid Amber

2.12 Installing the Power Supply

There are **Three (3)** power connectors on your S7106 motherboard. The S7106 supports EPS 12V power supply.

PWR1: ATX 24-Pin Power Connector

1 12

13 24

Signal	Pin	Pin	Signal
+3.3V	**1**	**13**	+3.3V
+3.3V	**2**	**14**	-12V
GND	**3**	**15**	GND
+5V	**4**	**16**	PS-ON#
GND	**5**	**17**	GND
+5V	**6**	**18**	GND
GND	**7**	**19**	GND
PWR_OK	**8**	**20**	RES
5VSB	**9**	**21**	+5V
+12V	**10**	**22**	+5V
+12V	**11**	**23**	+5V
+3.3V	**12**	**24**	GND

PWR2: 8-pin CPU0 Power Connector

1 4

5 8

Signal	Pin	Pin	Signal
GND	**1**	**5**	P0_P12V
GND	**2**	**6**	P0_P12V
GND	**3**	**7**	P0_MEM_P12V
GND	**4**	**8**	P0_MEM_P12V

PWR3: 8-Pin CPU1Power Connector

1 4

5 8

Signal	Pin	Pin	Signal
GND	**1**	**5**	P1_P12V
GND	**2**	**6**	P1_P12V
GND	**3**	**7**	P1_MEM_P12V
GND	**4**	**8**	P1_MEM_P12V

2.13 Finishing Up

Congratulations on making it this far! You have finished setting up the hardware aspect of your computer. Before closing up your chassis, make sure that all cables and wires are connected properly, especially SATA cables and most importantly, jumpers. You may have difficulty powering on your system if the motherboard jumpers are not set correctly.

In the rare circumstance that you have experienced difficulty, you can find help by asking your vendor for assistance. If they are not available for assistance, please find setup information and documentation online at our website or by calling your vendor's support line.

Chapter 3: BIOS Setup

3.1 About the BIOS

The BIOS is the basic input/output system, the firmware on the motherboard that enables your hardware to interface with your software. The BIOS determines what a computer can do without accessing programs from a disk. The BIOS contains all the code required to control the keyboard, display screen, disk drives, serial communications, and a number of miscellaneous functions. This chapter describes the various BIOS settings that can be used to configure your system.

The BIOS section of this manual is subject to change without notice and is provided for reference purposes only. The settings and configurations of the BIOS are current at the time of print and are subject to change, and therefore may not match exactly what is displayed on screen.

This section describes the BIOS setup program. The setup program lets you modify basic configuration settings. The settings are then stored in a dedicated, battery-backed memory (called NVRAM) that retains the information even when the power is turned off.

To start the BIOS setup utility:

1. Turn on or reboot your system.
2. Press <**F2**> or <**Del>** during POST (<**Tab**> on remote console) to start the BIOS setup utility.

3.1.1 Setup Basics

The table below shows how to navigate in the setup program using the keyboard.

Key	Function
↑ ↓ →←	Move cursor
<Enter>	Execute command or select submenu
<->/<+>	Select the previous or next value/setting of the field
<ESC>	Exit current menu
<F1>	General help
<F2>	Previous values
<F3>	Load the Optimal default configuration values of the menu
<F4>	Save and exit
<K>	Scroll help area upwards
<M>	Scroll help area downwards
<PgUp> / <PgDn>	Move cursor to next/previous page

3.1.2 Getting Help

Pressing [**F1**] will display a small help window that describes the appropriate keys to use and the possible selections for the highlighted item. To exit the Help Window, press [**ESC**] or the [**Enter**] key again.

3.1.3 In Case of Problems

If you have trouble booting your computer after making and saving the changes with the BIOS setup program, you can restart the computer by holding the power button down until the computer shuts off (usually within 4 seconds); resetting by pressing CTRL-ALT-DEL; or clearing the CMOS.
The best advice is to only alter settings that you thoroughly understand. In particular, do not change settings in the Chipset section unless you are absolutely sure of what you are doing. The Chipset defaults have been carefully chosen either by MiTAC or your system manufacturer for best performance and reliability. Even a seemingly small change to the Chipset setup options may cause the system to become unstable or unusable.

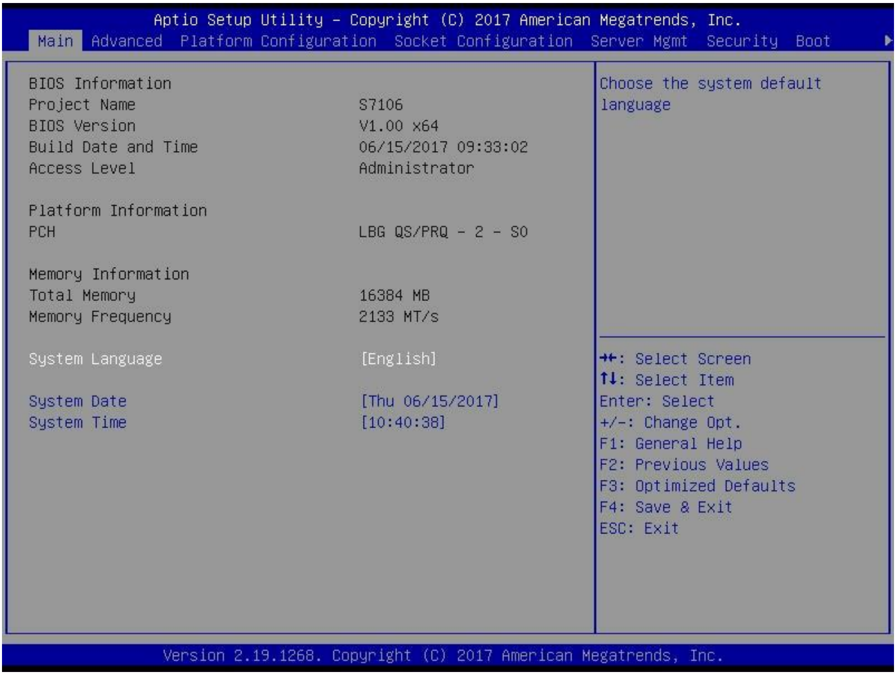
3.1.4 Setup Variations

Not all systems have the same BIOS setup layout or options. While the basic look and function of the BIOS setup remains more or less the same for most systems, the appearance of your Setup screen may differ from the charts shown in this section. Each system design and chipset combination requires a custom configuration. In addition, the final appearance of the Setup program depends on the system designer. Your system designer may decide that certain items should not be available for user configuration, and remove them from the BIOS setup program.

NOTE: The following pages provide the details of BIOS menu. Please be aware that the BIOS menus are continually changing due to continual BIOS updates over the product lifespan of the motherboard. The BIOS menus provided are current as of the date when this manual was written. Please visit TYAN's website at http://www.tyan.com for information on BIOS updates available for this specific motherboard.

3.2 Main Menu

In this section, you can alter general features such as the date and time.
Note that the options listed below are for options that can directly be changed within the Main Setup screen.



BIOS Information
It displays BIOS related information.

Platform Information
It displays Platform information.

Memory Information
This displays the total memory size.

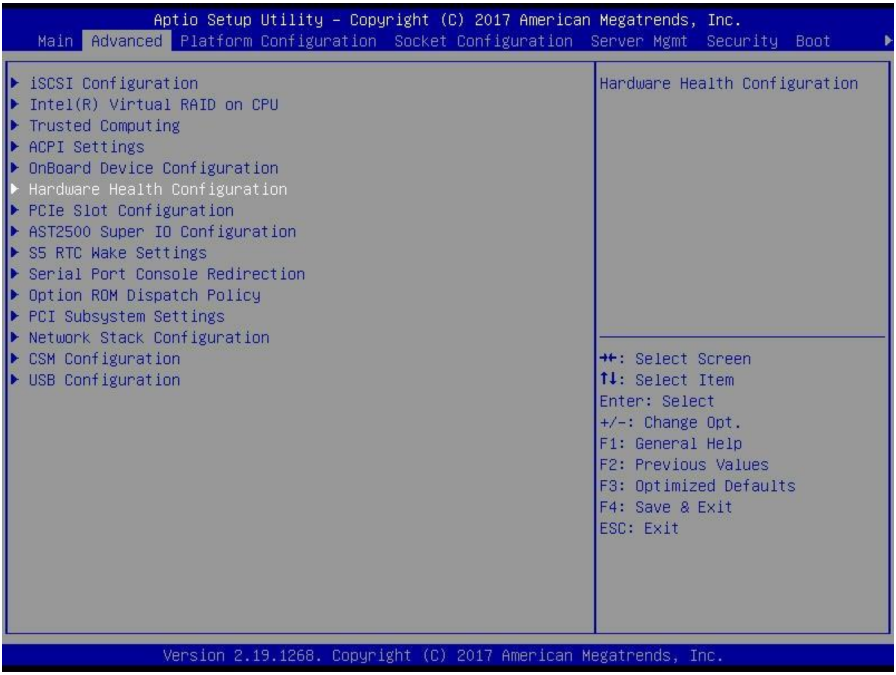
System Language
Choose the system default language

System Date
Adjust the system date.
MM (Months): DD (Days): YYYY (Years)

System Time
Adjust the system clock.
HH (24 hours format): MM (Minutes): SS (Seconds)

3.3 Advanced Menu

This section facilitates configuring advanced BIOS options for your system.



iSCSI Configuration
Configure the iSCSI parameters

Intel® Virtual RAID on CPU
This formset allows the user to manage Intel® Virtual RAID on CPU

Trusted Computing
Trusted Computing settings.

ACPI Settings
System ACPI Parameters.

Onboard Device Configuration
Onboard Device and Function Configuration.

Hardware Health Configuration
Hardware Health Configuration

PCIe Slot Configuration
Onboard PCIe Slot Configuration

AST2500 Super IO Configuration
System Super IO Chip Parameters.

S5 RTC Wake Settings
S5 RTC Wake Settings

Serial Port Console Redirection
Serial Port Console Redirection

Option ROM Dispatch Policy
Option ROM Dispatch Policy

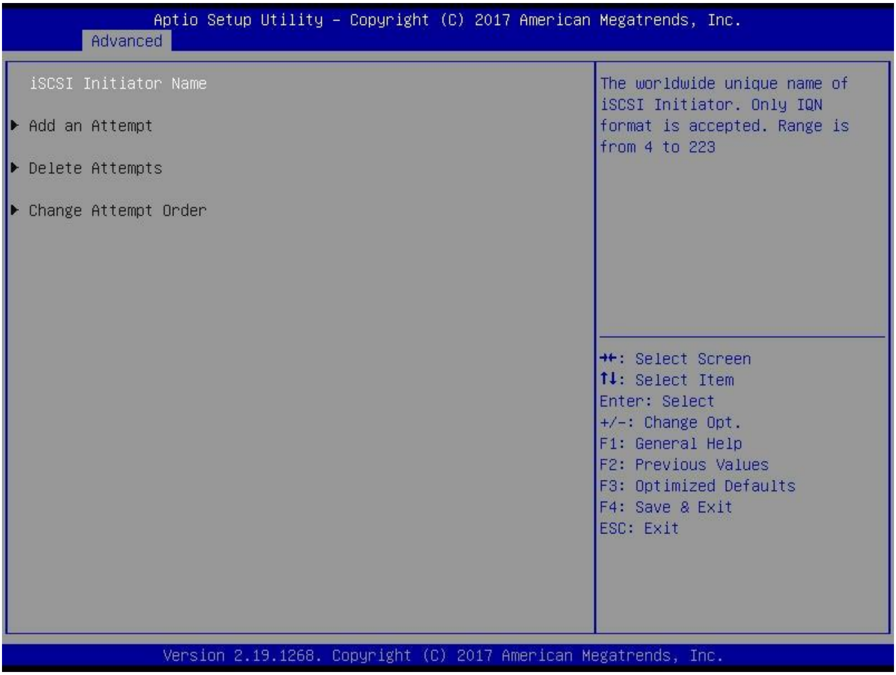
PCI Subsystem Settings
PCI, PCI-X and PCI Express Settings

Network Stack Configuration
Network Stack Settings

CSM Configuration
CSM Configuration, Enable/Disable Option ROM execution setting, etc

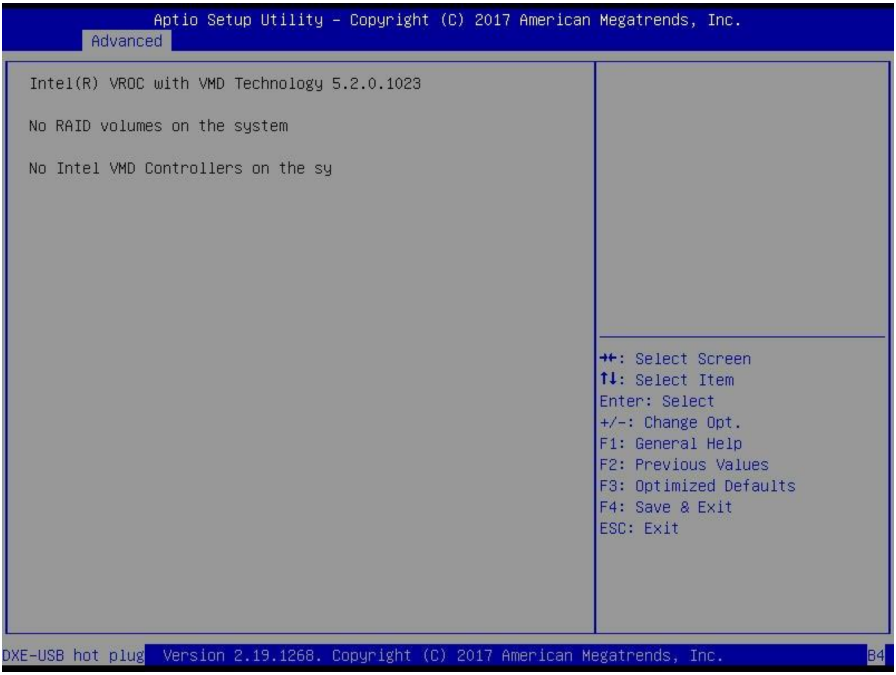
USB Configuration
USB Configuration Parameters.

3.3.1 iSCSI Configuration

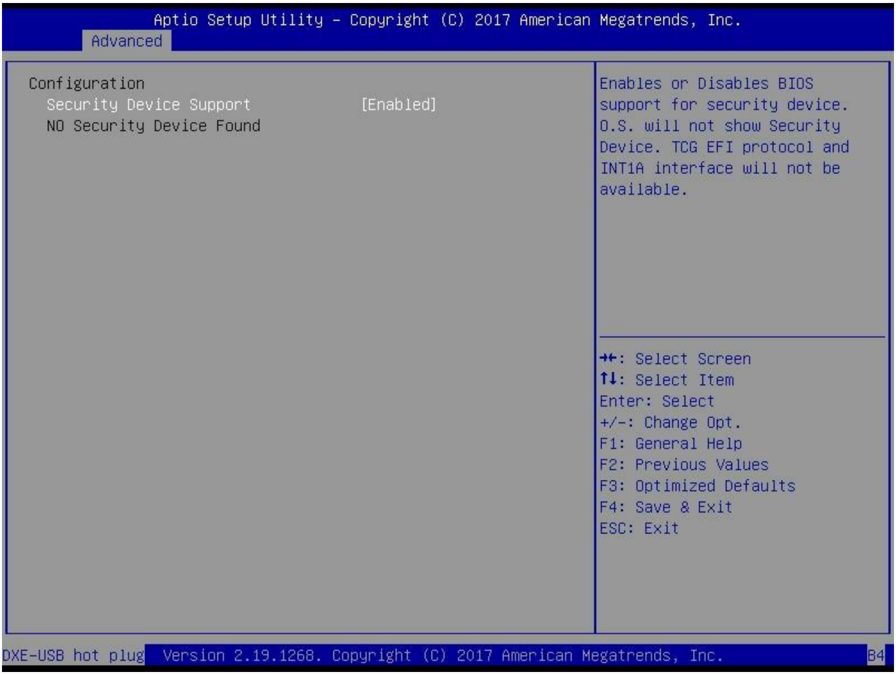


iSCSI Name
The worldwide unique name of iSCSI Initiator. Only IQN format is accepted. Range is from 4 to 223.

3.3.2 Intel(R) Virtual RAID on CPU



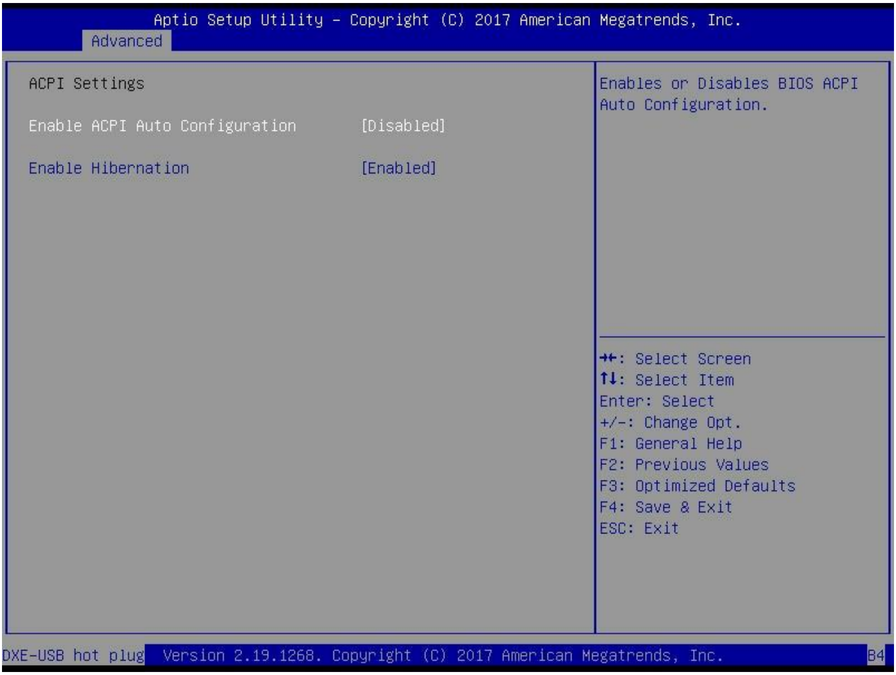
3.3.3 Trusted Computing



Security Device Support
Enable or disable BIOS support for security device. O.S. will not show Security device. O.S. will not show Security Device. TCG EFI protocol and INT1A interface will not be available.

**Enabled / **Disabled

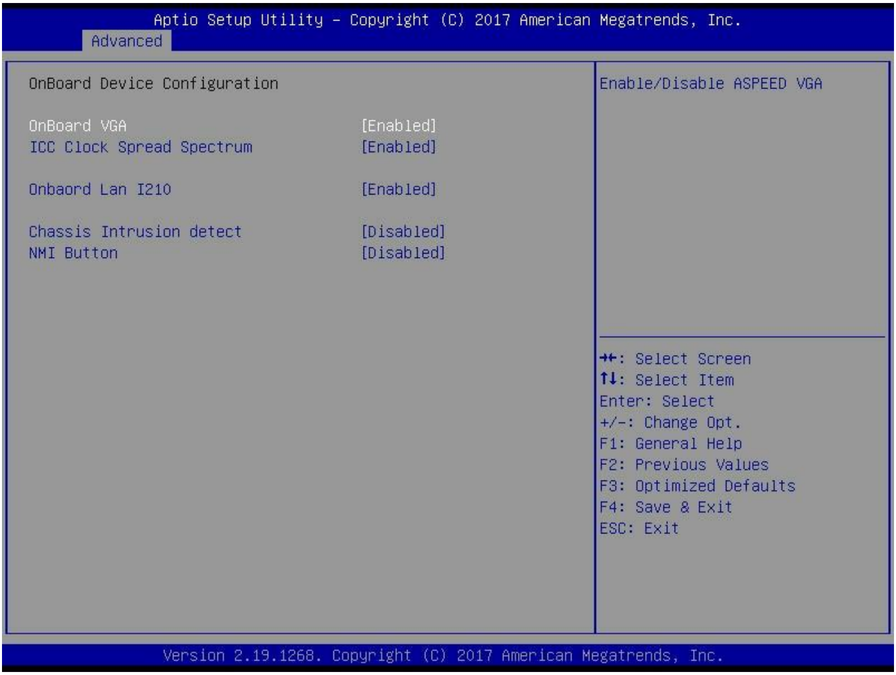
3.3.4 ACPI Settings



Enable ACPI Auto Configuration
Enable or disable BIOS ACPI Auto Configuration.
Disabled / Enabled

Enable Hibernation
Enables or disables System ability to Hibernate (OS/S4 Sleep State). This option may not be effective with some OS.
Disabled / **Enabled**

3.3.5 Onboard Device Configuration



Onboard VGA
Enable/Disable ASPEED VGA
Disabled / **Enabled**

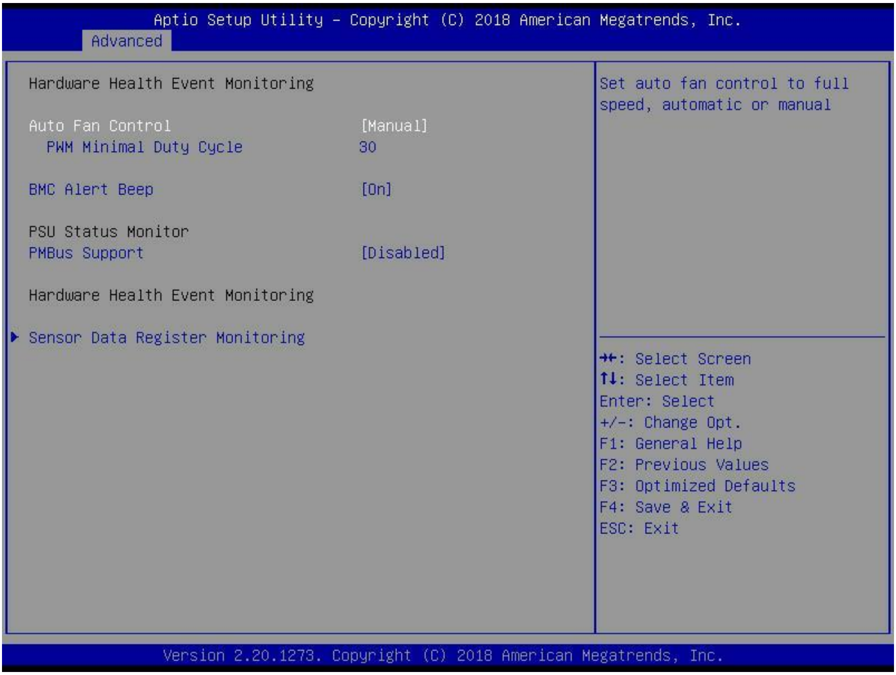
ICC Clock Spread Spectrum
Turn on / off Spread Spectrum Setting for IsCLK.
Enabled / Disabled

Onboard Lan I210
Enable/Disable Onboard Lan I210
Disabled / **Enabled**

Chassis Intrusion detect
Enabled: When a chassis open event is detected, the BIOS will record the event.
Disabled / Enabled

NMI Button
Enable or disable NMI button.
Disabled / Enabled

3.3.5 Hardware Health Configuration



Auto Fan Control
Auto Fan Control help.
Full speed **/ Manual**

NOTE: Auto Fan Control must be set to [Manual] PWM Minimal Duty Cycle menu will appear.

PWM Minimal Duty Cycle

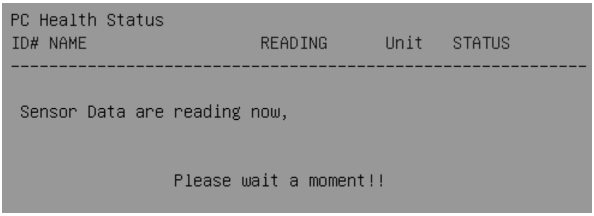
PWM Minimal Duty Cycle
30

BMC Alert Beep
Enable/Disable BMC Alert Beep.
On / Off

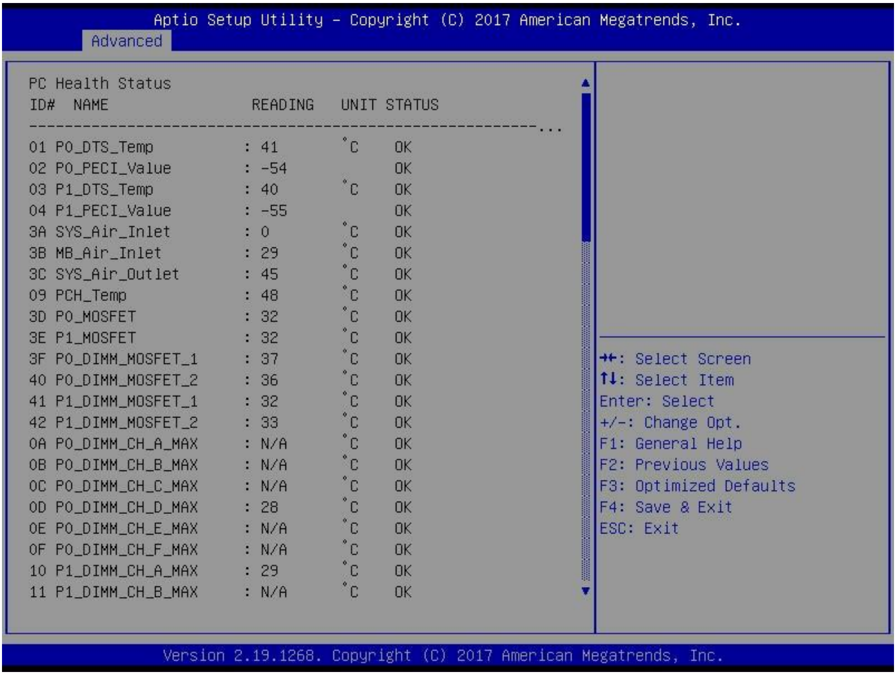
PMBus support
PMBus Support
Disabled / Enabled

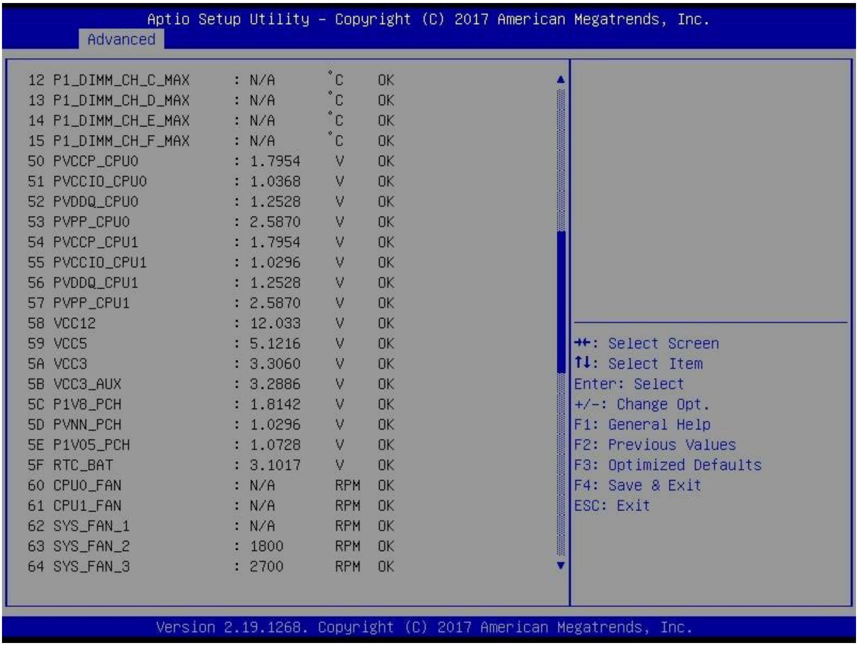
3.3.5.1 Sensor Data Register Monitoring

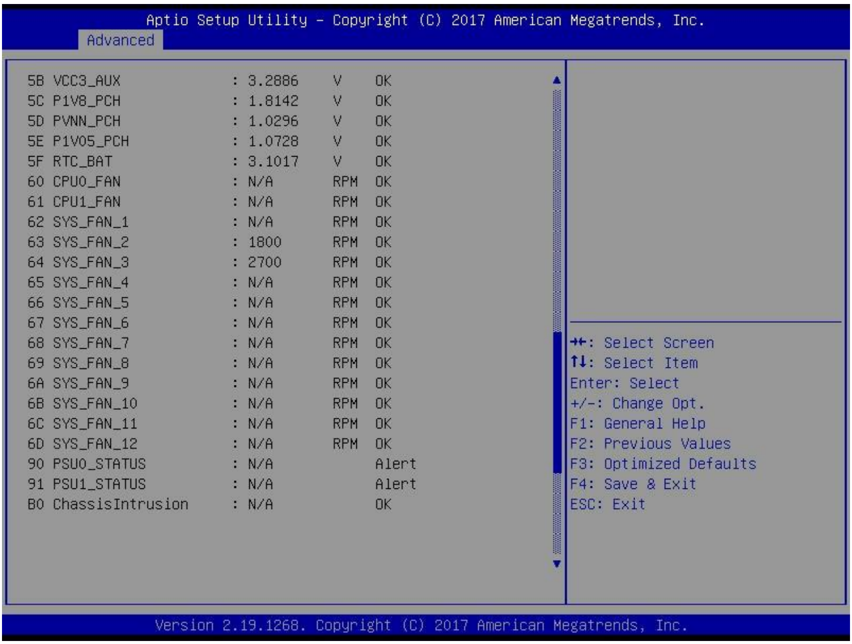
When you enter the **Sensor Data Register Monitoring** submenu, you will see the following dialog window pop out. Please wait 8~10 seconds.



NOTE 1: SDR can not be modified. Read only.


Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Advanced
PC Health Status
ID# NAME READING UNIT STATUS
01 P0_DTS_Temp : 41 °C OK
02 P0_PECI_Value : -54 OK
03 P1_DTS_Temp : 40 °C OK
04 P1_PECI_Value : -55 OK
3A SYS_Air_Inlet : 0 °C OK
3B MB_Air_Inlet : 29 °C OK
3C SYS_Air_Outlet : 45 °C OK
09 PCH_Temp : 48 °C OK
3D P0_MOSFET : 32 °C OK
3E P1_MOSFET : 32 °C OK
3F P0_DIMM_MOSFET_1 : 37 °C OK
40 P0_DIMM_MOSFET_2 : 36 °C OK
41 P1_DIMM_MOSFET_1 : 32 °C OK
42 P1_DIMM_MOSFET_2 : 33 °C OK
0A P0_DIMM_CH_A_MAX : N/A °C OK
0B P0_DIMM_CH_B_MAX : N/A °C OK
0C P0_DIMM_CH_C_MAX : N/A °C OK
0D P0_DIMM_CH_D_MAX : 28 °C OK
0E P0_DIMM_CH_E_MAX : N/A °C OK
0F P0_DIMM_CH_F_MAX : N/A °C OK
10 P1_DIMM_CH_A_MAX : 29 °C OK
11 P1_DIMM_CH_B_MAX : N/A °C OK
→←: Select Screen
↑↓: Select Item
Enter: Select
+/-: Change Opt.
F1: General Help
F2: Previous Values
F3: Optimized Defaults
F4: Save & Exit
ESC: Exit
Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.


Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Advanced
12 P1_DIMM_CH_C_MAX : N/A °C OK
13 P1_DIMM_CH_D_MAX : N/A °C OK
14 P1_DIMM_CH_E_MAX : N/A °C OK
15 P1_DIMM_CH_F_MAX : N/A °C OK
50 PVCCP_CPU0 : 1.7954 V OK
51 PVCCIO_CPU0 : 1.0368 V OK
52 PVDDQ_CPU0 : 1.2528 V OK
53 PVPP_CPU0 : 2.5870 V OK
54 PVCCP_CPU1 : 1.7954 V OK
55 PVCCIO_CPU1 : 1.0296 V OK
56 PVDDQ_CPU1 : 1.2528 V OK
57 PVPP_CPU1 : 2.5870 V OK
58 VCC12 : 12.033 V OK
59 VCC5 : 5.1216 V OK
5A VCC3 : 3.3060 V OK
5B VCC3_AUX : 3.2886 V OK
5C P1V8_PCH : 1.8142 V OK
5D PVNN_PCH : 1.0296 V OK
5E P1V05_PCH : 1.0728 V OK
5F RTC_BAT : 3.1017 V OK
60 CPU0_FAN : N/A RPM OK
61 CPU1_FAN : N/A RPM OK
62 SYS_FAN_1 : N/A RPM OK
63 SYS_FAN_2 : 1800 RPM OK
64 SYS_FAN_3 : 2700 RPM OK
→←: Select Screen
↑↓: Select Item
Enter: Select
+/-: Change Opt.
F1: General Help
F2: Previous Values
F3: Optimized Defaults
F4: Save & Exit
ESC: Exit
Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.

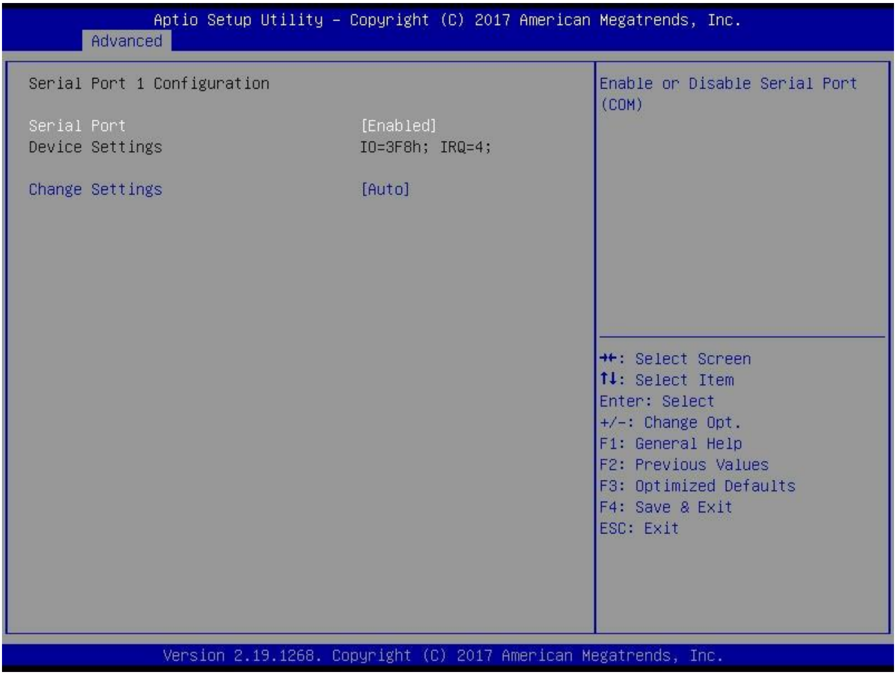

Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Advanced
5B VCC3_AUX : 3.2886 V OK
5C P1V8_PCH : 1.8142 V OK
5D PVNN_PCH : 1.0296 V OK
5E P1V05_PCH : 1.0728 V OK
5F RTC_BAT : 3.1017 V OK
60 CPU0_FAN : N/A RPM OK
61 CPU1_FAN : N/A RPM OK
62 SYS_FAN_1 : N/A RPM OK
63 SYS_FAN_2 : 1800 RPM OK
64 SYS_FAN_3 : 2700 RPM OK
65 SYS_FAN_4 : N/A RPM OK
66 SYS_FAN_5 : N/A RPM OK
67 SYS_FAN_6 : N/A RPM OK
68 SYS_FAN_7 : N/A RPM OK
69 SYS_FAN_8 : N/A RPM OK
6A SYS_FAN_9 : N/A RPM OK
6B SYS_FAN_10 : N/A RPM OK
6C SYS_FAN_11 : N/A RPM OK
6D SYS_FAN_12 : N/A RPM OK
90 PSU0_STATUS : N/A Alert
91 PSU1_STATUS : N/A Alert
B0 ChassisIntrusion : N/A OK
→←: Select Screen
↑↓: Select Item
Enter: Select
+/-: Change Opt.
F1: General Help
F2: Previous Values
F3: Optimized Defaults
F4: Save & Exit
ESC: Exit
Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.

3.3.7 AST2500 Super IO Configuration

Serial Port 1 Configuration
Set Parameters of Serial Port 1 (COMA)

Serial Port 2 Configuration
Set Parameters of Serial Port 2 (COMA)

3.3.7.1 Serial Port 1/2 Configuration

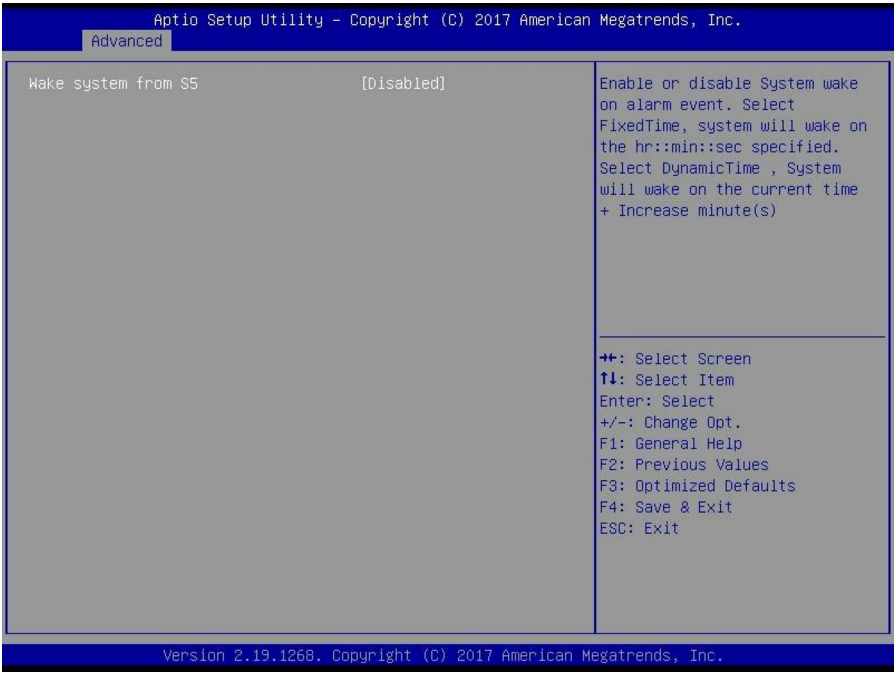


Serial Port
Enable or disable Serial Port (COM).
Enabled / Disabled

Device Settings
Read only.

Change Settings
Select an optimal setting for Super IO Device.
Auto / IO=2F8h; IRQ=3;
/ IO=3F8h, IRQ=3, 4, 5, 6, 7, 9, 10, 11, 12;
/ IO=2F8h; IRQ=3, 4, 5, 6, 7, 9, 10, 11, 12;
/ IO=3E8h, IRQ=3, 4, 5, 6, 7, 9, 10, 11, 12;
/ IO=2E8h, IRQ=3, 4, 5, 6, 7, 9, 10, 11, 12;

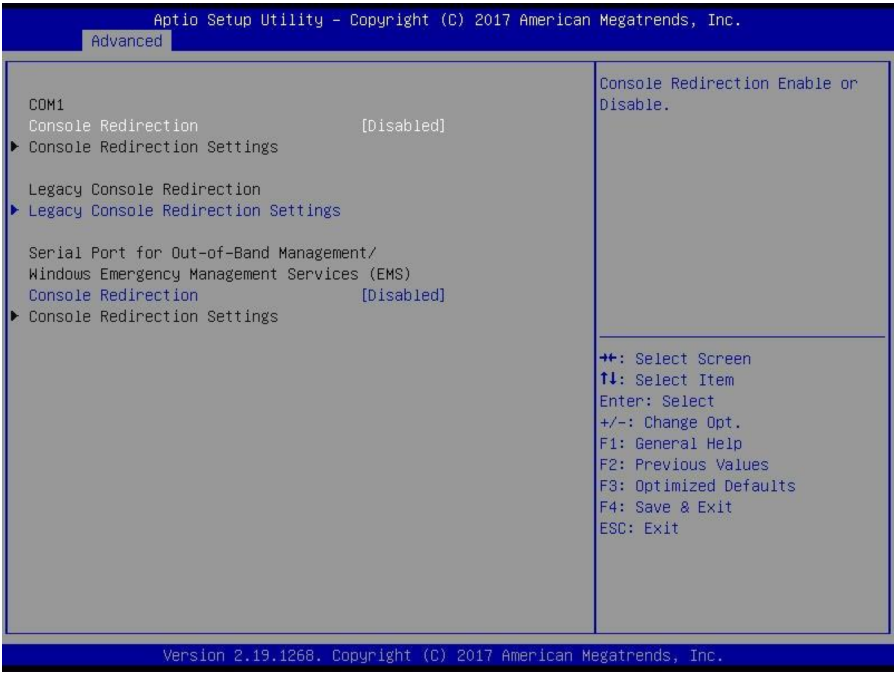
3.3.8 S5 RTC Wake Settings



Wake system from S5
Enable or disable system wake on alarm event. Select Fixed time, system will wake on the hr::min::sec specified. Select dynamic time, system will wake on the current time+ increase minute(s)

Disabled / Fixed time / Dynamic time

3.3.9 Serial Port Console Redirection



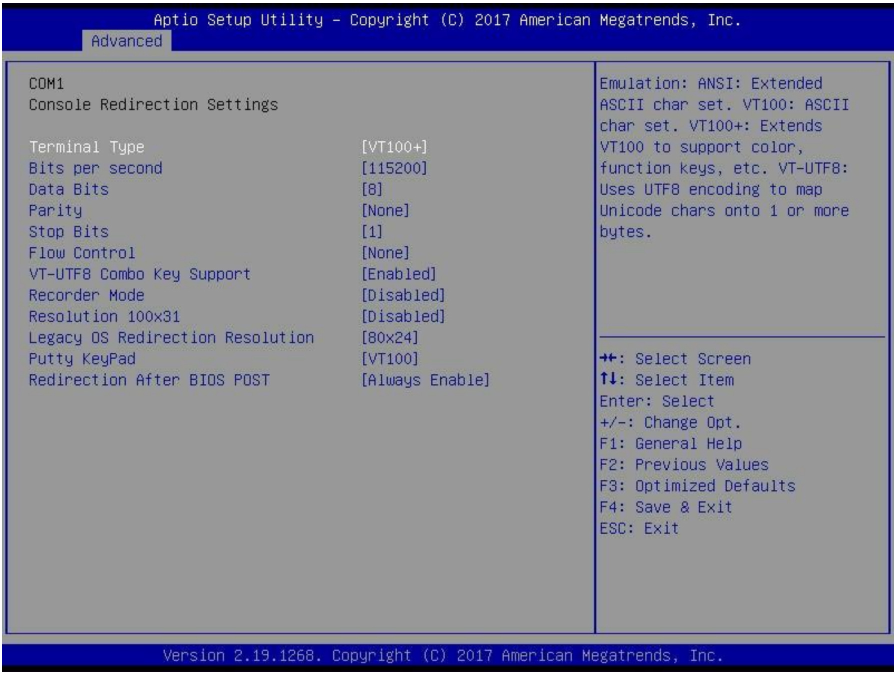
Console Redirection
Console redirection enable or disable.
Disabled **/ Enabled**

Legacy Console Redirection
Legacy Console Redirection Settings

Serial Port for Out-Of-Band Management/Windows Emergency Services (EMS)
Console Redirection
Console redirection enable or disable.
Disabled **/ Enabled**

Console Redirection Settings
The settings specify how the host computer (which the user is using) will exchange data. Both computers should have the same or compatible settings.
NOTE: Console Redirection Settings menu only appear when **Console Redirection** was set to **[Enabled].**

3.3.9.1 Console Redirection Settings



Terminal Type

Emulation: ANSI: Extended ASCII char set.
VT100: ASCII char set.
VT100+: Extends VT100 to support color function keys, etc.
VT-UTF8: Uses UTF8 encoding to map Unicode chars onto 1 or more bytes.

VT-UTF8 **/** VT100 / **VT100+** / ANSI

Bits per Second

Select serial port transmission speed. The speed must be matched on the other side. Long or noisy lines may require lower speeds.

38400 **/** 9600 / 19200 / **115200** / 57600

Data Bits

8 / 7

Parity

A parity bit can be sent with the data bits to detect some transmission errors. Even: parity bit is 0 if the num of 1’s in the data bits is even. Odd: parity bit is 0 if the num of 1’s in the data bits is odd. Mark: parity bit is always 1. Space: parity bit is always 0. Mark and Space parity do not allow for error detection.

None / Even / Odd / Mark / Space

Stop Bits
Stop bits indicate the end of a serial data packet. (A start bit indicates the beginning). The standard setting is 1 stop bit. Communication with slow devices may require more than 1 stop bit.

1 / 2

Flow Control
Flow Control can prevent data loss from buffer overflow. When sending data, if the receiving buffers are full, a 'stop' signal can be sent to stop the data flow. Once the buffers are empty, a 'start' signal can be sent to re-start the flow. Hardware flow control uses two wires to send start/stop signal.

None / Hardware RTS/CTS

VT-UTF8 Combo Key Support
Enable VT-UTF8 Combination Key Support for ANSI/VT100 terminals.

Enabled / Disabled

Recorder Mode
With this mode enabled only text will be sent. This is to capture Terminal data.

Disabled / Enabled

Resolution 100x31
Enable or disable extended terminal resolution.

Disabled / Enabled

Legacy OS Redirection Resolution
On Legacy OS, the number of rows and columns supported redirection.

80x24 / 80x25

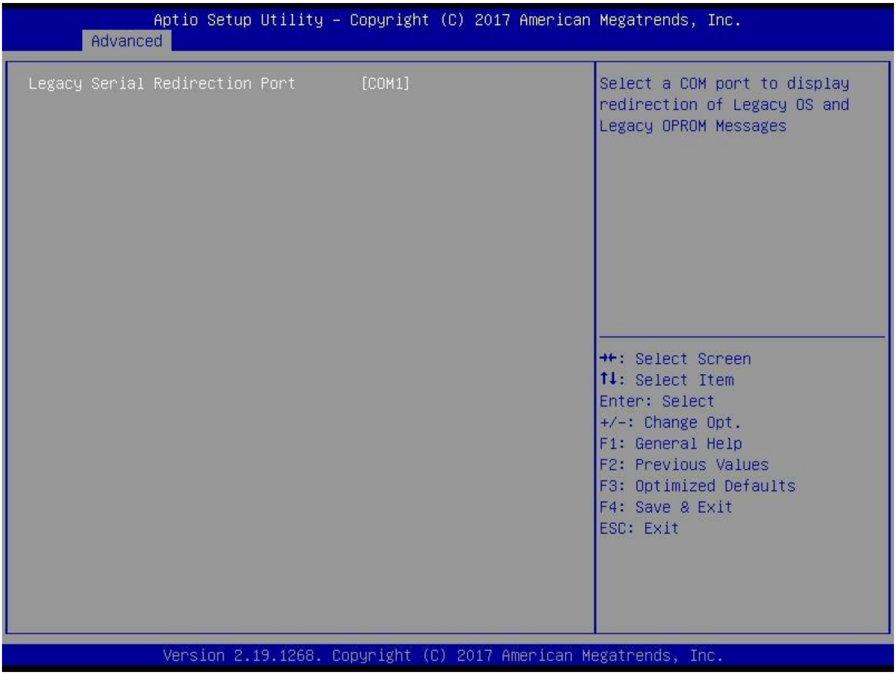
Putty KeyPad
Select FunctionKey and KeyPad on Putty.

VT100 / LINUX / XTERMR6 / SCO / ESCN / VT400

Redirection After BIOS POST
The settings specify if BootLoader is selected than Legacy console redirection is disabled before booting to Legacy OS. Default value is always enable means Legacy.

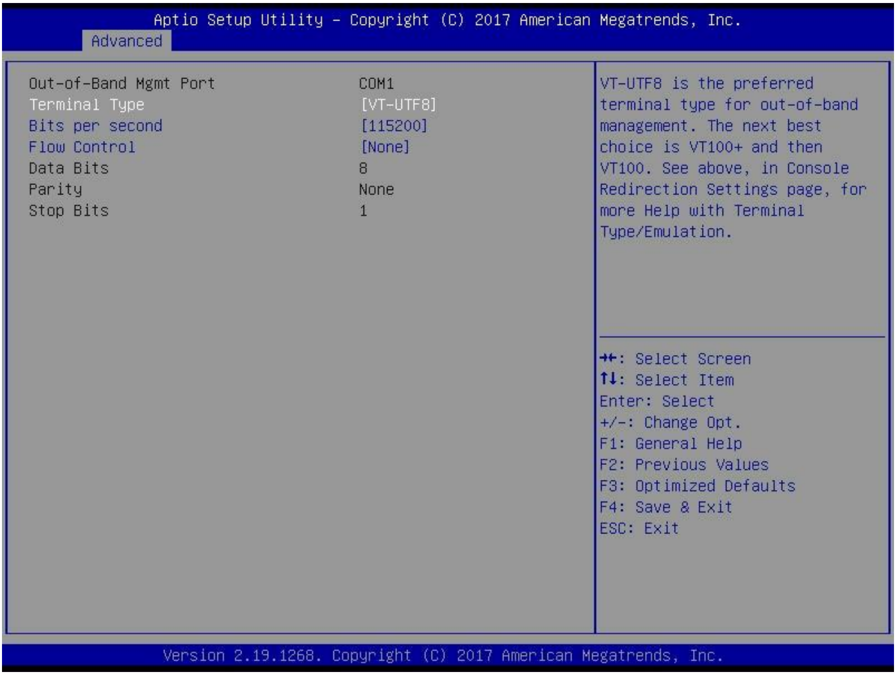
Always Enable / Bootloader

3.3.9.2 Legacy Console Redirection Settings



Legacy Serial Redirected Port
Select a COM port to display redirection of Legacy OS and Legacy OPROM Messages
COM1

3.3.9.3 Serial Port for Out-Of-Band Management/Windows Emergency Services (EMS) Console Redirection Settings



Out-of Band Mgmt Port
Microsoft Windows Emergency Management Services (EMS) allows for remote management of a Windows Server OS through a serial port.
COM1

Terminal Type
VT-UTF8 is the preferred terminal type for out-of-band management. The next best choice is VT100+ and then VT100. See above, in Console Redirection Settings page, for more Help with Terminal Type/Emulation.
VT-UTF8 / VT100 / VT100+ / ANSI

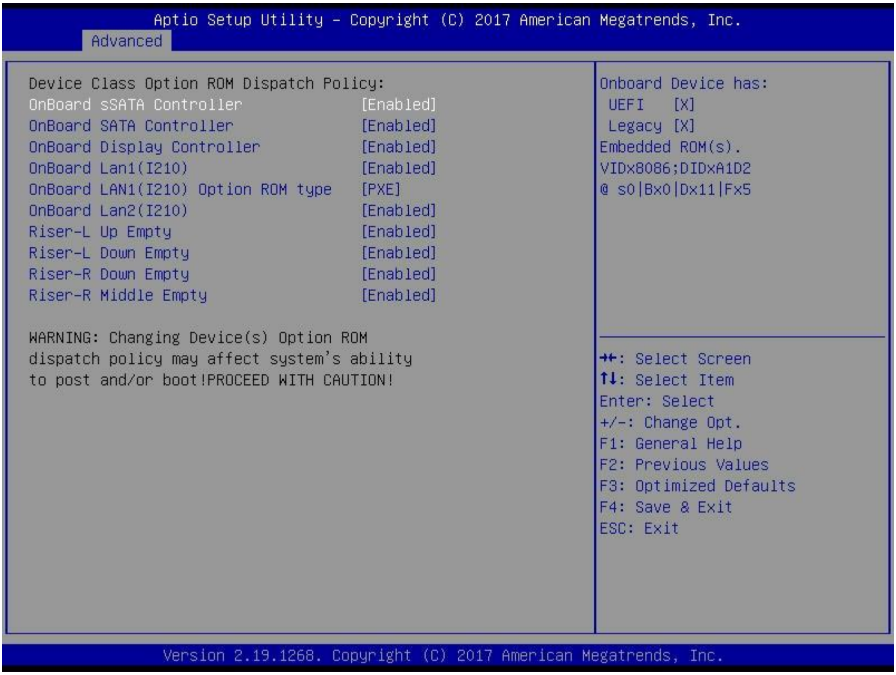
Bits per Second
Select serial port transmission speed. The speed must be matched on the other side. Long or noisy lines may require lower speeds.
115200 / 9600 / 19200 / 38400 / 57600

Flow Control
Flow Control can prevent data loss from buffer overflow. When sending data, if the receiving buffers are full, a 'stop' signal can be sent to stop the data flow. Once the buffers are empty, a 'start' signal can be sent to restart the flow. Hardware flow control uses two wires to send start/stop signal.

None / Hardware RTS/CTS

Data Bits / Parity / Stop Bits
Read only.

3.3.10 Option ROM Dispatch Subsystem



OnBoard sSATA Controller
Onboard Device has:
UEFI [X]
Legacy [X]
Embedded ROM(s).
VIDx8086:DIOxA1D2
@ s0(Bx0) Dx11/ Fx5
Disabled **/ Enabled**

OnBoard SATA Controller
Onboard Device has:
UEFI [X]
Legacy [X]
Embedded ROM(s).
VIDx8086:DIOxA182
@ s0(Bx0) Dx17/ Fx0
Disabled **/ Enabled**

OnBoard Display Controller
Onboard Device has:
UEFI [X]
Legacy [X]
Embedded ROM(s).
VIDx1A03;DIOx2000
@ s0(Bx5) Dx0/ Fx0
Disabled **/ Enabled**

OnBoard Lan1(I210)
Onboard Device has:
UEFI [X]
Legacy [X]
Embedded ROM(s).
VIDx8086;DIOx1533
@ s0(Bx2) Dx0/ Fx0
Disabled **/ Enabled**

OnBoard LAN1(I210) Option ROM type
Select PXE or iSCSI OpROM for OnBoard LAN1(I210)
PXE / ISCSI

OnBoard Lan2(I210)
Onboard Device has:
UEFI [X]
Legacy [X]
Embedded ROM(s).
VIDx8086;DIOx1533
@ s0(Bx2) Dx0/ Fx0
Disabled **/ Enabled**

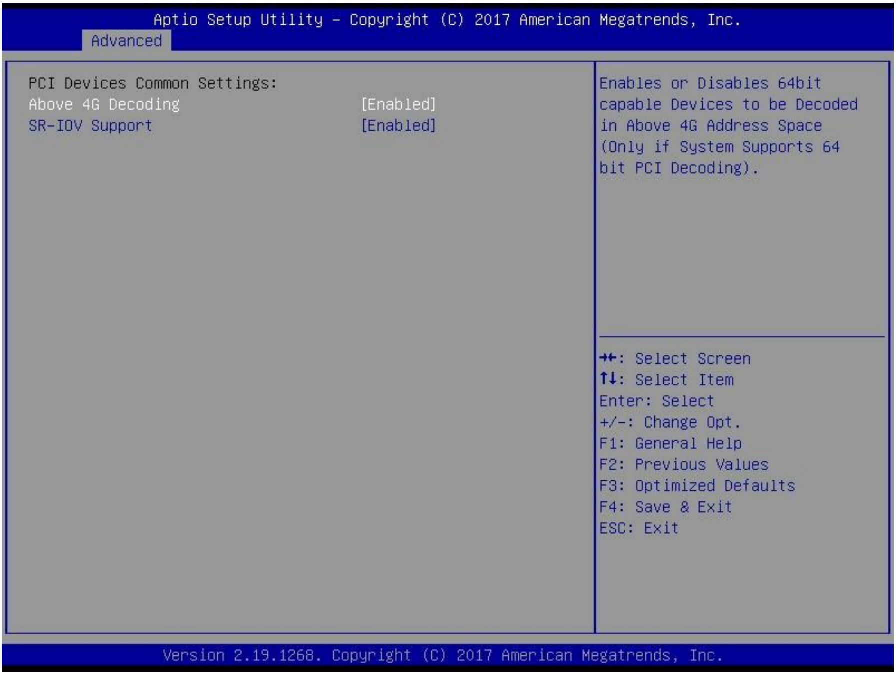
Riser-L Up Empty
Enable or Disable Option ROM execution for selected Slot.
Disabled **/ Enabled**

Riser-L Down Empty
Enable or Disable Option ROM execution for selected Slot.
Disabled **/ Enabled**

Riser-R Down Empty
Enable or Disable Option ROM execution for selected Slot.
Disabled **/ Enabled**

Riser-R Middle Empty
Enable or Disable Option ROM execution for selected Slot.
Disabled **/ Enabled**

3.3.11 PCI Subsystem



Above 4G Decoding

Enables or Disables 64bit capable Devices to be decoded in Above 4G Address Space(Only if System supports 64 bit PCI decoding).

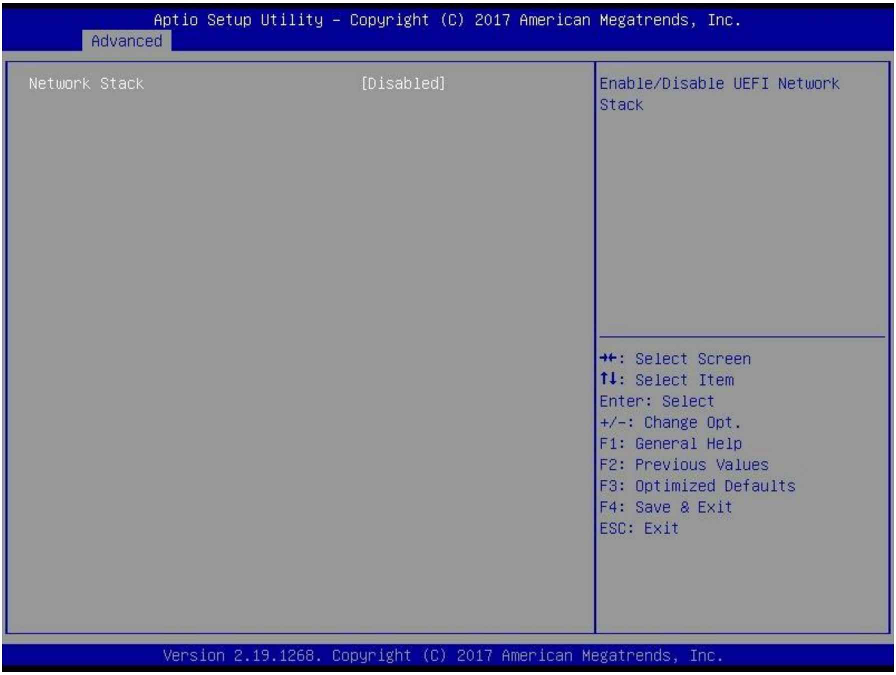
Enabled / Disabled

SR-IOV Support

If system has SR-IOV capable PCIe devices, this option Enable or Disable Single root IO virtualization Support

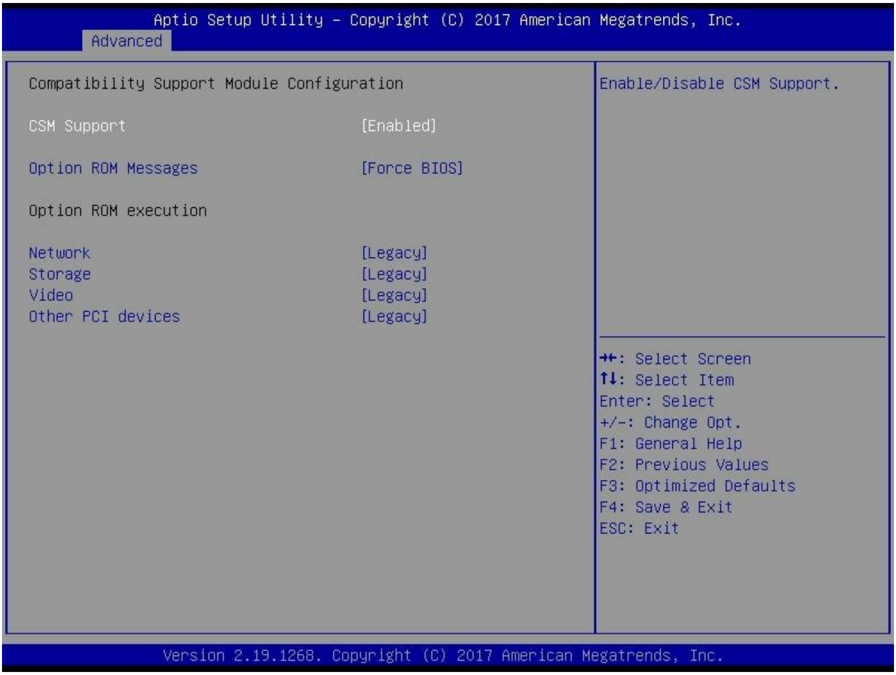
Enabled / Disabled

3.3.12 Network Stack Configuration



Network Stack
Enable / Disable UEFI Network Stack.
Disabled / Enabled

3.3.13 CSM Configuration



CSM support
Enable/Disable CSM Support
Enabled / Disabled

Option ROM Messages
Set display mode for Option ROM
Force BIOS / Keep Current

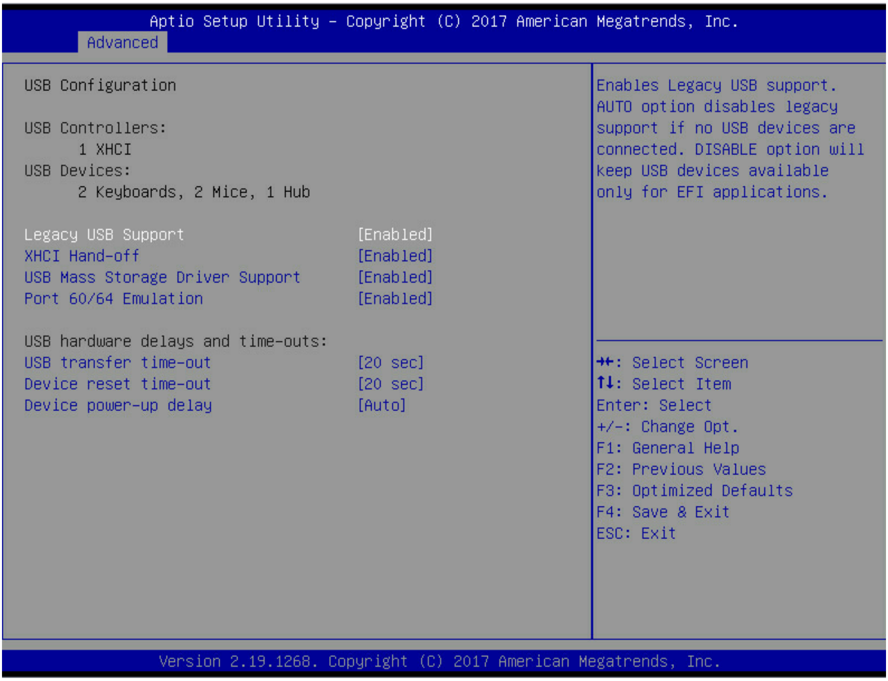
Network
Controls the execution of UEFI and legacy PXE OpROM
Do not launch / UEFI / **legacy**

Storage
Controls the execution of UEFI and legacy PXE OpROM
Do not launch / UEFI / **legacy** /

Video
Controls the execution of UEFI and legacy PXE OpROM
Do not launch / UEFI / **legacy**

Other PCI devices
Determines OpRom execution policy for devices other than network, storage, or video
legacy / UEFI

3.3.14 USB Configuration



Legacy USB Support
Enables USB legacy support. AUTO option disables legacy support if no USB devices are connected. DISABLE option will keep USB devices available only for EFI applications.
Enabled / Disabled / Auto

XHCI Hand-off
This is a workaround for OSes without XHCI hand-off support. The XHCI ownership change should be claimed by XHCI driver.
Enabled / Disabled

USB Mass Storage Driver Support
Enable/Disable USB Mass Storage Driver Support.
Disabled **/ Enabled**

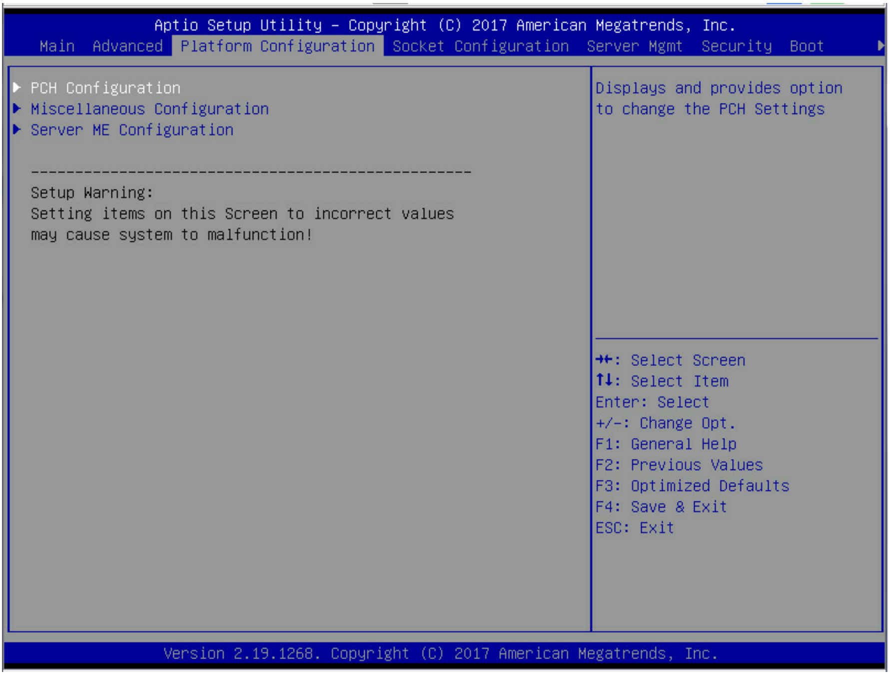
Port 60/64 Emulation
Enable I/O port 60h/64h emulation support. This should be enabled for the complete USB keyboard legacy support for non-USB aware OSes.
Disabled **/ Enabled**

USB transfer time-out
The time-out value for Control, Bulk and Interrupt transfers.
20 sec / 10 sec / 5 sec / 1 sec

Device reset time-out
USB mass storage device Start Unit command time-out.
20 sec / 10 sec / 30 sec / 40 sec

Device power-up delay
Maximum time the device will take before it properly reports itself to the Host Controller. AUTO uses default value: for a Root port it is 100 ms, for a Hub port the delay is taken from Hub descriptor.
Auto / Manual

3.4 Platform Configuration Menu

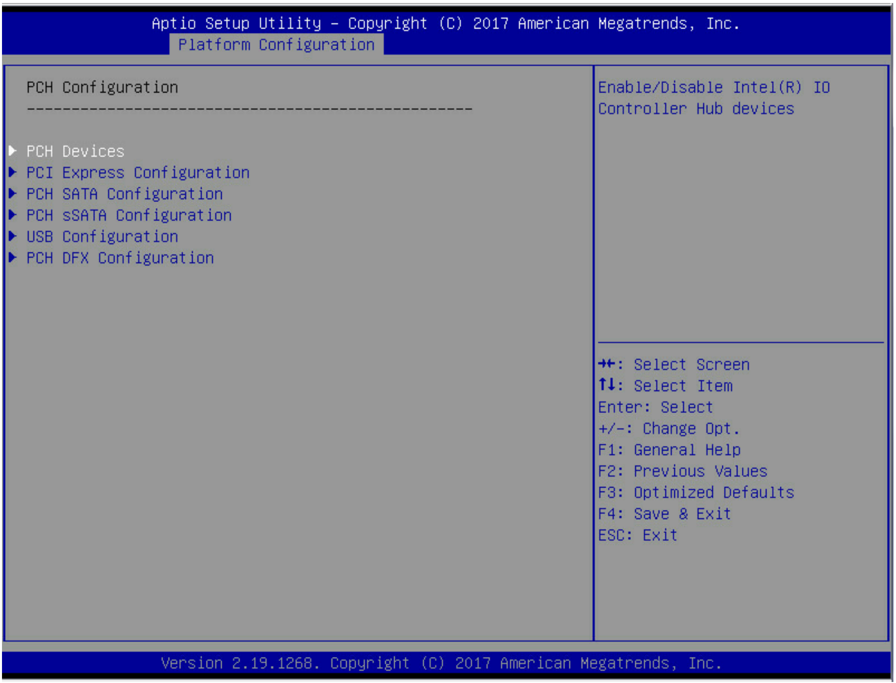


PCH Configuration
Displays and provides option to change the PCH Settings.

Miscellaneous Configuration
Miscellaneous Configuration

Server ME Configuration
Configure Server ME technology Parameters

3.4.1 PCH Configuration



PCH Devices
Enable/Disable Intel(R) IO Controller Hub devices

PCI Express Configuration
PCI Express Configuration settings

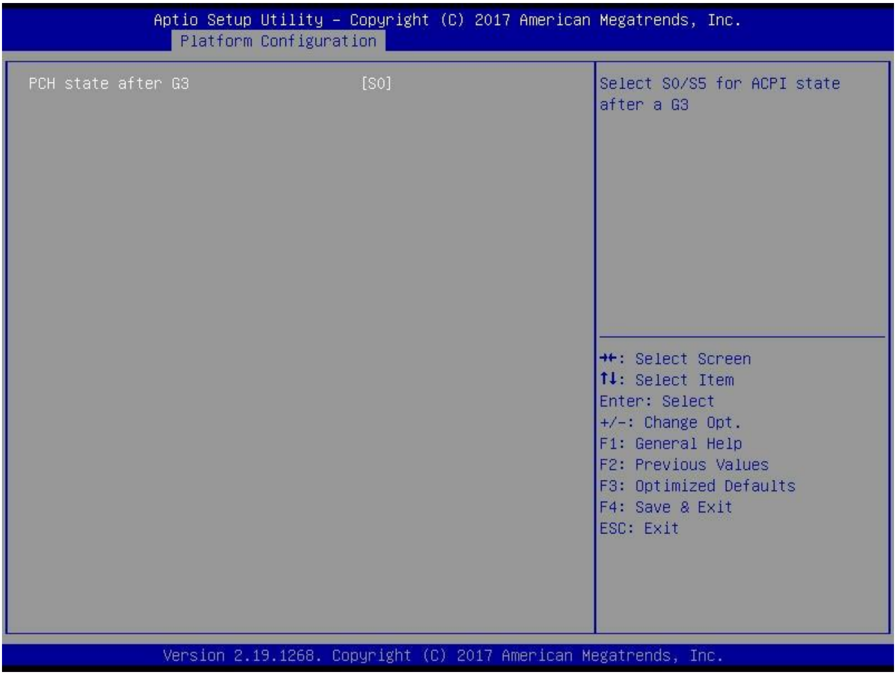
PCH SATA Configuration
SATA devices and settings

PCH sSATA Configuration
sSATA devices and settings

USB Configuration
USB Configuration Settings

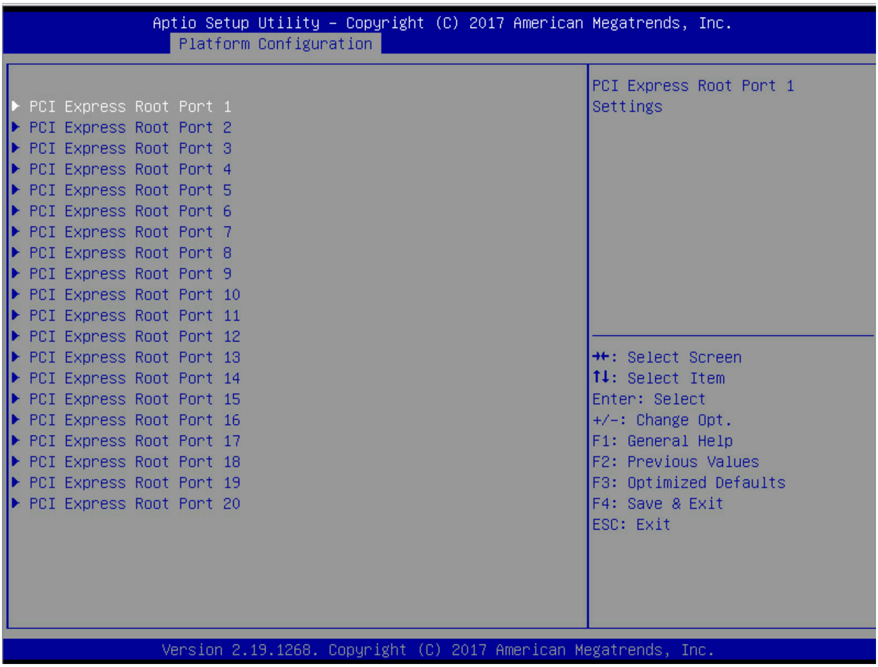
PCH DFX Configuration
PCH DFX Configuration Options

3.4.1.1 PCH Devices Configuration



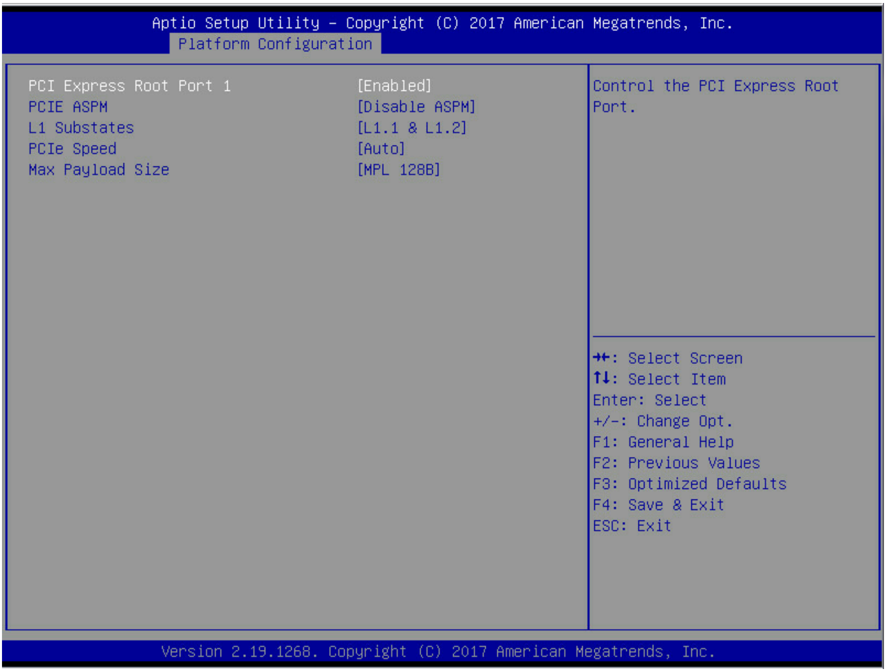
PCH state after G3
Select S0/S5 for ACPI state after a G3
S0 **/**S5 **/** Leave power state unchanged

3.4.1.2 PCI Express Configuration



PCI Express Root Port 1/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20
PCI Express Root Port Settings

3.4.1.2.1 PCI Express Root Port 1/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20 Configuration



PCI Express Root Port 1/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20
Control the PCI Express Root Port
Disabled **/ Enabled**

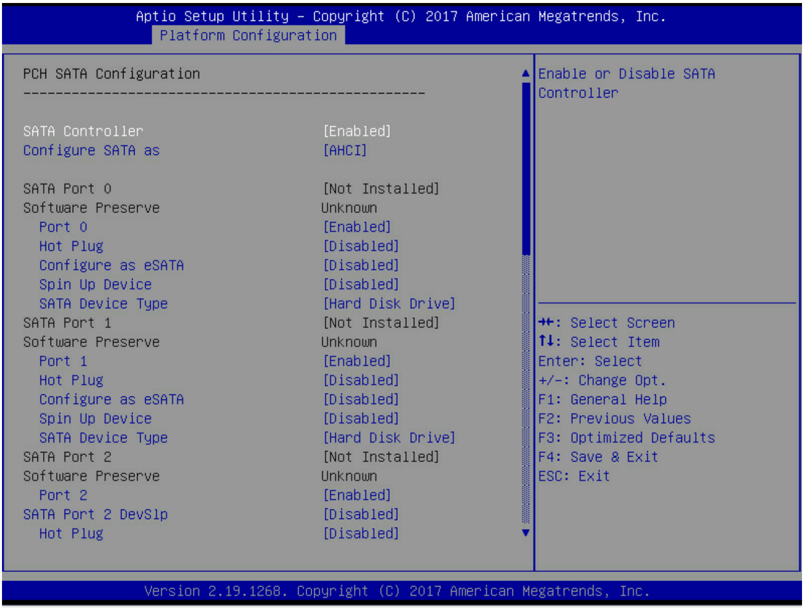
PCIE ASPM
PCI Express Root port ASPM Setting
Disable ASPM / ASPM L1 / ASPM Auto

L1 Substates
PCI Express L1 Substates settings.
Disabled / L1.1 / L1.2 / **L1.1 & L1.2**

PCIe Speed
PCI Express Root Port Completion Timer TO settings
Auto / Gen1 / Gen2 / Gen3

Max Payload Size
PCIE Max Payload Size Selection.

3.4.1.3 PCH SATA Configuration Submenu



```
Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Platform Configuration

    Hot Plug                      [Disabled]          Configures port as External
    Configure as eSATA            [Disabled]          SATA (eSATA)
    Spin Up Device                [Disabled]
    SATA Device Type              [Hard Disk Drive]
  SATA Port 3                     [Not Installed]
  Software Preserve               Unknown
    Port 3                        [Enabled]
    Hot Plug                      [Disabled]
    Configure as eSATA            [Disabled]
    Spin Up Device                [Disabled]
    SATA Device Type              [Hard Disk Drive]
  SATA Port 4                     [Not Installed]
  Software Preserve               Unknown
    Port 4                        [Enabled]           →←: Select Screen
    Hot Plug                      [Disabled]          ↑↓: Select Item
    Configure as eSATA            [Disabled]          Enter: Select
    Spin Up Device                [Disabled]          +/-: Change Opt.
    SATA Device Type              [Hard Disk Drive]   F1: General Help
  SATA Port 5                     [Not Installed]     F2: Previous Values
  Software Preserve               Unknown             F3: Optimized Defaults
    Port 5                        [Enabled]           F4: Save & Exit
    Hot Plug                      [Disabled]          ESC: Exit
    Configure as eSATA            [Disabled]
    Spin Up Device                [Disabled]
    SATA Device Type              [Hard Disk Drive]

Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.
```

```
Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Platform Configuration

    Hot Plug                      [Disabled]          Identify the SATA port is
    Configure as eSATA            [Disabled]          connected to Solid State Drive
    Spin Up Device                [Disabled]          or Hard Disk Drive
    SATA Device Type              [Hard Disk Drive]
  SATA Port 5                     [Not Installed]
  Software Preserve               Unknown
    Port 5                        [Enabled]
    Hot Plug                      [Disabled]
    Configure as eSATA            [Disabled]
    Spin Up Device                [Disabled]
    SATA Device Type              [Hard Disk Drive]
  SATA Port 6                     [Not Installed]
  Software Preserve               Unknown
    Port 6                        [Enabled]           →←: Select Screen
    Hot Plug                      [Disabled]          ↑↓: Select Item
    Configure as eSATA            [Disabled]          Enter: Select
    Spin Up Device                [Disabled]          +/-: Change Opt.
    SATA Device Type              [Hard Disk Drive]   F1: General Help
  SATA Port 7                     WDC WD1200BEVS - 120....  F2: Previous Values
  Software Preserve               Unknown             F3: Optimized Defaults
    Port 7                        [Enabled]           F4: Save & Exit
    Hot Plug                      [Disabled]          ESC: Exit
    Configure as eSATA            [Disabled]
    Spin Up Device                [Disabled]
    SATA Device Type              [Hard Disk Drive]

Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.
```

SATA Controller
Enable or Disable SATA Controller
Disabled **/ Enabled**

Configure SATA as
Indentify the SATA port is connected to Solid State Drive or Hard Disk Drive
AHCI / RAID

SATA Port 0/1/2/3/4/5/6/7
Port 0/1/2/3/4/5/6/7
Disabled **/ Enabled**

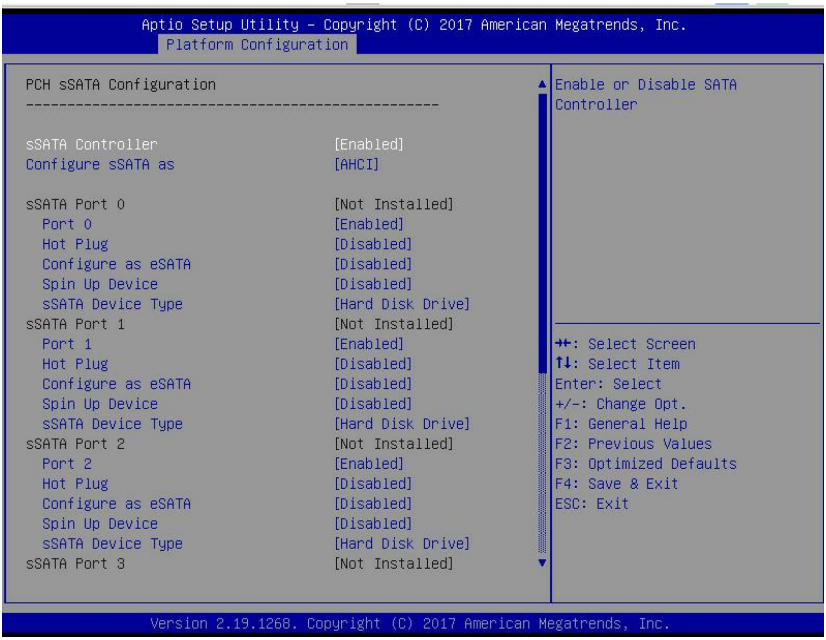
Hot Plug
Enable/Disable SATA Ports Hot Plug Support.
Disabled / Enabled

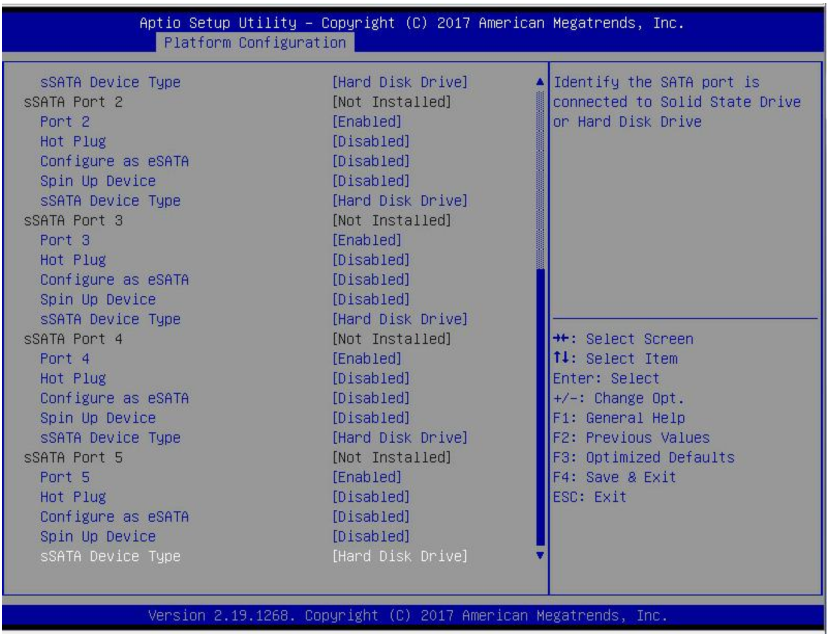
Configure as eSATA
Configures port as External SATA (eSATA)
Disabled / Enabled

Spin Up Device
AHCI Supports Staggered Spin-up
Disabled / Enabled

SATA Device Type
Indentify the SATA port is connected to Solid State Drive or Hard Disk Drive
Hard Disk Drive / Solid State Drive

3.4.1.4 PCH sSATA Configuration Submenu





sSATA Controller
Enable or Disable SATA Controller
Disabled **/ Enabled**

Configure sSATA as
Indentify the SATA port is connected to Solid State Drive or Hard Disk Drive
AHCI / RAID

sSATA Port 0/1/2/3/4/5
Enable or Disable SATA Port
Disabled **/ Enabled**

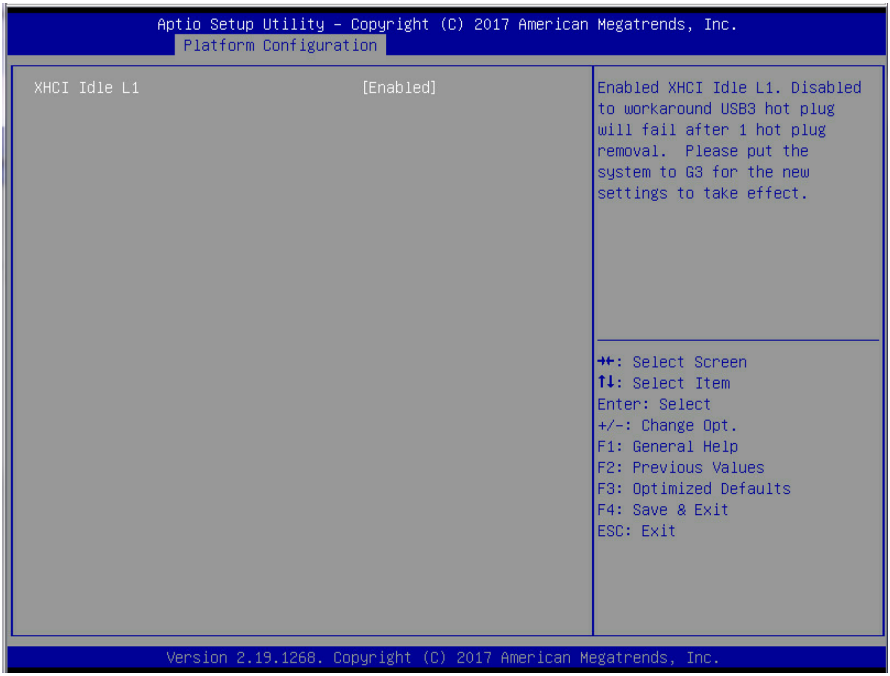
Hot Plug
Designates this port as Hot Pluggable
Disabled / Enabled

Configure as eSATA
Configures port as External SATA (eSATA)
Disabled / Enabled

Spin Up Device
If enabled for any of ports Staggerred Spin Up will be performed and only the drives witch have this option enabled will spin up at boot. Otherwise all drives spin up at boot.
Disabled / Enabled

sSATA Device Type
Indentify the SATA port is connected to Solid State Drive or Hard Disk Drive
Hard Disk Drive / Solid State Drive

3.4.1.5 USB Configuration Submenu

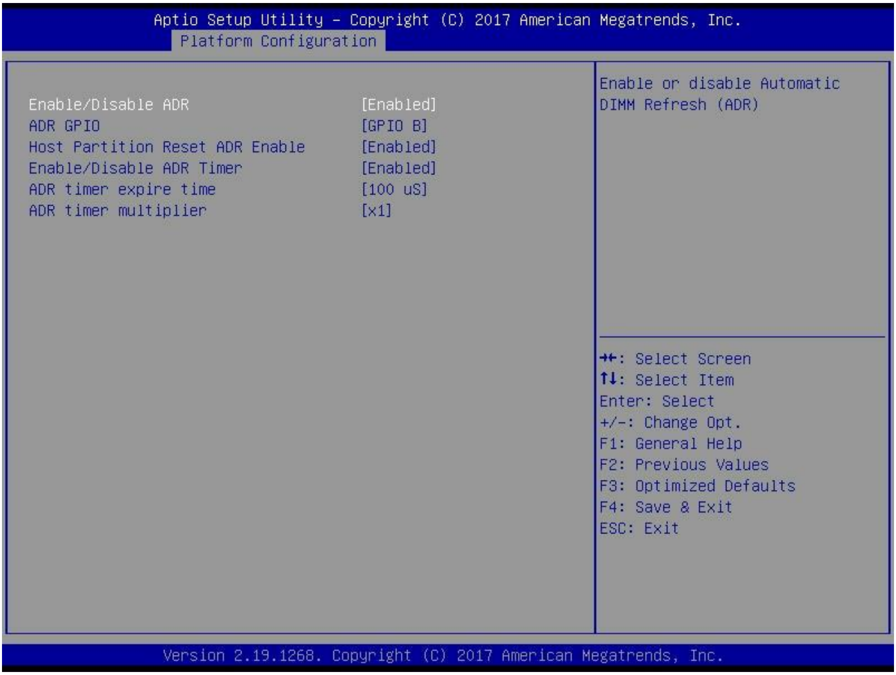


XHCI Idle L1

Enabled XHCI Idle L1. Disabled to workaround USB3 hot plug will fail after 1 hot plug removal. Please put the system to G3 for the new settings to take effect.

Disable **/ Enabled**

3.4.1.6 PCH DFX Configuration Submenu



Enable/Disable ADR
Enable or disable Automatic DIMM Refresh (ADR)
Enabled / Disabled

ADR GPIO
Select between GPIO_B or GPIO_C
GPIO B / GPIO C

Host Partition Reset ADR Enable
Enables/Disables ADR on Host Partition Reset
Enabled / Disabled

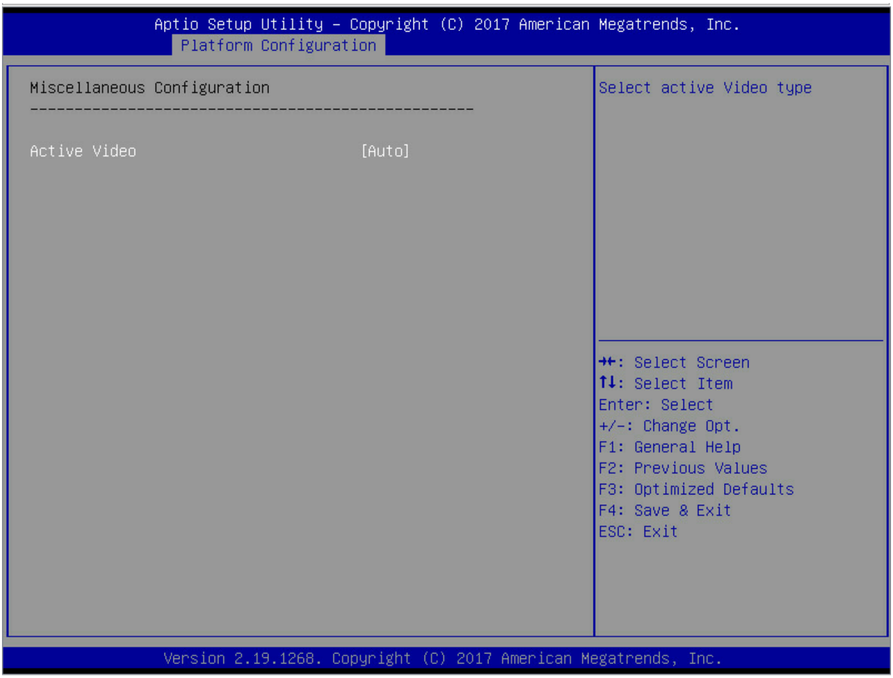
Enable/Disable ADR Timer
Held-off for DEBUG PURPOSES ONLY!
Enabled / Held-off

ADR timer expire time
Select proper ADR timer value: 25uS,50uS,100uS or 0.
25 uS / 50 uS / **100 uS** / 0 uS

ADR timer multiplier
Select proper ADR timer multiplier: x1,8,24,40,56,64,72,80,88,96.

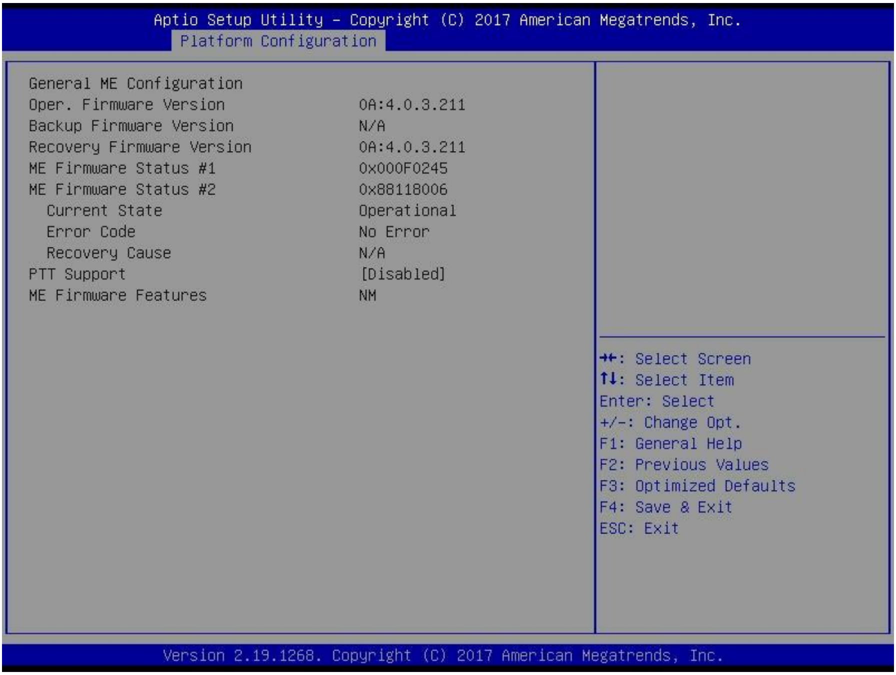
x1 / x8 / x24 / x40 / x56 / x64 / x72 / x80 / x88 /x96

3.4.2 Miscellaneous Configuration Submenu



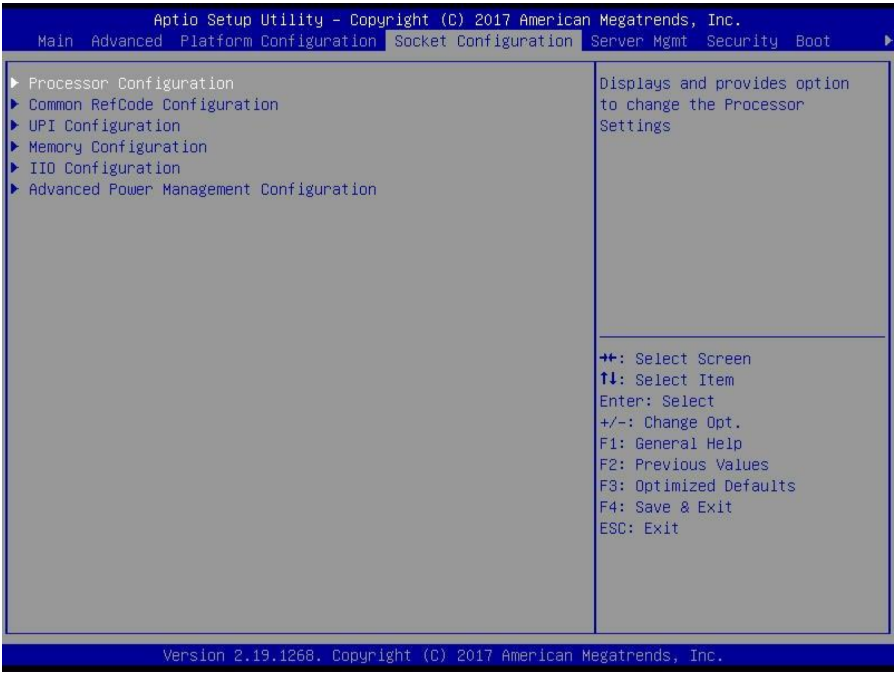
Active Video
Select active Video type
Auto / Onboard Device **/** PCIE Device

3.4.3 Server ME Configuration Submenu



Read Only

3.5 Socket Configuration



Processor Configuration
Displays and provides option to change the Processor Settings.

Common RefCode Configuration
Displays and provides option to change the Common RefCode Settings

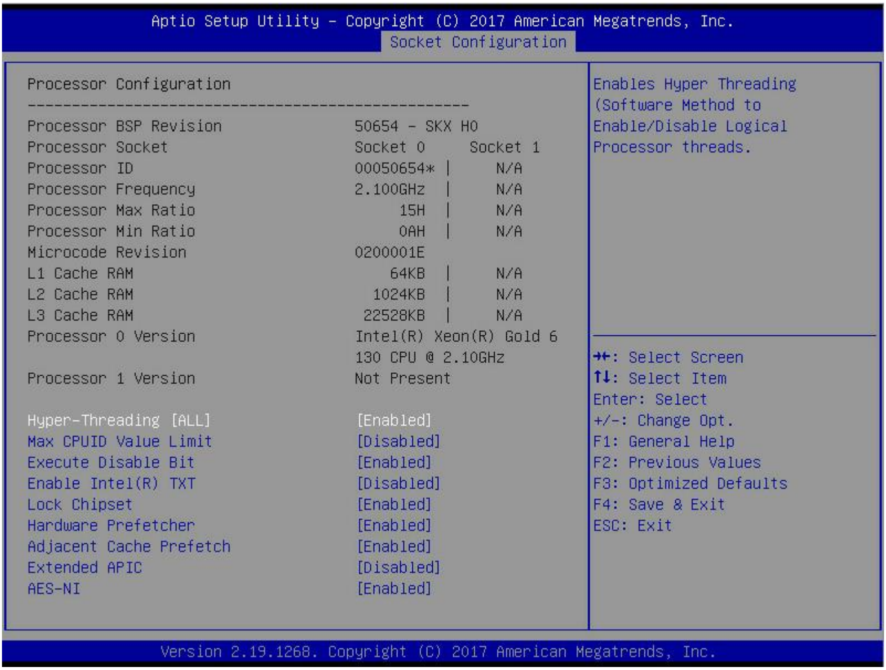
UPI Configurtion
Displays and provides option to change the UPI Settings

Memory Configuration
Displays and provides option to change the Memory Settings

IIO Configuration
Displays and provides option to change the IIO Settings

Advance Power Management Configuration
Displays and provides option to change the Power Management Settings

3.5.1 Processor Configuration Submenu



Hyper-Threading [ALL]
Enables Hyper Threading (Software Method to Enable/Disable logical Processor threads.
Disabled **/ Enabled**

Max CPUID Value Limit
This should be enabled in order to boot legacy OSes that cannot support CPUs with extended CPUID functions.
Disabled / Enable

Execute Disable Bit
When disabled, forces the XD feature flag to always return 0.
Disabled **/ Enabled**

Enable Intel(R) TXT
Enables Intel(R) TXT
Disabled / Enabled

Lock Chipset
Lock or Unlock chipset

Disabled / **Enabled**

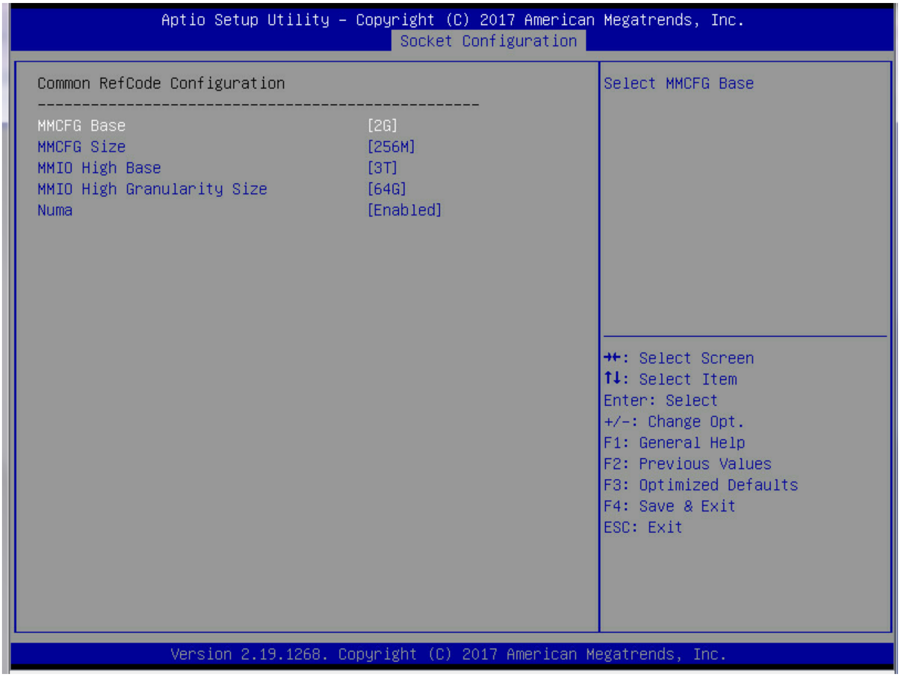
Hardware Prefetcher
MLC Streamer Prefetcher (MSR 1A4h Bit [0])
Disabled / **Enabled**

Adjacent Cache Prefetch
MLC Spatial Prefetcher (MSR 1A4h Bit [1])
Disabled / **Enabled**

Extended APIC
Enable/disable extended APIC support
Disabled / Enabled

AES-NI
Enable/disable AES-NI support
Disabled / **Enabled**

3.5.2 Common RefCode Configuration Submenu



MMCFG Base

Select MMCFG Base

1G /1.5G / 1.75G / **2G** / 2.25G /3G

MMCFG Size

Select MMCFG Size

64M / 128M / **256M** / 512M / 1G / 2G

MMIO High Base

Select MMIO High Base

56T / 40T / 24T /16T / 4T/ **3T**/ 2T/ 1T

MMIO High Granularity Size

Selects the allocation size used to assign mmioh resources. Total mmioh space can be up to 32x granularity.

Per stack mmioh resource assignments are multiples of the granularity where 1 unit Per stack is the default allocation.

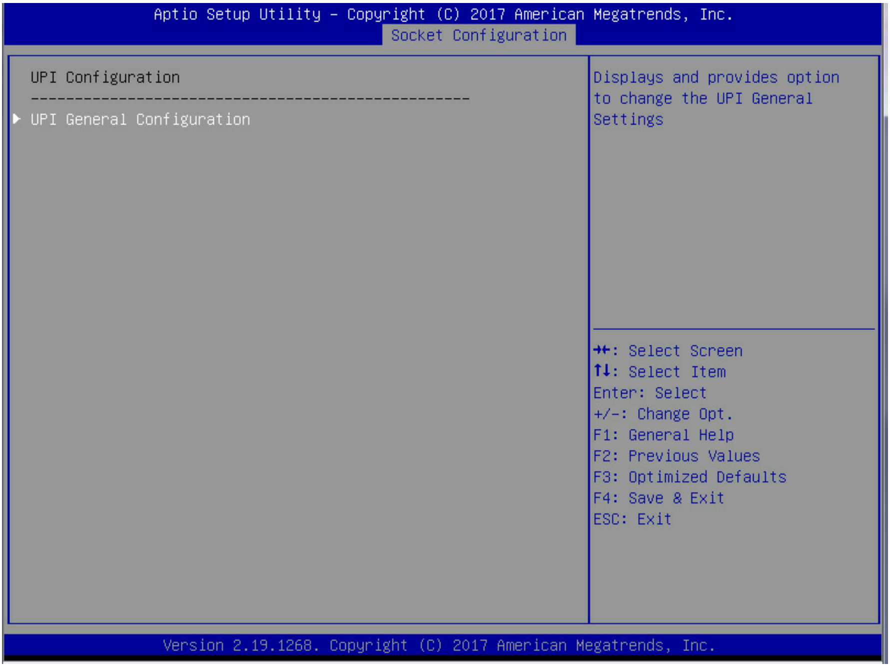
1G / 4G / 16G / **64G** / 256G / 1024G

Numa

Enable or Disable non uniform Memory Access. (NUMA)

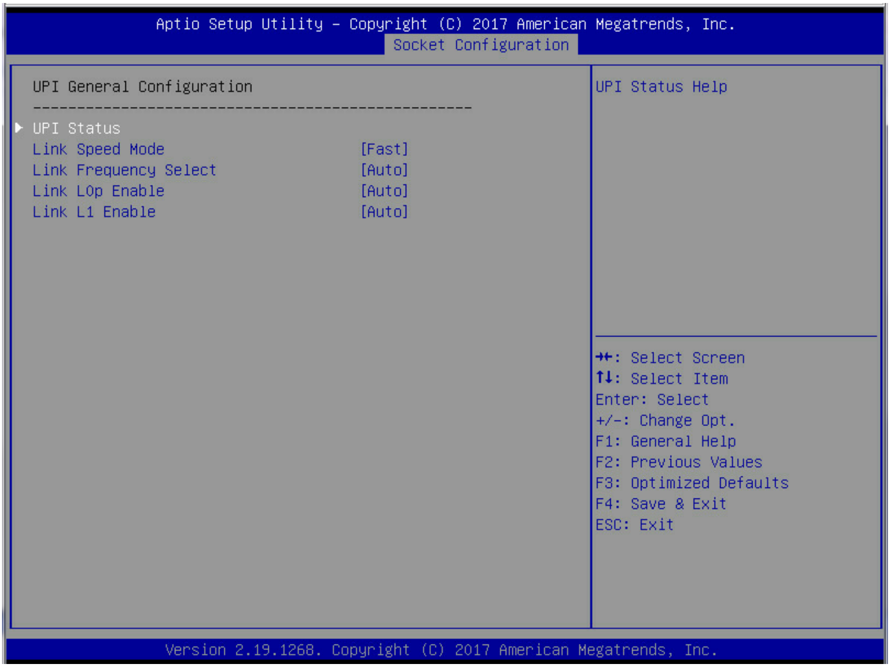
Enabled / Disabled

3.5.3 UPI Configuration Submenu



UPI General Configuration
Displays and provides option to change the UPI General Settings

3.5.3.1 UPI General Configuration Submenu



UPI Status
UPI Status Help.

Link Speed Mode
Select the UPI link speed as either the POR speed (Fast) or default speed (Slow)
Slow / **Fast**

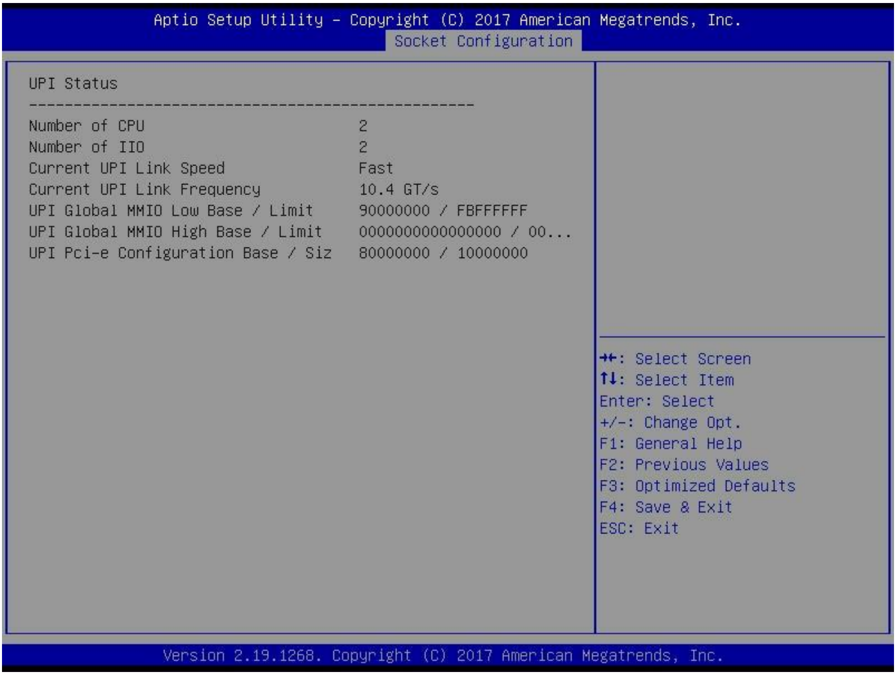
Link Frequency Select
Allows for selecting the UPI Link frequency
9.6GB/s /10.4GB/s / **Auto** / Use Per Link Setting

Link L0p Enable
Enable – Set the c_L0p_en, Disable – Reset it, Auto – Auto decides based on Si Compatibility
Disable/Enable/**Auto**

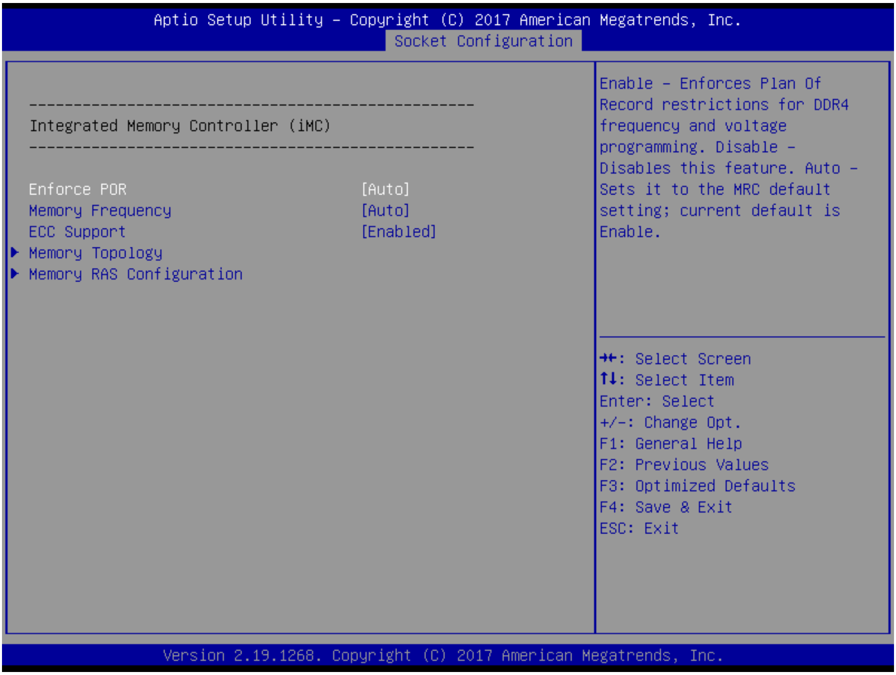
Link L1 Enable
Enable – Set the c_L1_en, Disable – Reset it, Auto – Auto decides based on Si Compatibility

Disable/Enable/**Auto**

3.5.3.2 UPI Status Configuration Submenu



3.5.4 Memory Configuration Submenu



Enforce POR
Enable – Enforce Plan Of Record restrictions for DDR4 frequency and voltage programming .Disable – Disables this feature. Auto – Sets it to the MRC default setting; current default is Enable.
Auto / POR / Disable

Memory Frequency
Maximum Memory Frequency Selections in Mnz. Do not select Reserved
Auto / 2133 / 2400 / 2666

ECC Support
Enable – Enables DDR ECC Support. Disable – Disables this feature. Auto – Sets it to MRC default setting; current default is Enable.
Disabled **/ Enabled**

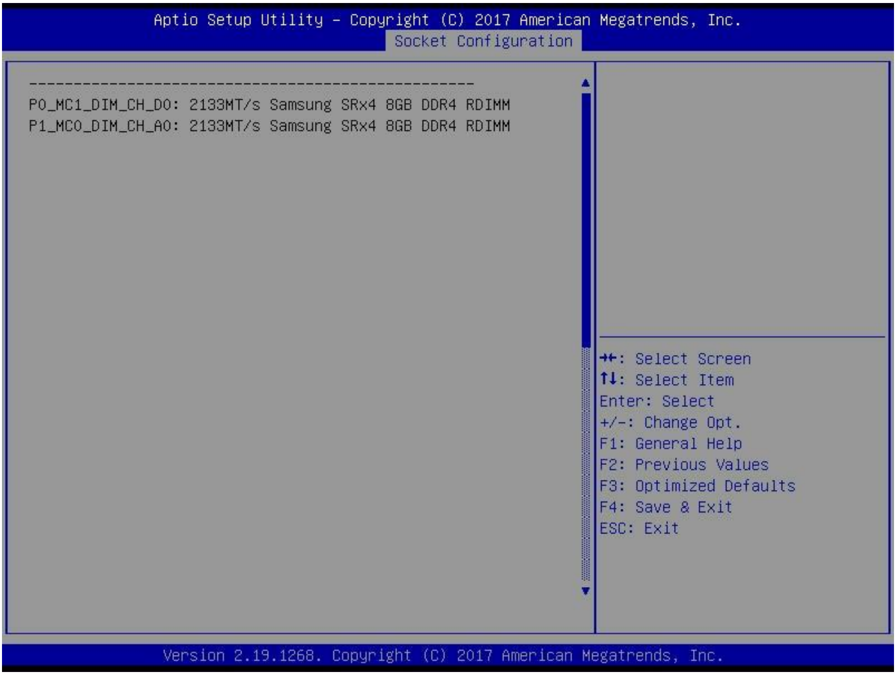
Memory Topology

Displays memory topology with Dimm population information

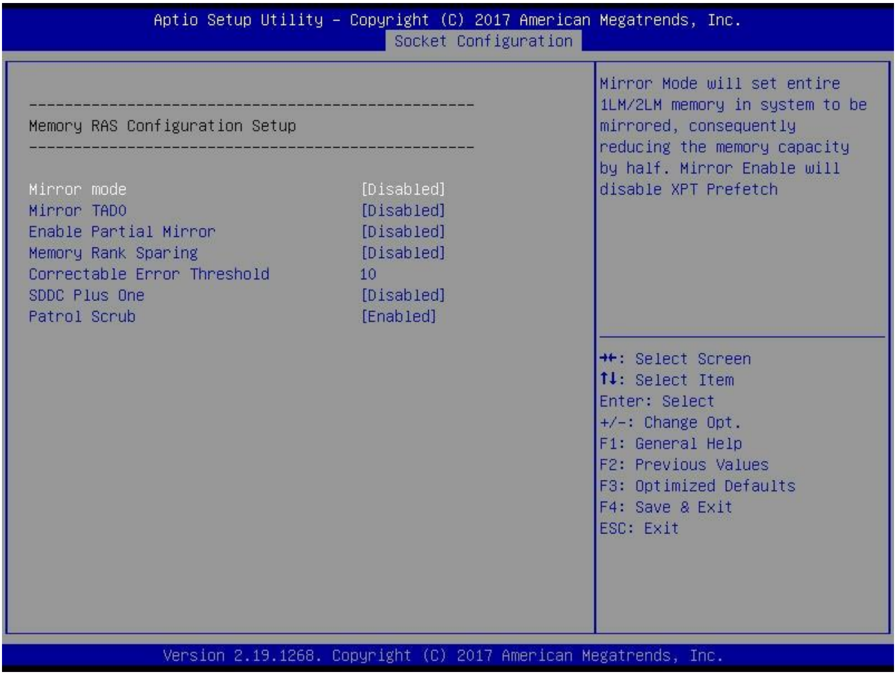
Memory RAS Configuration

Displays and provides option to change the Memory Ras Settings

3.5.4.1 Memory Topology Configuration Submenu



3.5.4.2 Memory RAS Configuration Submenu



Mirror mode
Mirror Mode will set entire 1LM/2LM memory in system to be mirrored, consequently reducing the memory capacity by half. Mirror Enable will disable XPT Prefetch.

Disabled / Enabled

Mirror TAD0
Enable Mirror on entire memory for TAD0

Disabled / Enabled

Enable Partial Mirror
Partial mirror mode will enable the required size of memory to be mirrored. If rank sparing is enabled partial mirroring will not take effect. Mirror Enable will disable XPT Prefetch.

Disabled / Enabled

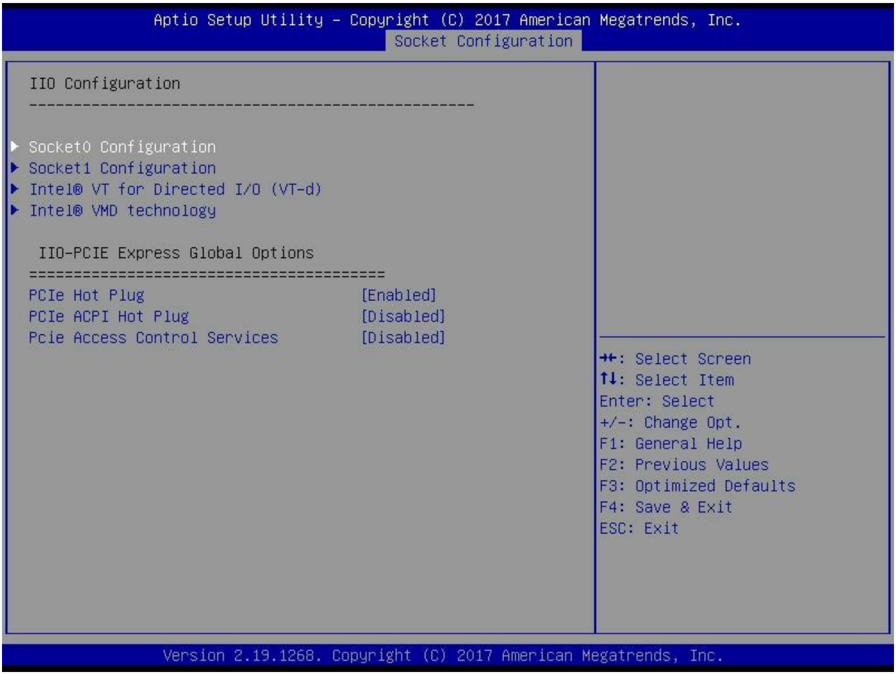
Memory Rank Sparing
Enable/Disable Memory Rank Sparing
Disabled / Enabled

Correctable Error Threshold
Enable/ Disable Memory Rank Sparing

SDDC Plus One
Selects the address mode between System Physical Address (or) Reverse Address
Disabled / Enabled

Patrol Scrub
Enable/Disable Patrol Scrub
Disabled / **Enabled**

3.5.5 IIO Configuration



Socket0 Configuration
Socket 0 Configuration

Socket1 Configuration
Socket 1 Configuration

Intel® VT for Directed I/O (VT-d)
Press <Enter> to bring up the Intel® VT for Directed I/O (VT-d) Configuration menu.

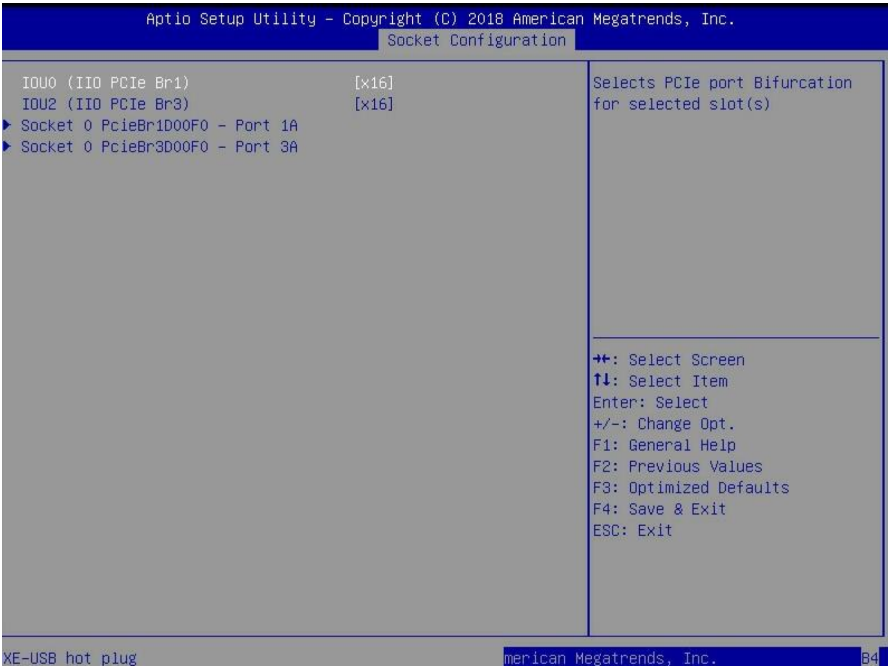
Intel® VMD technology
Press <Enter> to bring up the Intel® VMD for Volume Management Device Configuration menu.

PCIe Hot Plug
Enable/Disable PCIe Hot Plug globally.
Disabled / **Enabled**

PCIe ACPI Hot Plug
Enable/Disable PCIe ACPI Hot Plug globally, or allow per-port control. When Disabled, MSI is generated on HP event. When Enabled, _HPGPE message is generated.
Disabled / Enabled

PCIe Access Control Services
Enable/Disable Access Control Service
Disabled / Enabled

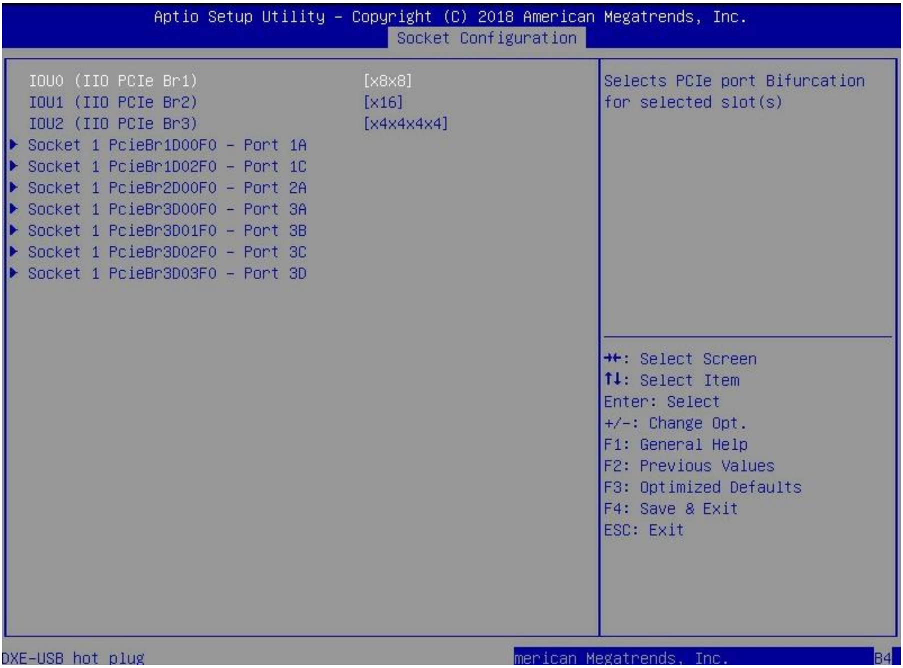
3.5.5.1 Socket0 Submenu



IOU0 (IIO PCIe Br1)
Select PCIe port Bifuncation for selected slot (s)
x4x4x4x4 / X4x4x8 / x8x4x4 / x8x8 /**x16** / Auto

IOU2 (IIO PCIe Br3)
Selects PCIe port Bifuncation for selected slot (s)
x4x4x4x4 / X4x4x8 / x8x4x4 / x8x8 / **x16** / Auto

3.5.5.2 Socket1 Submenu

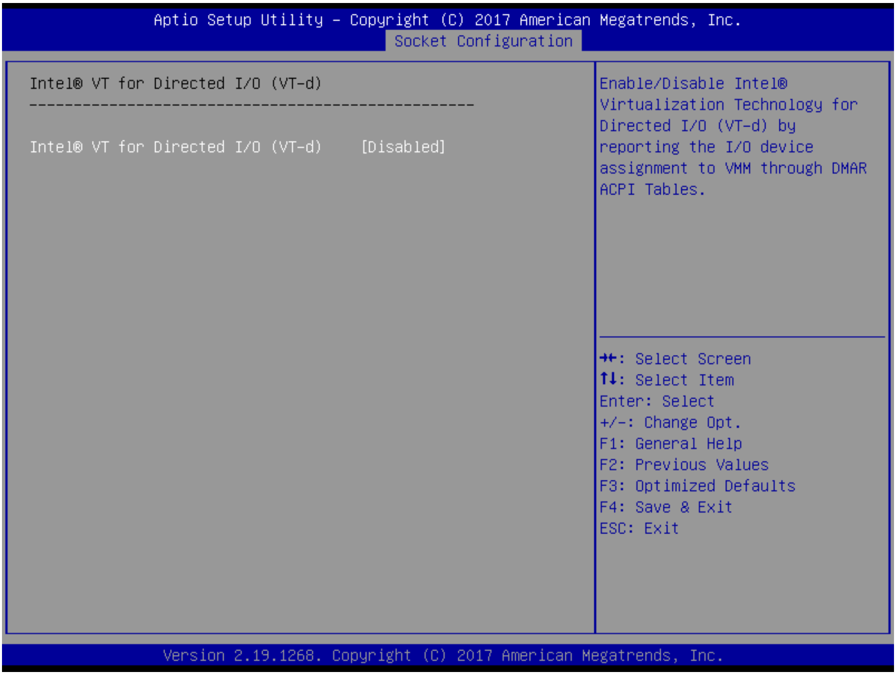


IOU0 (IIO PCIe Br1)
Select PCIe port Bifuncation for selected slot (s)
x4x4x4x4 / X4x4x8 / x8x4x4 / **x8x8** / x16 / Auto

IOU1 (IIO PCIe Br2)
Selects PCIe port Bifurcation for selected slot (s)
x4x4x4x4 / X4x4x8 / x8x4x4 / x8x8 **/ x16** / Auto

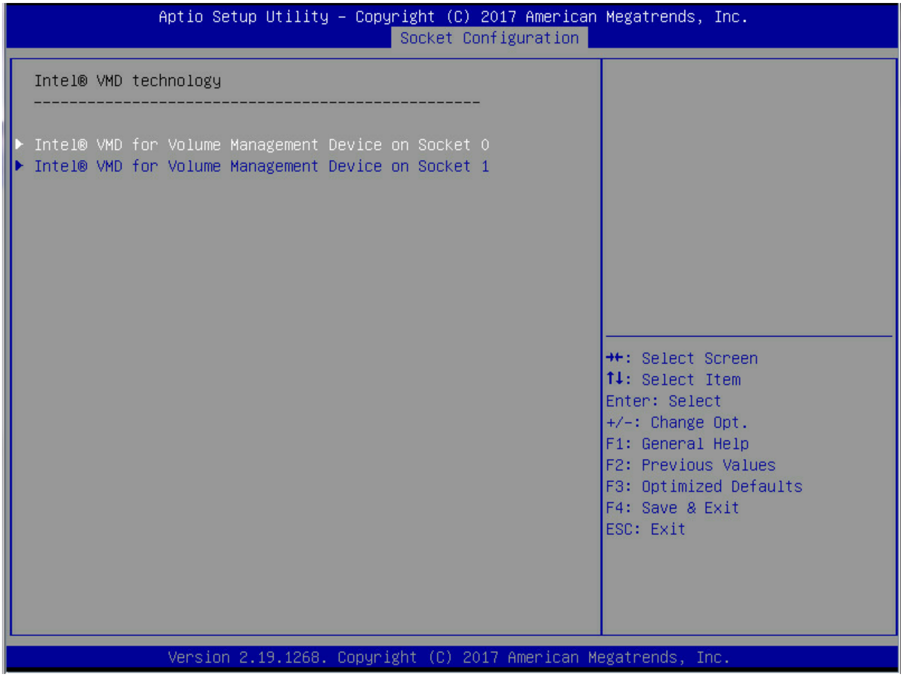
IOU2 (IIO PCIe Br3)
Selects PCIe port Bifuncation for selected slot (s)
x4x4x4x4 / X4x4x8 / x8x4x4 / x8x8 /x16 / Auto

3.5.5.3 Intel® VT for Directed I/O (VT-d) Submenu

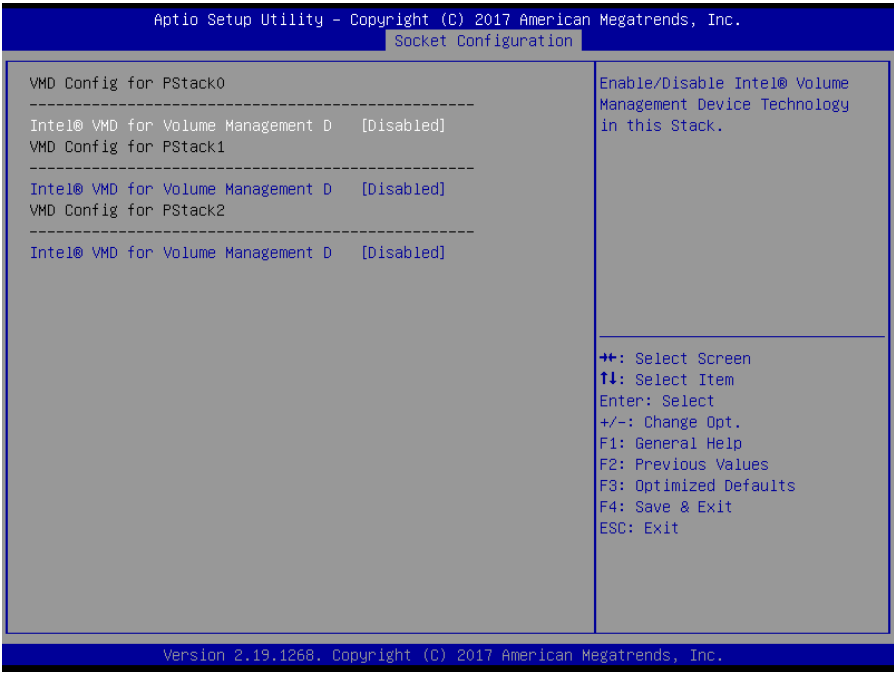


Intel® VT for Directed I/O (VT-d)
Enable/Disable Intel® Virtualization Technology for Directed I/O (VT-d) by reporting the I/O device assignment to VMM through DMAR ACPI Tables.
Disabled / Enabled

3.5.5.4 Intel® VMD technology Submenu



3.5.5.4.1 Intel® VMD for Volume Management Device on Socket 0/1 Submenu



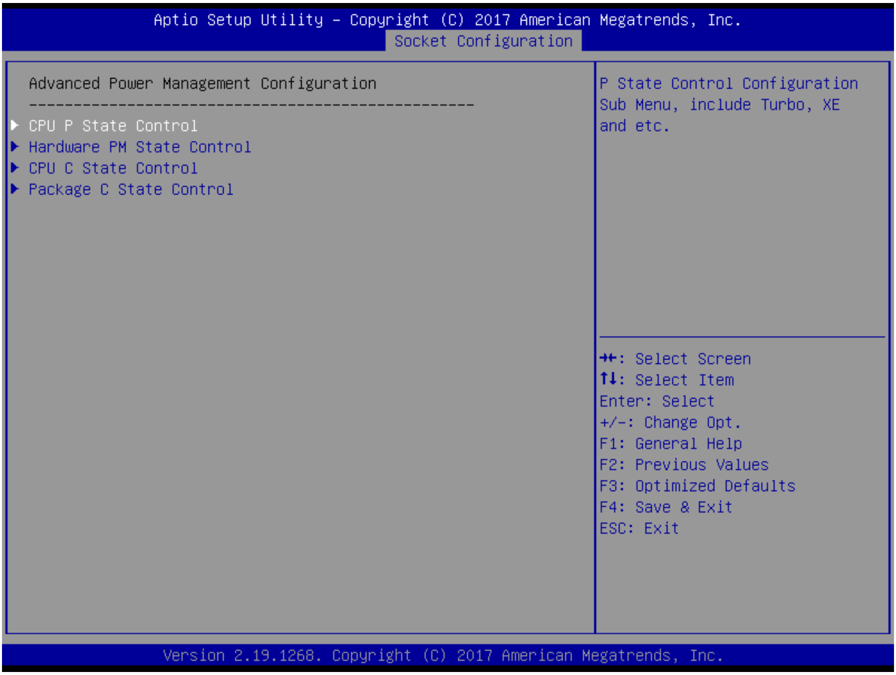
VMD Config for PStack 0/1/2

Intel® VMD for Volume Management D

Enable/Disable Intel® Volume Management Device Technology in this Stack.

Disabled / Enabled

3.5.5 Advance Power Management Configuration



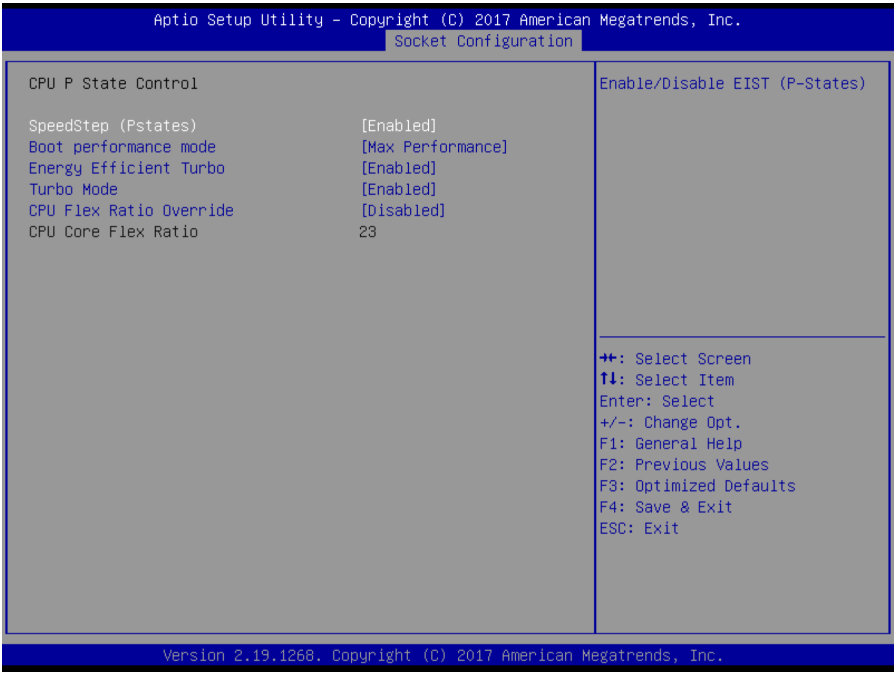
CPU P State Control
P State Control Configuration Sub Menu, include Turbo, XE and etc.

Hardware PM State Control
Hardware P-State setting

CPU C State Control
CPU C State setting

Package C State Control
Package C State setting

3.5.5.1 CPU P State Control Submenu



SpeedStep (Pstates)
Enable/Disable EIST (P-States)
Disabled / **Enabled**

Boot performance mode
Select the performance state that the BIOS will set before OS hand off.
Max Performance / Max Efficient / Set by Intel Node

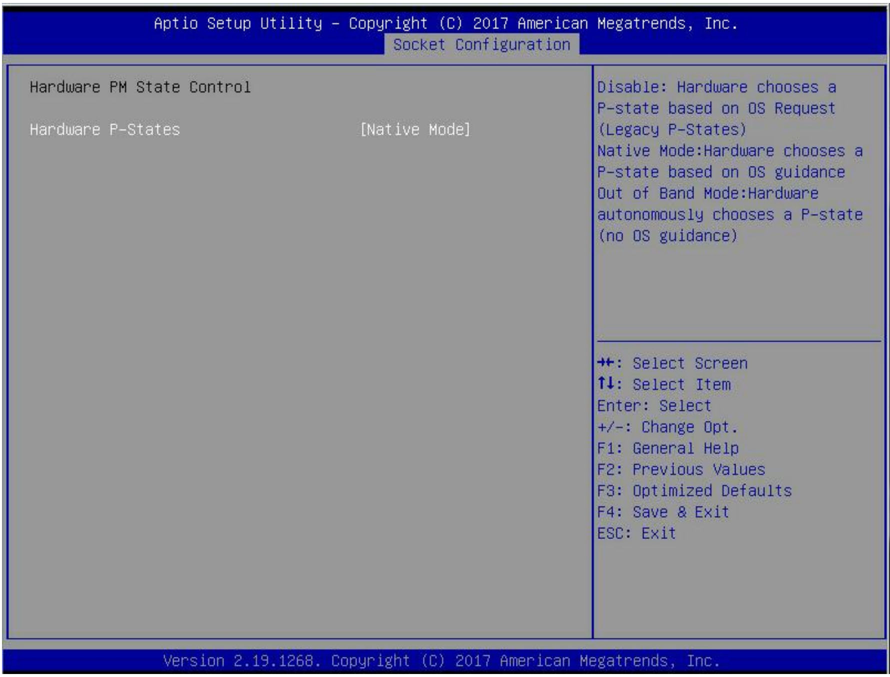
Energy Efficient Turbo
Energy Efficient Turbo Disable, MSR 0x1FC [19]
Disabled / **Enabled**

Turbo Mode
Enable/Disable processor Turbo Mode (requires EMTTM enabled too).
Disabled / **Enabled**

CPU Flex Ratio Override
Enable/Disable CPU Flex Ratio Programming

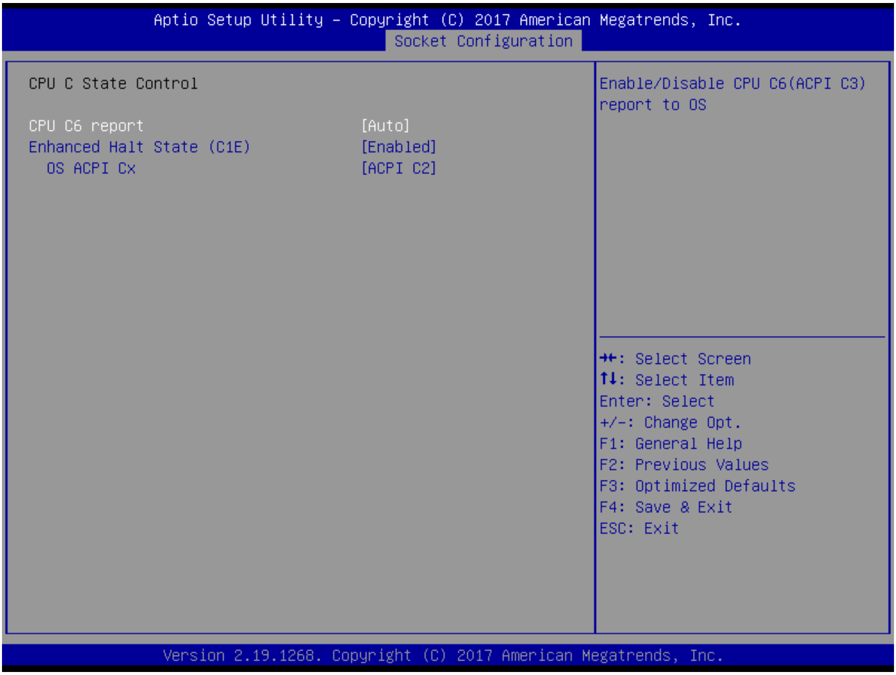
Disabled / Enabled

3.5.5.2 Hardware PM State Control Submenu



Hardware P-States
Disable: Hardware choose a P-state based on OS Request (Legacy P-States)
Native Mode: Hardware choose a P-state based on OS guidance
Out of Band Mode: Hardware autonomously choose a P-state (No OS guidance)
Disable / **Native Mode** / Out of Band Mode / Native Mode with No Legacy Support

3.5.5.3 CPU C State Control Submenu

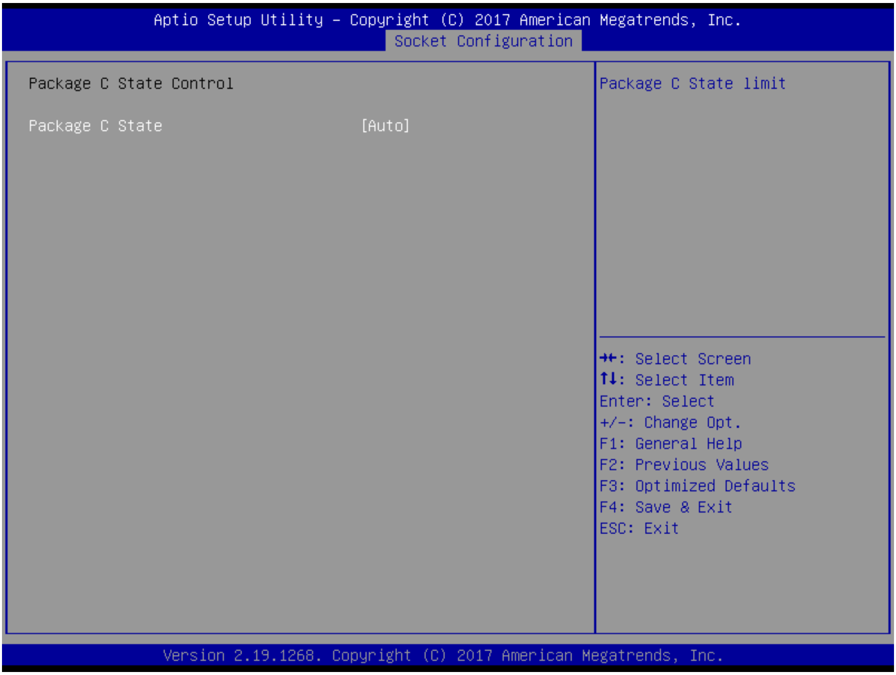


CPU C6 report
Enable/Disable CPU C6(ACPI C3) report to OS
Auto / Enable / Disable

Enhanced Halt State (C1E)
Enables the Enhanced C1E state of the CPU, takes effect after reboot
Enabled / Disabled

OS ACPI Cx
Report CC3/CC6 to OS ACPI C2 or ACPI C3
ACPI C2 / ACPI C3

3.5.5.4 Package C State Control Submenu

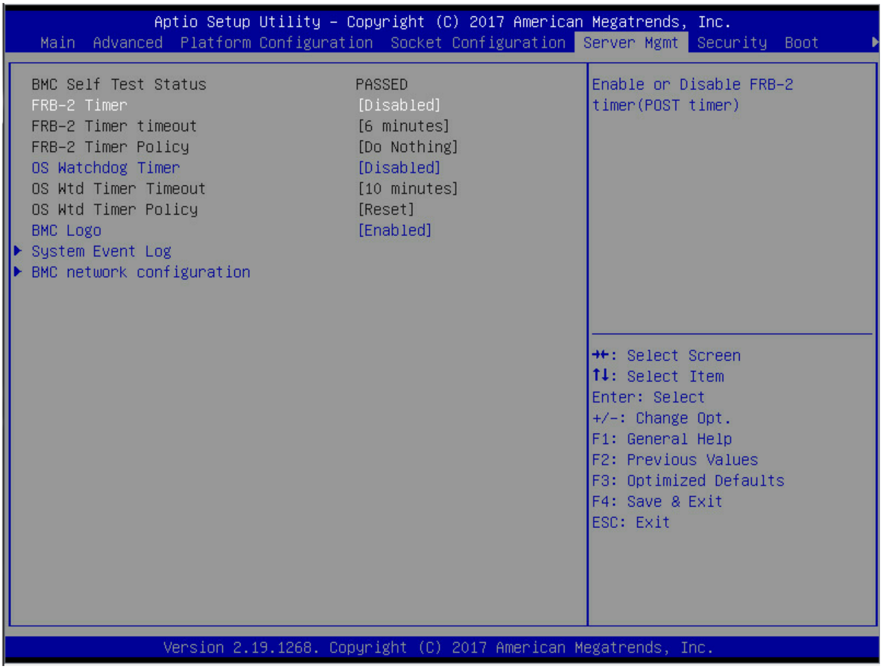


Package C State Control

Package C State Limit

C0/C1 state / C2 state / C6(non Retention) state / C6(Retention) state / No Limit / **Auto**

3.6 Server Management



FRB-2 Timer
Enable or Disable FRB-2 timer (POST timer)
Disabled / Enabled

NOTE: When [FRB-2 Timer] is set to [Enabled], the following items will be available.

FRB-2 Timer timeout
Enter value Between 3 to 6 min for FRB-2 Timer Expiration value
3 minutes **/** 4 minutes **/** 5 minutes **/** **6 minutes**

FBR-2 Timer Policy
Configure how the system should respond if the FRB-2 Timer expires. Not available if FRB-2 Timer is disabled.
Do Nothing **/** Reset **/** Power Down **/** Power Cycle

OS Watchdog Timer

If enabled, starts a BIOS timer which can only be shut off by management Software after the OS loads. Helps determine that the OS successfully loaded or follows the OS Boot Watchdog Timer policy.

Disabled / Enabled

NOTE: When [OS Watchdog Timer] is set to [Enabled], the following items will be available.

OS Wtd Timer Timeout

Configure the length of the OS Boot Watchdog Timer. Not available if OS Boot Watchdog Timer is disabled.

5 minutes **/ 10 minutes / 1**5 minutes **/** 20 minutes

OS Wtd Timer Policy

Configure how the system should respond if the OS Boot Watchdog Timer expires. Not available if OS Boot Watchdog Timer is disabled.

Do Nothing **/ Reset /** Power Down **/** Power Cycle

BMC Logo

Enable or Disable BMC logo

Disabled / **Enabled**

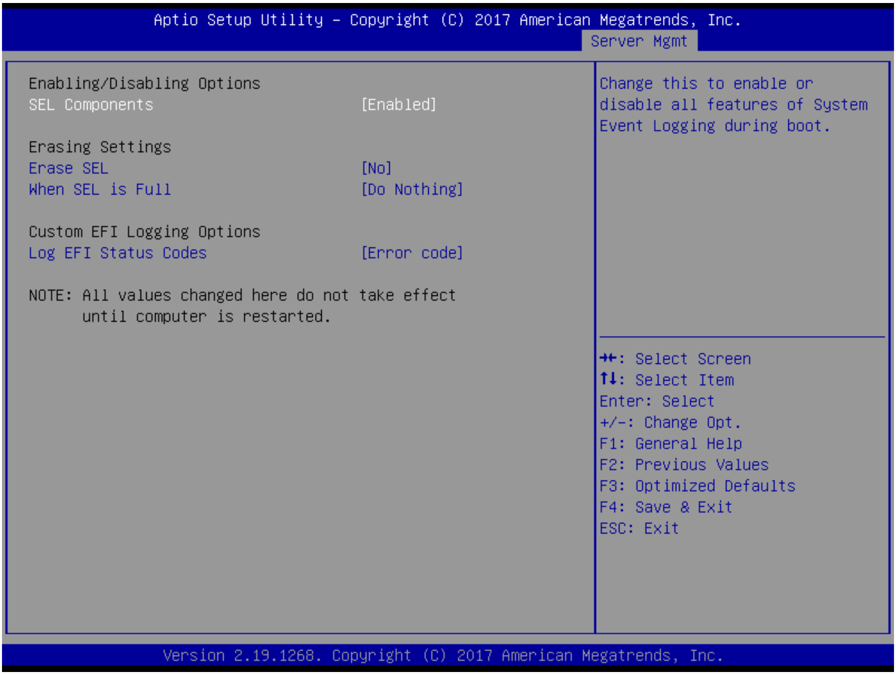
System Event Log

Press<Enter> to change the SEL event log configuration.

BMC network configuration

Configure BMC network parameters

3.6.1 System Event Log Submenu



SEL Components
Change this to enable or disable all features of System Event Logging during boot.
Enabled / Disabled

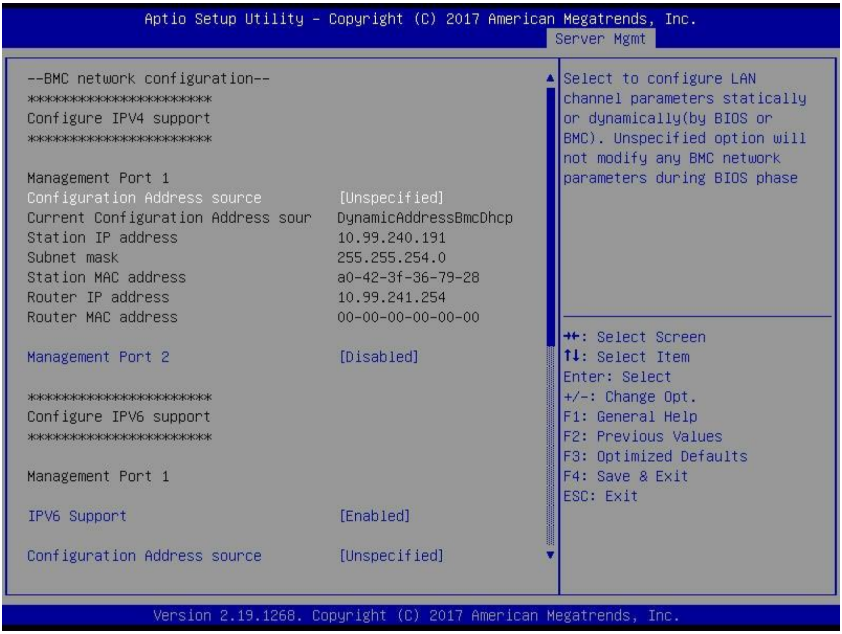
Erase SEL
Choose options for erasing SEL.
No / Yes, on next reset / No, on every reset

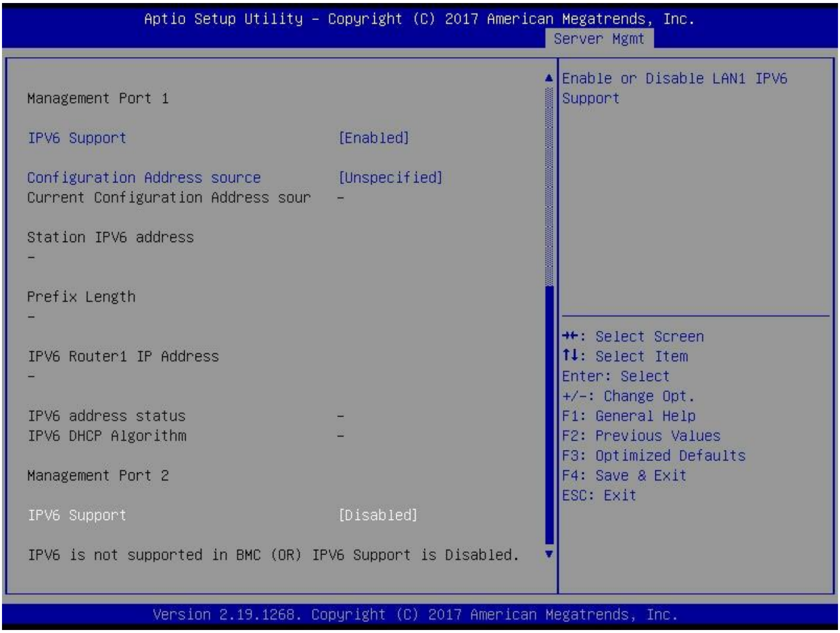
When SEL is Full
Choose options for reactions to a full SEL.
Do Nothing / Erase Immediately

Log EFI Status Codes
Disable the logging of EFI Status Codes or log only error code or only progress code or both.

Both **/** Disabled / **Error Code** / Progress Code

3.6.2 BMC Network Configuration Submenu





Configuration Address Source
Select the configure LAN channel parameters statically or dynamically (by BIOS or BMC). Unspecified option will not modify any BMC network parameters during BIOS phase.

Unspecified / Static / DynamicBmcDhcp / DynamicBmcNonDhcp

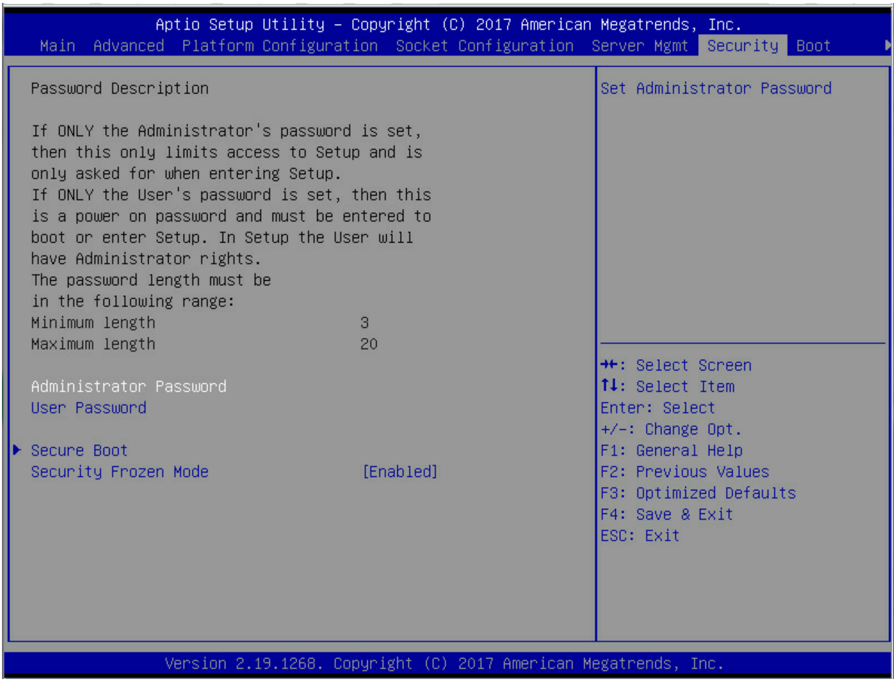
Management Port 2
Enable/Disable BMC Share NIC

Disabled / Enabled

IPV6 Support
Enable or Disable LAN1 IPV6 Support

Disabled / Enabled

3.7 Security



Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Main Advanced Platform Configuration Socket Configuration Server Mgmt Security Boot
Password Description
If ONLY the Administrator's password is set, then this only limits access to Setup and is only asked for when entering Setup.
If ONLY the User's password is set, then this is a power on password and must be entered to boot or enter Setup. In Setup the User will have Administrator rights.
The password length must be in the following range:
Minimum length 3
Maximum length 20
Administrator Password
User Password
Secure Boot
Security Frozen Mode [Enabled]
Set Administrator Password
→←: Select Screen
↑↓: Select Item
Enter: Select
+/-: Change Opt.
F1: General Help
F2: Previous Values
F3: Optimized Defaults
F4: Save & Exit
ESC: Exit
Version 2.19.1268. Copyright (C) 2017 American Megatrends, Inc.

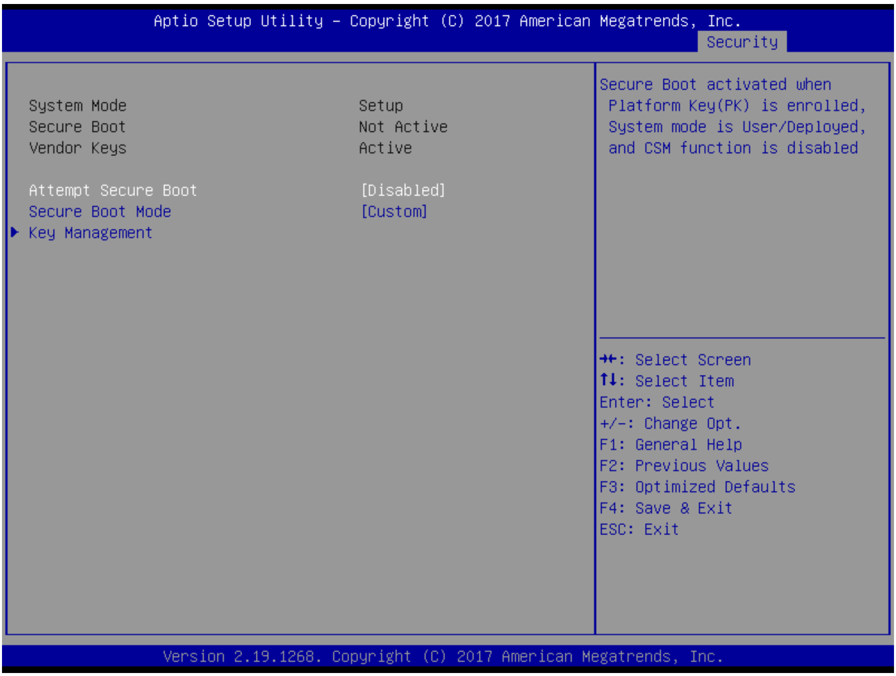
Administrator Password
Set Administrator Password.

User Password
Set User Password.

Secure Boot
Customizable Secure Boot settings

Security Frozen Mode
Enable or disable HDD security freeze lock.
Disable to support secure erase function.
Disabled / **Enabled**

3.7.1 Secure Boot Configuration Submenu

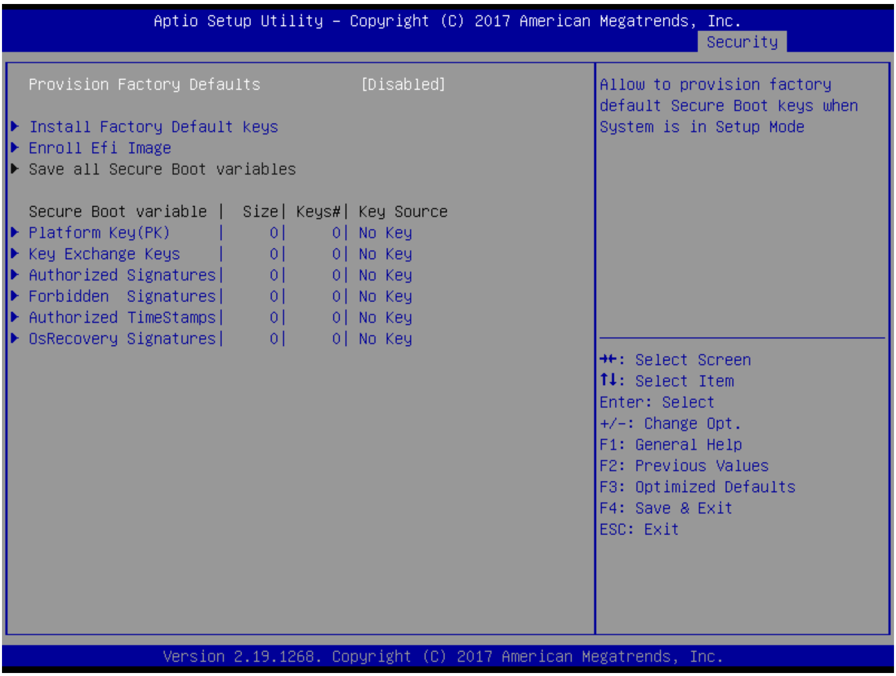


Attempt Secure Boot
Secure boot activated when Platform Key(PK) is enrolled,
System mode is User/Deployed, and CSM function is disabled
Enabled / **Disabled**

Secure Boot Mode
Secure Boot mode selector. 'Custom' mode enables users to change Image execution policy and manage secure boot keys.
Standard / **Custom**

Key Management
Enables experienced users to modify Secure Boot variables

3.7.2 Key Management



Provision Factory Default Keys
Install factory default Secure Boot Keys when System is in Setup Mode.
Enabled / **Disabled**

Install Factory Default Keys
Force System to User Mode-install all Factory Default keys

Enroll Efi Image
Allow the image to run in Secure Boot mode. Enroll SHA256 hash of the binary into Autorized Signature Database (db)

Platform Key (PK)
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)
Key Source:
Default, External, Mixed, Test
Set New

Key Exchange Keys
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)
Key Source:
Default, External, Mixed, Test

Authorized Signatures
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)

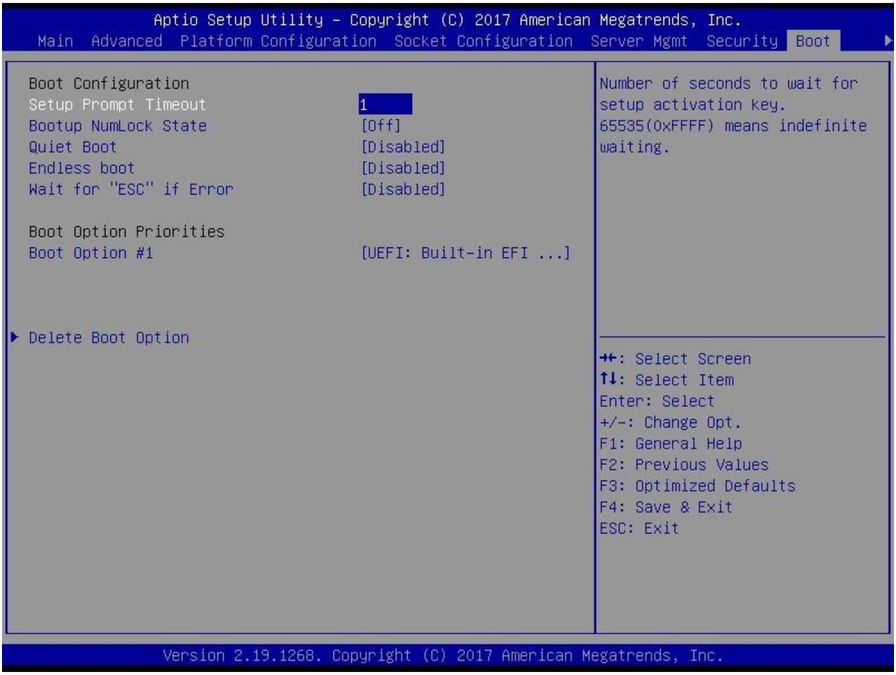
Key Source:
Default, External, Mixed, Test

Forbidden Signatures
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)
Key Source:
Default, External, Mixed, Test

Authorized TimeStamps
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)
Key Source:
Default, External, Mixed, Test

OsRecovery Signatures
Enroll Factory Defaults or load certificates from a file:
1. Public Key Certificate in:
a) EFI_SIGNATURE_LIST
b) EFI_CERT_X509 (DER encoded)
c) EFI_CERT_RSA2048 (bin)
d) EFI_CERT_SHA256,384,512
2. Authenticated UEFI Variable
3. EFI PE/COFF Image(SHA256)
Key Source:
Default, External, Mixed, Test

3.8 Boot



Setup Prompt Timeout
Number of seconds to wait for setup activation key.
65535 (0xFFFF) means indefinite waiting.

Bootup NumLock State
Select the keyboard NumLock state.
Off / On

Quiet Boot
Enable or disable Quiet Boot option.
Disabled / Enabled

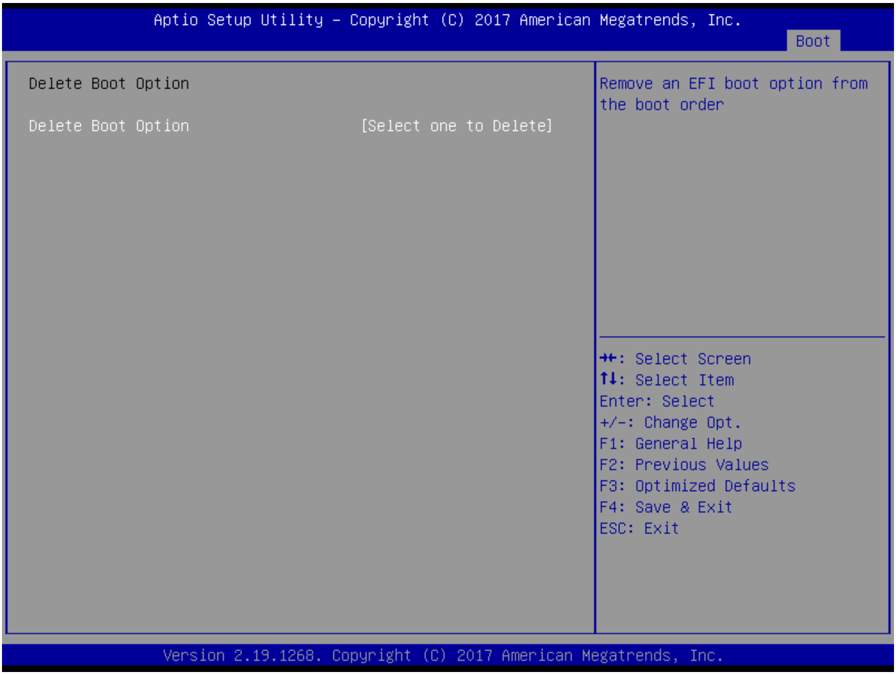
Endless Boot
Enabled or Disabled Endless boot
Disabled / Enabled

Wait for “ESC” if Error
Wait for ESC stop when BIOS has error appeared.
Disabled / Enabled

Boot Option Priorities
Boot Option #1
Sets the system boot order.
Device Name **/** Disabled

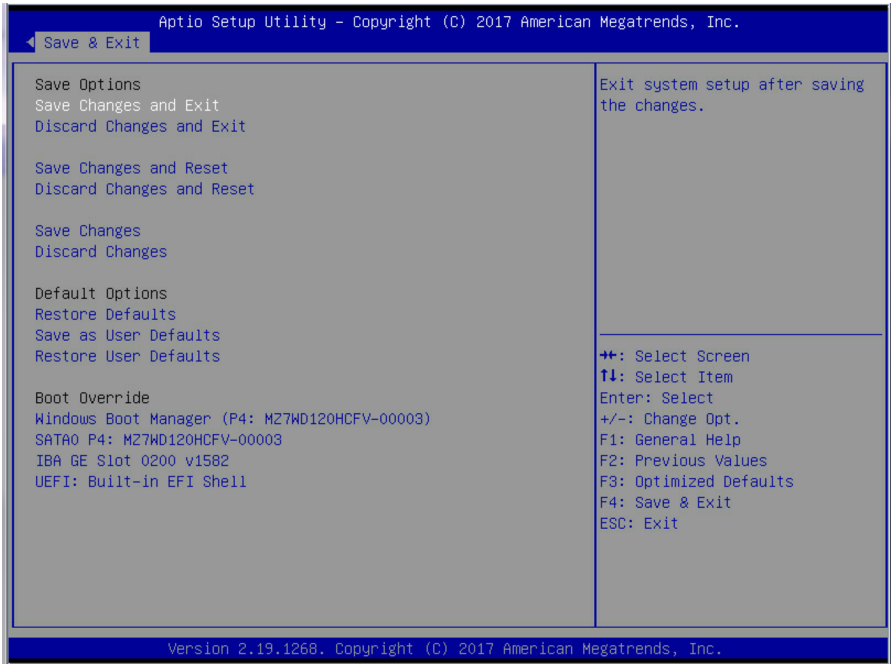
Delete Boot Option
Remove an EFI boot option from the boot order

3.8.1 Delete Boot Option Configuration



Delete Boot Option
Sets the system boot order.
Device Name / Select one to Delete

3.9 Save & Exit



Save Changes and Exit
Exit system setup after saving the changes.

Discard Changes and Exit
Exit system setup without saving any changes.

Save Changes and Reset
Reset the system after saving the changes.

Discard Changes and Reset
Reset system setup without saving any changes.

Save Changes
Save changes done so far to any of the setup options.

Discard Changes
Discard changes done so far to any of the setup options.

Restore Defaults
Restore/Load Default values for all the setup options.

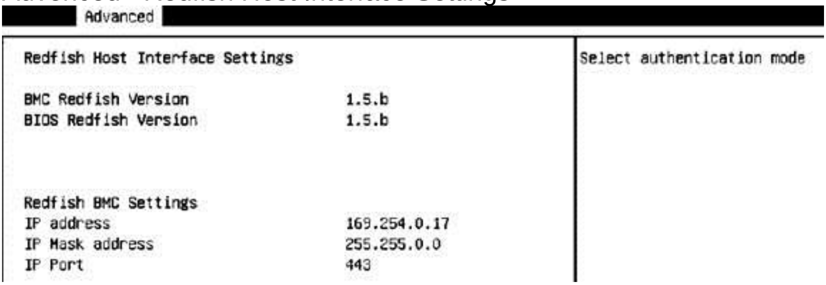
Save as User Defaults
Save the changes done so far as User Defaults.

Restore User Defaults
Restore the User Defaults to all the setup options.

Boot Override
Device Name

Bmc and BIOS 都要搭配有 Redfish 版本

Advenced >Redfish Host Interface Settings



- BMC Redfish Version

The version cannot be blank and needs to return the correct version.

- BIOS Redfish Version

The version cannot be blank and needs to return the correct version.

Redfish BMC Settings

- IP address	169.254.0.17
- IP Mask address	255.255.0.0
IP Port	443

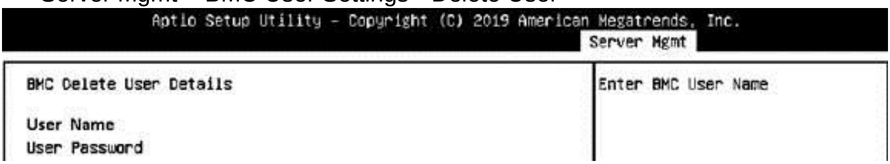
[BMC User Settings]

■ Server Mgmt > BMC User Settings

■ Server Mgmt > BMC User Settings->Add User

- User Name (to wait for typing by user)
- User Password (to wait for typing by user)
- User Access
[Enabled]\[Disabled]
- Channel No
[N/A]\[1]\[8]
- User Privilege Limit
[None]\[User]\[Operator]\[Administrator]

■ Server Mgmt > BMC User Settings->Delete User



- User Name (to wait for typing by user)
- User Password (to wait for typing by user)

Advanced ->Tls Auth Configuration

Advanced ->Tls Auth Configuration->Server CA Configuration

Advanced ->Tls Auth Configuration->Server CA Configuration->Enroll Cert

- Cert GUID

Input digit character in 11111111-2222-3333-4444-1234567890ab format.

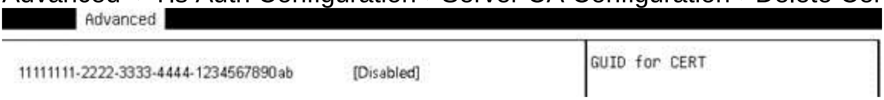
- Commit Changes and Exit

To commit the inputted digit file.

- Discard Changes and Exit

To discard the inputted digit file.

Advanced ->Tls Auth Configuration->Server CA Configuration->Delete Cert



- 11111111-2222-3333-4444-1234567890ab

[Disabled] / [Enabled]

If user selects enabled, the exited CA will be removed.

[SDDC]

■ Socket Configuration >Memory Configuration > Integrated Memory Controller (iMC)> Memory RAS Configuration

```
                                                    Socket Configuration
/--------------------------------------------------------+------------------------\
|                                                        |Mirror Mode will set    |
|  ---------------------------------------------...      |entire 1LM/2LM memory   |
|  Memory RAS Configuration Setup                        |in system to be         |
|  ---------------------------------------------...      |mirrored, consequently  |
|                                                        |reducing the memory     |
|  Mirror mode                 [Disabled]                |capacity by half.       |
|  Mirror TAD0                 [Disabled]                |Mirror Enable will      |
|  Enable Partial Mirror       [Disabled]                |disable XPT Prefetch    |
|  Memory Rank Sparing         [Disabled]                |                        |
|  Correctable Error Thr       10                        |------------------------|
|  SDDC                        [Disabled]                |><: Select Screen       |
|  Patrol Scrub                [Enabled]                 |^v: Select Item         |
|                                                        |Enter: Select           |
|                                                        |+/-: Change Opt.        |
|                                                        |F1: General Help        |
|                                                        |F2: Previous Values     |
|                                                        |F3: Optimized Defaults  |
|                                                        |F4: Save & Exit         |
|                                                        |ESC: Exit               |
\--------------------------------------------------------+------------------------/
```

- Mirror mode

[Enabled Mirror Mode (1LM)]/[Disabled]

- Mirror TAD0

[Enabled]/[Disabled]

- Enable Partial Mirror

[Enabled Mirror Mode (1LM)]/[Disabled]

- Memory Rank Sparing

[Enabled]/[Disabled]

- Correctable Error Threshold

*Default value is 10, Correctable Error Threshold (1 - 32767) used for sparing, tagging, and leaky bucket.

* “ 0-Disable Error ” need add on setup help to remind user.

- SDDC

[Enabled]/[Disabled]

- Patrol Scrub

[Enabled]/[Disabled]

NVME Information and Boot Option

NVME in the boot option shows following：

1. When the NVME populates on PCIE slot or riser card, the boot option item displays such as “NVMe #BBDF PLEXTOR PX-128M8PeY”.
2. When NVME populates on HDD BP Board of TYAN, the boot option item displays “NVMex #BBDF ADATA SX7000NP”, refer to chapter of HDD SEQUENCE ORDER BY FRONT SIDE VIEW of system engineering spec.

   Note: x is Tyan NVMe HD BP location, and 0 means first NVMe HD. #BBDF, BB means bus number, D is device number and F is function number.

For example,

```
Boot Option #1                                  [NVMe #8100 PLEXTOR...]
Boot Option #2                                  [NVMe1 #C400 ADATA ...]
```

Boot option #1, #8100 meaning BUS 81 Device 0 and Function 0, is located in a specific slot or riser card.

Boot option #2, #C400 meaning BUS C4 Device 0 and Function 0, is located in NVMe3 such as below picture from system engineering spec.

HDD SEQUENCE ORDER BY FRONT SIDE VIEW

NVMe 1	HDD 1	HDD 3	HDD 5	NVMe 3
NVMe 0	HDD 0	HDD 2	HDD 4	NVMe 2

The NVMe controller and drive information in Advance page corresponds with boot option.

```
NVMe controller and Drive information

NVMe Bus:81 Dev:0 Func:0                PLEXTOR PX-128M8PeY
Nvme Size                               128.0GB

NVMe1 Bus:C4 Dev:0 Func:0               ADATA SX7000NP
Nvme Size                               128.0GB
```

Chapter 4: Diagnostics

NOTE: if you experience problems with setting up your system, always check the following things in the following order:

Memory, Video, CPU

By checking these items, you will most likely find out what the problem might have been when setting up your system. For more information on troubleshooting, check the TYAN website at http://www.tyan.com.

4.1 Flash Utility

Every BIOS file is unique for the motherboard it was designed for. For Flash Utilities, BIOS downloads, and information on how to properly use the Flash Utility with your motherboard, please check the TYAN web site at http://www.tyan.com

NOTE: Please be aware that by flashing your BIOS, you agree that in the event of a BIOS flash failure, you must contact your dealer for a replacement BIOS. There are no exceptions. TYAN does not have a policy for replacing BIOS chips directly with end users. In no event will TYAN be held responsible for damages done by the end user.

4.2 AMIBIOS Post Code (Aptio)

The POST code checkpoints are the largest set of checkpoints during the BIOS pre-boot process. The following table describes the type of checkpoints that may occur during the POST portion of the BIOS:

Checkpoint Ranges

Status Code Range	Description
0x01 – 0x0B	SEC execution
0x0C – 0x0F	Sec errors
0x10 – 0x2F	PEI execution up to and including memory detection
0x30 – 0x4F	PEI execution after memory detection
0x50 – 0x5F	PEI errors
0x60 – 0x8F	DXE execution up to BDS
0x90 – 0xCF	BDS execution
0xD0 – 0xDF	DXE errors
0xE0 – 0xE8	S3 Resume (PEI)
0xE9 – 0xEF	S3 Resume errors (PEI)
0xF0 – 0xF8	Recovery (PEI)
0xF9 – 0xFF	Recovery errors (PEI)

Standard Checkpoints

SEC Phase

Status Code	Description
0x00	Note used
Progress Codes	
0x01	Power on. Reset type detection (soft/hard).
0x02	AP initialization before microcode loading
0x03	North Bridge initialization before microcode loading
0x04	South Bridge initialization before microcode loading
0x05	OEM initialization before microcode loading
0x06	Microcode loading
0x07	AP initialization after microcode loading
0x08	North Bridge initialization after microcode loading
0x09	South Bridge initialization after microcode loading
0x0A	OEM initialization after microcode loading
0x0B	Cache initialization

SEC Error Codes	
0x0C – 0x0D	Reserved for future AMI SEC error codes
0x0E	Microcode not found
0x0F	Microcode not found

SEC Phase

None

PEI Phase

Status Code	Description
Progress Codes	
0x10	PCI Core is started
0x11	Pre-memory CPU initialization is started
0x12	Pre-memory CPU initialization (CPU module specific)
0x13	Pre-memory CPU initialization (CPU module specific)
0x14	Pre-memory CPU initialization (CPU module specific)
0x15	Pre-memory North Bridge initialization is started
0x16	Pre-Memory North Bridge initialization (North Bridge module specific)
0x17	Pre-memory North Bridge initialization (North Bridge module specific)
0x18	Pre-Memory North Bridge initialization (North Bridge module specific)
0x19	Pre-memory South Bridge initialization is started
0x1A	Pre-Memory South Bridge initialization (South Bridge module specific)
0x1B	Pre-memory South Bridge initialization (South Bridge module specific)
0x1C	Pre-Memory South Bridge initialization (South Bridge module specific)
0x1D – 0x2A	OEM pre-memory initialization codes
0x2B	Memory initialization. Serial Presence Detect (SPD) data reading
0x2C	Memory initialization. Memory presence detection
0x2D	Memory initialization. Programming memory timing information
0x2E	Memory initialization. Configuring memory
0x2F	Memory initialization (other)
0x30	Reserved for ASL (see ASL Status Codes section below)
0x31	Memory Installed
0x32	CPU post-memory initialization is started.
0x33	CPU post-memory initialization. Cache initialization
0x34	CPU post-memory initialization. Application Processor(s) (AP) initialization

Status Code	Description
0x35	CPU post-memory initialization. Boot Strap Processor (BSP) selection
0x36	CPU post-memory initialization. System Management Mode (SMM) initialization
0x37	Post-Memory North Bridge initialization is started.
0x38	Post-Memory North Bridge initialization (North Bridge module specific)
0x39	Post-Memory North Bridge initialization (North Bridge module specific)
0x3A	Post-Memory North Bridge initialization (North Bridge module specific)
0x3B	Post-Memory South Bridge initialization is started
0x3C	Post-Memory South Bridge initialization (South Bridge module specific)
0x3D	Post-Memory South Bridge initialization (South Bridge module specific)
0x3E	Post-Memory South Bridge initialization (South Bridge module specific)
0x3F – 0x4E	OEM post memory initialization codes
0x4F	DXE PIL is started
PCI Error Codes	
0x50	Memory initialization error. Invalid memory type or incompatible memory speed
0x51	Memory initialization error. SPD reading has failed.
0x52	Memory initialization error. Invalid memory size or memory modules do not match.
0x53	Memory initialization error. No usable memory detected
0x54	Unspecified memory initialization error
0x55	Memory not installed
0x56	Invalid CPU type or speed
0x57	CPU mismatch
0x58	CPU self test failed or possible CPU cache error
0x59	CPU microcode is not found or microcode update is failed.
0x5A	Internal CPU error
0x5B	Reset PPI is not available.
0x5C – 0x5F	Reserved for future AMI error codes
S3 Resume Progress Codes	
0xE0	S3 Resume is started (S3 Resume PPI is called by the DXE IPL).
0xE1	S3 Boot Script execution
0xE2	Video repost
0xE3	OS S3 wake vector call
0xE4 – 0xE7	Reserved for future AMI progress codes

Status Code	Description
S3 Resume Error Codes	
0xE8	S3 Resume failed
0xE9	S3 Resume PPI not found
0xEA	S3 Resume Boot Script error
0xEB	S3 OS wake error
0xEC – 0xEF	Reserved for future AMI error codes
Recovery Progress Codes	
0xF0	Recovery condition triggered by firmware (Auto recovery)
0xF1	Recovery condition triggered by user (forced recovery)
0xF2	Recovery process started
0xF3	Recovery firmware image is found.
0xF4	Recovery firmware image is loaded.
0xF5 – 0xF7	Reserved for future AMI progress codes
Recovery Error Codes	
0xF8	Recovery PPI is not available.
0xF9	Recovery capsule is not found.
0xFA	Invalid recovery capsule
0xFB – 0xFF	Reserved for future AMI error codes

PEI Beep Codes

# of Beeps	Description
Progress Codes	
1	Memory not installed
1	Memory was installed twice (installPEIMemory routine in PEI Core called twice).
2	Recovery started
3	DXEIPL was not found.
3	DXE Core Firmware Volume was not found.
4	Recovery failed
4	S3 Resume failed
7	Reset PPI is not available.

DXE Phase

Status Code	Description
0x60	DXE Core is started.
0x61	NVRAM initialization
0x62	Installation of the South Bridge Runtime Services

Status Code	Description
0x63	CPU DXE initialization is started.
0x64	CPU DXE initialization (CPU module specific)
0x65	CPU DXE initialization (CPU module specific)
0x66	CPU DXE initialization (CPU module specific)
0x67	CPU DXE initialization (CPU module specific)
0x68	PCI host bridge initialization
0x69	North Bridge DXE initialization is started.
0x6A	North Bridge DXE SMM initialization is started.
0x6B	North Bridge DXE initialization (North Bridge module specific)
0x6C	North Bridge DXE initialization (North Bridge module specific)
0x6D	North Bridge DXE initialization (North Bridge module specific)
0x6E	North Bridge DXE initialization (North Bridge module specific)
0x6F	North Bridge DXE initialization (North Bridge module specific)
0x70	South Bridge DXE initialization is started.
0x71	South Bridge DXE SMM initialization is started.
0x72	South Bridge devices initialization
0x73	South Bridge DXE initialization (South Bridge module specific)
0x74	South Bridge DXE initialization (South Bridge module specific)
0x75	South Bridge DXE initialization (South Bridge module specific)
0x76	South Bridge DXE initialization (South Bridge module specific)
0x77	South Bridge DXE initialization (South Bridge module specific)
0x78	ACPI module initialization
0x79	CSM initialization
0x7A – 0x7F	Reserved for future AMI DXE codes
0x80 – 0x8F	OEM DXE initialization codes
0x90	Boot Device Selection (BDS) phase is started
0x91	Driver connecting is started
0x92	PCI Bus initialization is started
0x93	PCI Bus Hot Plug Controller initialization
0x94	PCI Bus Enumeration
0x95	PCI BUS Request Resources
0x96	PCI Bus Assign Resources
0x97	Console output devices connect
0x98	Console Input devices connect
0x99	Super IO initialization
0x9A	USB initialization is started.

Status Code	Description
0x9B	USB Reset
0x9C	USB Detect
0x9D	USB Enable
0x9E -0x9F	Reserved for future AMI codes
0xA0	IDE initialization is started
0xA1	IDE Reset
0xA2	IDE Detect
0xA3	IDE Enable
0xA4	SCSI initialization is started.
0xA5	SCSI Reset
0xA6	SCSI Detect
0xA7	SCSI Enable
0xA8	Setup Verifying Password
0xA9	Start of Setup
0xAA	Reserved for ASL (see ASL Status Codes section below)
0xAB	Setup Input Wait
0xAC	Reserved for ASL (see ASL Status Codes section below)
0xAD	Ready To Boot event
0xAE	Legacy Boot event
0xAF	Exit Boot Services event
0xB0	Runtime Set Virtual Address MAP Begin
0xB1	Runtime Set Virtual Address MAP End
0xB2	Legacy Option ROM initialization
0xB3	System Reset
0xB4	USB hot plug
0xB5	PCI bus hot plug
0xB6	Clean-up of NVRAM
0xB7	Configuration Reset (reset of NVRAM settings)
0xB8 – 0xBF	Reserved for future AMI codes
0xC0 – 0xCF	OEM BDS initialization codes
DXE Error Codes	
0xD0	CPU initialization error
0xD1	North Bridge initialization error
0xD2	South Bridge initialization error
0xD3	Some of the Architectural Protocols are not available
0xD4	PCI resource allocation error. Out of Resources

Status Code	Description
0xD5	No Space for Legacy Option ROM
0xD6	No Console Output Devices are found.
0xD7	No Console Input Devices are found.
0xD8	Invalid password
0xD9	Error loading Boot Option (LoadImage returned error)
0xDA	Boot Option is failed (StartImage returned error).
0xDB	Flash update is failed.
0xDC	Reset protocol is not available.

DXE Beep Codes

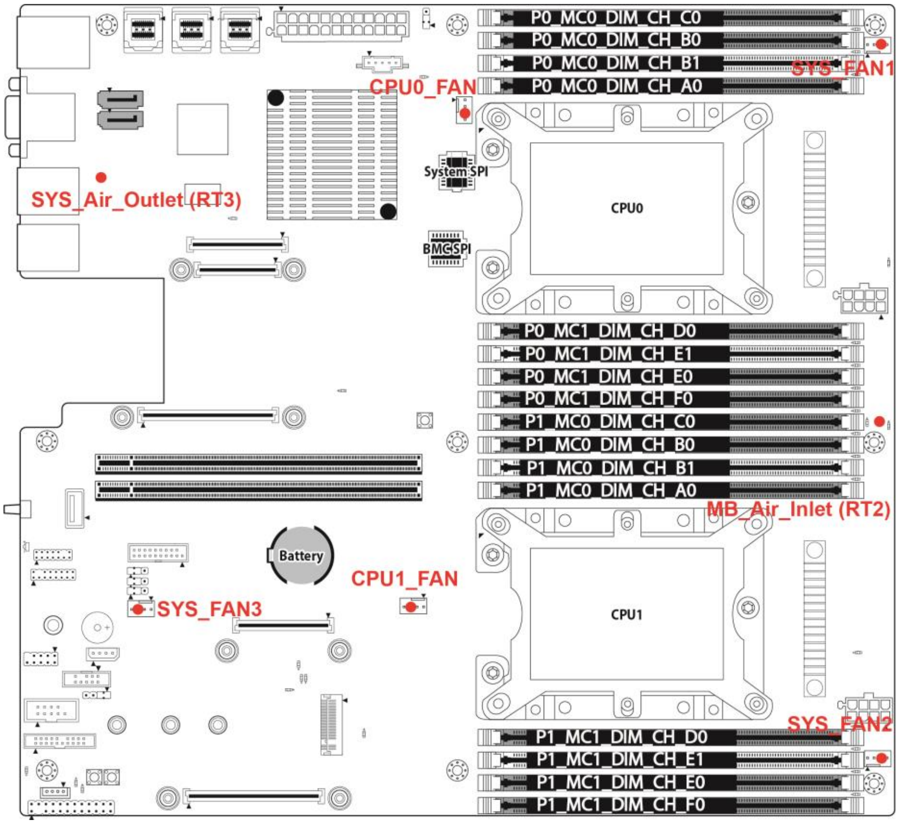
# of Beeps	Description
1	Invalid password
4	Some of the Architectural Protocols are not available.
5	No Console Output Devices are found.
5	No Console Input Devices are found.
6	Flash update is failed.
7	Reset protocol is not available.
8	Platform PCI resource requirements cannot be met.

ACPI/ASL Checkpoints

Status Code	Description
0x01	System is entering S1 sleep state.
0x02	System is entering S2 sleep state.
0x03	System is entering S3 sleep state.
0x04	System is entering S4 sleep state.
0x05	System is entering S5 sleep state.
0x10	System is waking up from the S1 sleep state.
0x20	System is waking up from the S2 sleep state.
0x30	System is waking up from the S3 sleep state.
0x40	System is waking up from the S4 sleep state.
0xAC	System has transitioned into ACPI mode. Interrupt controller is in APIC mode.
0xAA	System has transitioned into ACPI mode. Interrupt controller is in APIC mode.

Appendix I: Fan and Temp Sensors

This section aims to help readers identify the locations of some specific FAN and Temp Sensors on the motherboard. A table of BIOS Temp sensor name explanation is also included for readers' reference.

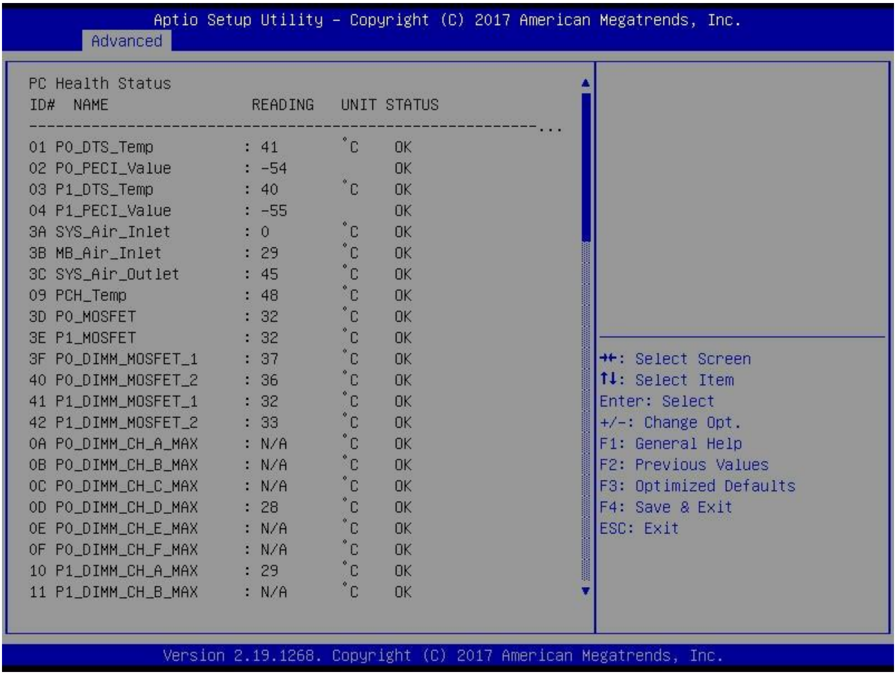


NOTE: The red dot indicates the sensor.

Fan and Temp Sensor Location:

1. Fan Sensor: It is located in the third pin of the fan connector, which detects the fan speed (rpm)
2. Temp Sensor: SYS_Air_Outlet (RT3),and MB_Air_Inlet (RT2). They detect the system temperature around.
   NOTE: The system temperature is measured in a scale defined by **Intel**, not in Fahrenheit or Celsius.

BIOS Temp Sensor Name Explanation:



```
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Advanced

12 P1_DIMM_CH_C_MAX      : N/A      °C    OK
13 P1_DIMM_CH_D_MAX      : N/A      °C    OK
14 P1_DIMM_CH_E_MAX      : N/A      °C    OK
15 P1_DIMM_CH_F_MAX      : N/A      °C    OK
50 PVCCP_CPU0            : 1.7954   V     OK
51 PVCCIO_CPU0           : 1.0368   V     OK
52 PVDDQ_CPU0            : 1.2528   V     OK
53 PVPP_CPU0             : 2.5870   V     OK
54 PVCCP_CPU1            : 1.7954   V     OK
55 PVCCIO_CPU1           : 1.0296   V     OK
56 PVDDQ_CPU1            : 1.2528   V     OK
57 PVPP_CPU1             : 2.5870   V     OK
58 VCC12                 : 12.033   V     OK
59 VCC5                  : 5.1216   V     OK      →←: Select Screen
5A VCC3                  : 3.3060   V     OK      ↑↓: Select Item
5B VCC3_AUX              : 3.2886   V     OK      Enter: Select
5C P1V8_PCH              : 1.8142   V     OK      +/-: Change Opt.
5D PVNN_PCH              : 1.0296   V     OK      F1: General Help
5E P1V05_PCH             : 1.0728   V     OK      F2: Previous Values
5F RTC_BAT               : 3.1017   V     OK      F3: Optimized Defaults
60 CPU0_FAN              : N/A      RPM   OK      F4: Save & Exit
61 CPU1_FAN              : N/A      RPM   OK      ESC: Exit
62 SYS_FAN_1             : N/A      RPM   OK
63 SYS_FAN_2             : 1800     RPM   OK
64 SYS_FAN_3             : 2700     RPM   OK

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Aptio Setup Utility – Copyright (C) 2017 American Megatrends, Inc.
Advanced

5B VCC3_AUX              : 3.2886   V     OK
5C P1V8_PCH              : 1.8142   V     OK
5D PVNN_PCH              : 1.0296   V     OK
5E P1V05_PCH             : 1.0728   V     OK
5F RTC_BAT               : 3.1017   V     OK
60 CPU0_FAN              : N/A      RPM   OK
61 CPU1_FAN              : N/A      RPM   OK
62 SYS_FAN_1             : N/A      RPM   OK
63 SYS_FAN_2             : 1800     RPM   OK
64 SYS_FAN_3             : 2700     RPM   OK
65 SYS_FAN_4             : N/A      RPM   OK
66 SYS_FAN_5             : N/A      RPM   OK
67 SYS_FAN_6             : N/A      RPM   OK
68 SYS_FAN_7             : N/A      RPM   OK      →←: Select Screen
69 SYS_FAN_8             : N/A      RPM   OK      ↑↓: Select Item
6A SYS_FAN_9             : N/A      RPM   OK      Enter: Select
6B SYS_FAN_10            : N/A      RPM   OK      +/-: Change Opt.
6C SYS_FAN_11            : N/A      RPM   OK      F1: General Help
6D SYS_FAN_12            : N/A      RPM   OK      F2: Previous Values
90 PSU0_STATUS           : N/A            Alert   F3: Optimized Defaults
91 PSU1_STATUS           : N/A            Alert   F4: Save & Exit
B0 ChassisIntrusion      : N/A            OK      ESC: Exit

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```

P0_DTS_Temp	Temperature of the P0_DTS
P0_PECI_Value	Temperature of the P0_PECI_Value
P1_DTS_Temp	Temperature of the P1_DTS
P1_PECI_Value	Temperature of the P1_PECI_Value
SAS_Air_Inlet	Temperature of the SAS_Air_Inlet Area
MB_Air_Inlet	Temperature of the MB_Air_Inlet Area
SAS_Air_Outlet	Temperature of the SAS_Air_Outlet Area
PCH_Temp	Temperature of the PCH
P0_MOSFET	Temperature of the CPU0 MOSFET
P1_MOSFET	Temperature of the CPU1 MOSFET
P0_DIMM_MOSFET_1	Temperature of the CPU0 DIMM Channel ABC MOSFET
P0_DIMM_MOSFET_2	Temperature of the CPU0 DIMM Channel DEF MOSFET
P1_DIMM_MOSFET_1	Temperature of the CPU1 DIMM Channel ABC MOSFET
P1_DIMM_MOSFET_2	Temperature of the CPU1 DIMM Channel DEF MOSFET
P0_DIMM_CH_A_MAX	The highest temperature of CPU0 DIMM channel A slot
P0_DIMM_CH_B_MAX	The highest temperature of CPU0 DIMM channel B slot
P0_DIMM_CH_C_MAX	The highest temperature of CPU0 DIMM channel C slot
P0_DIMM_CH_D_MAX	The highest temperature of CPU0 DIMM channel D slot
P0_DIMM_CH_E_MAX	The highest temperature of CPU0 DIMM channel E slot
P0_DIMM_CH_F_MAX	The highest temperature of CPU0 DIMM channel F slot
P1_DIMM_CH_A_MAX	The highest temperature of CPU1 DIMM channel A slot
P1_DIMM_CH_B_MAX	The highest temperature of CPU1 DIMM channel B slot
P1_DIMM_CH_C_MAX	The highest temperature of CPU1 DIMM channel C slot
P1_DIMM_CH_D_MAX	The highest temperature of CPU1 DIMM channel D slot
P1_DIMM_CH_E_MAX	The highest temperature of CPU1 DIMM channel E slot
P1_DIMM_CH_F_MAX	The highest temperature of CPU1 DIMM channel F slot

BIOS FAN Sensor	**Name Explanation**
CPU0_FAN	Fan speed of CPU0_FAN
CPU1_FAN	Fan speed of CPU1_FAN
SYS_FAN_1	Fan speed of SYS_FAN_1
SYS_FAN_2	Fan speed of SYS_FAN_2
SYS_FAN_3	Fan speed of SYS_FAN_3
SYS_FAN_4	Fan speed of SYS_FAN_4
SYS_FAN_5	Fan speed of SYS_FAN_5
SYS_FAN_6	Fan speed of SYS_FAN_6
SYS_FAN_7	Fan speed of SYS_FAN_7
SYS_FAN_8	Fan speed of SYS_FAN_8
SYS_FAN_9	Fan speed of SYS_FAN_9

SYS_FAN_10	Fan speed of SYS_FAN_10
SYS_FAN_11	Fan speed of SYS_FAN_11
SYS_FAN_12	Fan speed of SYS_FAN_12

Appendix II: How to recover UEFI BIOS

Important Notes:

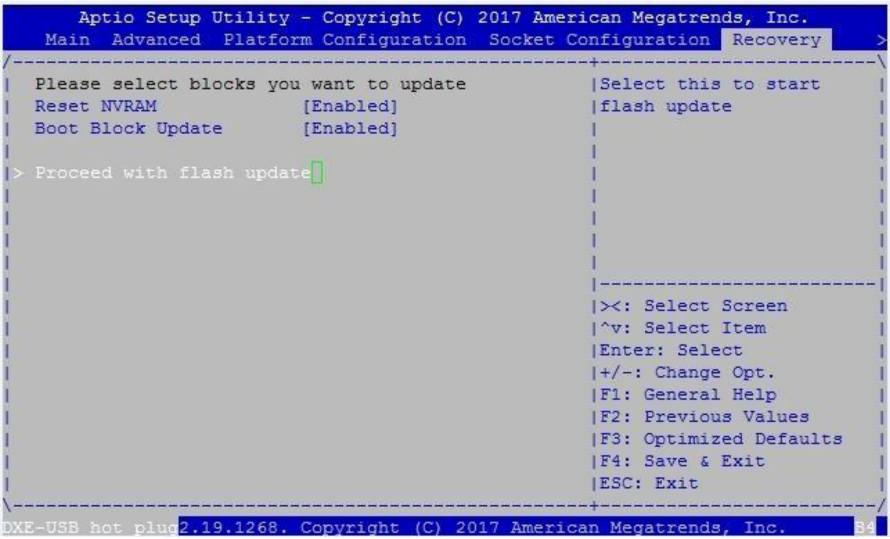
The emergency UEFI BIOS Recovery process is only used to rescue a system with a failed or corrupted BIOS image that fails to boot to an OS. It is not intended to be used as a general purpose BIOS flashing procedure and should not be used as such. Please do not shutdown or reset the system while the BIOS recovery process is underway or there is risk of damage to the UEFI recovery bootloader that would prevent the recovery process itself from working. In no event shall Tyan be liable for direct, indirect, incidental, special or consequential damages arising from the BIOS update or recovery.

The BIOS Recovery file is named xxxx.cap, where the 'xxxx' portion is the motherboard model number. Examples: 5630.cap, 7106.cap, 7109.cap, etc. Please make sure that you are using the correct BIOS Recovery file from Tyan's web site.

BIOS Recovery Process

1.Place the recovery BIOS file (xxxx.cap) in the root directory of a USB disk.
2.Ensure that the system is powered off.
3.Insert the USB disk to any USB port on the motherboard or chassis.
4.Power the system on while pressing “Ctrl” and “Home” simultaneously on the keyboard. Continue to hold these keys down until the following Tyan screen is displayed on the monitor:

5.The system will boot to BIOS setup. A new menu item will appear at the far right of the screen. Scroll to the 'Recovery' tab, move the curser to “Proceed with flash update” and press the "Enter" key on the keyboard to start the BIOS recovery process.



6.IMPORTANT: Do not power off or reboot the server during the BIOS recovery process. This can damage the BIOS recovery bootloader and prevent it from loading a subsequent time.

7.Wait for the BIOS recovery procedure to complete. Completion is signified with the message “Flash update completed. Press any key to reset the system” displayed on screen.

8.Remove the USB disk and reboot.

If your system does not have video output or the POST code halts at “FF” on the right-lower portion of the screen, please contact Tyan representatives for RMA service.

Appendix III: PCIE Slot Location and Setup Items Corresponding List

VMD with PCIE Slot

BIOS Setup → Socket Configuration → VMD config for PStackN (CPUN)		PCIE Slot Location	Reference on UG
PStack0(CPU0)	VMD Port 1A	A+C OCP	Chapter 2.4 Board Parts, Jumpers and Connectors and Motherboard Components
PStack0(CPU0)	VMD Port 1B	A+C OCP	
PStack0(CPU0)	VMD Port 1C	A+C OCP	
PStack0(CPU0)	VMD Port 1D	A+C OCP	
PStack2(CPU1)	VMD Port 3A	F+G OCP	
PStack2(CPU1)	VMD Port 3B	F+G OCP	
PStack2(CPU1)	VMD Port 3C	F+G OCP	
PStack2(CPU1)	VMD Port 3D	F+G OCP	

* PStack0 corresponds to IOU0(PCIE PORT 1A1B1C1D)
PStack1 corresponds to IOU1(PCIE PORT 2A2B2C2D)
PStack2 corresponds to IOU2(PCIE PORT 3A3B3C3D)

2U SKU PCIE slot populates riser card.

BIOS Setup- IIO Configuration		PCIE PORT	PCIE Slot Location	Riser Type	Reference on UG
Socket0 Configuration	IOU2(IIO PCIe Br3)	3A3B3C3D	E.PCIE	M7106-L24-3F(up)	Chapter 2.4 Board Parts, Jumpers and Connectors and Motherboard Components
Socket0 Configuration	IOU2(IIO PCIe Br3)	3C3D	E.PCIE	M7106-L24-3F(middle)	
Socket1 Configuration	IOU0(IIO PCIe Br1)	1A1B	E.PCIE	M7106-L24-3F(down)	
Socket1 Configuration	IOU1 (IIO PCIe Br2)	2C2D	D.PCIE	M7106-R24-3F(up)	
Socket1 Configuration	IOU1 (IIO PCIe Br2)	2A2B2C2D	D.PCIE	M7106-R24-3F(middle)	
Socket1 Configuration	IOU0 (IIO PCIe Br1)	1C1D	D.PCIE	M7106-R24-3F(down)	
Socket0 Configuration	IOU0 (IIO PCIe Br1)	1A1B1C1D	N/A	LAN OCP	
Socket1 Configuration	IOU2 (IIO PCIe Br3)	3A3B3C3D	N/A	STORAGE OCP	

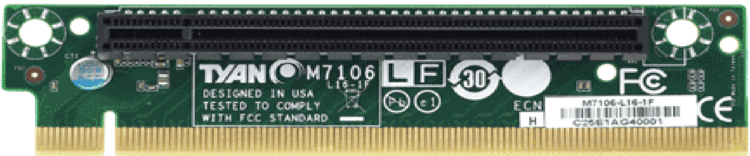
M7106-L24-3F (Sold separately) M7106-R24-3F (Sold separately)

1U SKU PCIE slot populates riser card.

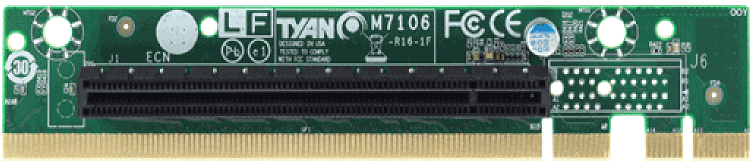
BIOS Setup- IIO Configuration		PCIE PORT	PCIE Slot Location	Riser Type	Reference on UG
Socket0 Configuration	IOU2(IIO PCIe Br3)	3A3B3C3D	E.PCIE	M7106-L16-1F	chapter 2.4 Board Parts, Jumpers and Connectors and Motherboard Components
Socket1 Configuration	IOU1(IIO PCIe Br2)	2A2B2C2D	D.PCIE	M7106-R16-1F	

PCIe x16 FH/HL cards (In Left and Right 1U riser cards)

M7106-L16-1F (Sold separately)



M7106-R16-1F (Sold separately)



NOTE

Glossary

ACPI (Advanced Configuration and Power Interface): a power management specification that allows the operating system to control the amount of power distributed to the computer's devices. Devices not in use can be turned off, reducing unnecessary power expenditure.

AGP (Accelerated Graphics Port): a PCI-based interface which was designed specifically for demands of 3D graphics applications. The 32-bit AGP channel directly links the graphics controller to the main memory. While the channel runs only at 66 MHz, it supports data transmission during both the rising and falling ends of the clock cycle, yielding an effective speed of 133 MHz.

ATAPI (AT Attachment Packet Interface): also known as IDE or ATA; a drive implementation that includes the disk controller on the device itself. It allows CD-ROMs and tape drives to be configured as master or slave devices, just like HDDs.

ATX: the form factor designed to replace the AT form factor. It improves on the AT design by rotating the board 90 degrees, so that the IDE connectors are closer to the drive bays, and the CPU is closer to the power supply and cooling fan. The keyboard, mouse, USB, serial, and parallel ports are built-in.

Bandwidth: refers to carrying capacity. The greater the bandwidth, the more data the bus, phone line, or other electrical path can carry. Greater bandwidth results in greater speed.

BBS (BIOS Boot Specification): a feature within the BIOS that creates, prioritizes, and maintains a list of all Initial Program Load (IPL) devices, and then stores that list in NVRAM. IPL devices have the ability to load and execute an OS, as well as provide the ability to return to the BIOS if the OS load process fails. At that point, the next IPL device is called upon to attempt loading of the OS.

BIOS (Basic Input/Output System): the program that resides in the ROM chip, which provides the basic instructions for controlling your computer's hardware. Both the operating system and application software use BIOS routines to ensure compatibility.

Buffer: a portion of RAM which is used to temporarily store data; usually from an application though it is also used when printing and in most keyboard drivers. The CPU can manipulate data in a buffer before copying it to a disk drive. While this improves system performance (reading to or writing from a disk drive a single time is much faster than doing so repeatedly) there is the possibility of losing your data should the system crash. Information in a buffer is temporarily stored, not permanently saved.

Bus: a data pathway. The term is used especially to refer to the connection between the processor and system memory, and between the processor and PCI or ISA local buses.

Bus mastering: allows peripheral devices and IDEs to access the system memory without going through the CPU (similar to DMA channels).

Cache: a temporary storage area for data that will be needed often by an application. Using a cache lowers data access times since the information is stored in SRAM instead of slower DRAM. Note that the cache is also much smaller than your regular memory: a typical cache size is 512KB, while you may have as much as 4GB of regular memory.

Closed and open jumpers: jumpers and jumper pins are active when they are "on" or "closed", and inactive when they are "off" or "open".

CMOS (Complementary Metal-Oxide Semiconductors): chips that hold the basic startup information for the BIOS.

COM port: another name for the serial port, which is called as such because it transmits the eight bits of a byte of data along one wire, and receives data on another single wire (that is, the data is transmitted in serial form, one bit after another). Parallel ports transmit the bits of a byte on eight different wires at the same time (that is, in parallel form, eight bits at the same time).

DDR (Double Data Rate): a technology designed to double the clock speed of the memory. It activates output on both the rising and falling edge of the system clock rather than on just the rising edge, potentially doubling output.

DIMM (Dual In-line Memory Module): faster and more capacious form of RAM than SIMMs, and do not need to be installed in pairs.

DIMM bank: sometimes called DIMM socket because the physical slot and the logical unit are the same. That is, one DIMM module fits into one DIMM socket, which is capable of acting as a memory bank.

DMA (Direct Memory Access): channels that are similar to IRQs. DMA channels allow hardware devices (like soundcards or keyboards) to access the main memory without involving the CPU. This frees up CPU resources for other tasks. As with IRQs, it is vital that you do not double up devices on a single line. Plug-n-Play devices will take care of this for you.

DRAM (Dynamic RAM): widely available, very affordable form of RAM which looses data if it is not recharged regularly (every few milliseconds). This refresh requirement makes DRAM three to ten times slower than non-recharged RAM such as SRAM.

ECC (Error Correction Code or Error Checking and Correcting): allows data to be checked for errors during run-time. Errors can subsequently be corrected at the same time that they're found.

EEPROM (Electrically Erasable Programmable ROM): also called Flash BIOS, it is a ROM chip which can, unlike normal ROM, be updated. This allows you to keep up with changes in the BIOS programs without having to buy a new chip. TYAN®'s BIOS updates can be found at http://**www.tyan.com**

ESCD (Extended System Configuration Data): a format for storing information about Plug-n-Play devices in the system BIOS. This information helps properly configure the system each time it boots.

Firmware: low-level software that controls the system hardware.

Form factor: an industry term for the size, shape, power supply type, and external connector type of the Personal Computer Board (PCB) or motherboard. The standard form factors are the AT and ATX.

Global timer: onboard hardware timer, such as the Real-Time Clock (RTC).

HDD: stands for Hard Disk Drive, a type of fixed drive.

H-SYNC: controls the horizontal synchronization/properties of the monitor.

HyperTransport™: a high speed, low latency, scalable point-to-point link for interconnecting ICs on boards. It can be significantly faster than a PCI bus for an equivalent number of pins. It provides the bandwidth and flexibility critical for today's networking and computing platforms while retaining the fundamental programming model of PCI.

IC (Integrated Circuit): the formal name for the computer chip.

IDE (Integrated Device/Drive Electronics): a simple, self-contained HDD interface. It can handle drives up to 8.4 GB in size. Almost all IDEs sold now are in fact Enhanced IDEs (EIDEs), with maximum capacity determined by the hardware controller.

IDE INT (IDE Interrupt): Hardware interrupt signal that goes to the IDE.

I/O (Input/Output): the connection between your computer and another piece of hardware (mouse, keyboard, etc.)

IRQ (Interrupt Request): an electronic request that runs from a hardware device to the CPU. The interrupt controller assigns priorities to incoming requests and delivers them to the CPU. It is important that there is only one device hooked up to each IRQ line; doubling up devices on IRQ lines can lock up your system. Plug-n-Play operating systems can take care of these details for you.

Latency: the amount of time that one part of a system spends waiting for another part to catch up. This occurs most commonly when the system sends data out to a peripheral device and has to wait for the peripheral to spread (peripherals tend to be slower than onboard system components).

NVRAM: ROM and EEPROM are both examples of Non-Volatile RAM, memory that holds its data without power. DRAM, in contrast, is volatile.

Parallel port: transmits the bits of a byte on eight different wires at the same time.

PCI (Peripheral Component Interconnect): a 32 or 64-bit local bus (data pathway) which is faster than the ISA bus. Local buses are those which operate within a single system (as opposed to a network bus, which connects multiple systems).

PCI PIO (PCI Programmable Input/Output) modes: the data transfer modes used by IDE drives. These modes use the CPU for data transfer (in contrast, DMA channels do not). PCI refers to the type of bus used by these modes to communicate with the CPU.

PCI-to-PCI Bridge: allows you to connect multiple PCI devices onto one PCI slot.

Pipeline burst SRAM: a fast secondary cache. It is used as a secondary cache because SRAM is slower than SDRAM, but usually larger. Data is cached first to the faster primary cache, and then, when the primary cache is full, to the slower secondary cache.

PnP (Plug-n-Play): a design standard that has become ascendant in the industry. Plug-n-Play devices require little set-up to use. Devices and operating systems that are not Plug-n-Play require you to reconfigure your system each time you add or change any part of your hardware.

PXE (Preboot Execution Environment): one of four components that together make up the Wired for Management 2.0 baseline specification. PXE was designed to define a standard set of preboot protocol services within a client with the goal of allowing networked-based booting to boot using industry standard protocols.

RAID (Redundant Array of Independent Disks): a way for the same data to be stored in different places on many hard drives. By using this method, the data is stored redundantly and multiple hard drives will appear as a single drive to the operating system. RAID level 0 is known as striping, where data is striped (or overlapped) across multiple hard drives, but offers no fault-tolerance. RAID level 1 is known as mirroring, which stores the data within at least two hard drives, but does not stripe. RAID level 1 also allows for faster access time and fault-tolerance, since either hard drive can be read at the same time. RAID level 0+1 is striping and mirroring, providing fault-tolerance, striping, and faster access all at the same time.

RAIDIOS: RAID I/O Steering (Intel)

RAM (Random Access Memory): technically refers to a type of memory where any byte can be accessed without touching the adjacent data and is often referred to the system's main memory. This memory is available to any program running on the computer.

ROM (Read-Only Memory): a storage chip which contains the BIOS; the basic instructions required to boot the computer and start up the operating system.

SDRAM (Synchronous Dynamic RAM): called as such because it can keep two sets of memory addresses open simultaneously. By transferring data alternately from one set of addresses and then the other, SDRAM cuts down on the delays associated with non-synchronous RAM, which must close one address bank before opening the next.

Serial port: called as such because it transmits the eight bits of a byte of data along one wire, and receives data on another single wire (that is, the data is transmitted in serial form, one bit after another).

SCSI Interrupt Steering Logic (SISL): Architecture that allows a RAID controller, such as AcceleRAID 150, 200 or 250, to implement RAID on a system board-embedded SCSI bus or a set of SCSI busses. SISL: SCSI Interrupt Steering Logic (LSI) (only on LSI SCSI boards)

Sleep/Suspend mode: in this mode, all devices except the CPU shut down.

SDRAM (Static RAM): unlike DRAM, this type of RAM does not need to be refreshed in order to prevent data loss. Thus, it is faster and more expensive.

SLI (Scalable Link Interface): NVIDIA SLI technology links two graphics cards together to provide scalability and increased performance. NVIDIA SLI takes advantage of the increased bandwidth of the PCI Express bus architecture, and features hardware and software innovations within NVIDIA GPUs (graphics processing units) and NVIDIA MCPs (media and communications processors). Depending on the application, NVIDIA SLI can deliver as much as two times the performance of a single GPU configuration.

Standby mode: in this mode, the video and hard drives shut down; all other devices continue to operate normally.

UltraDMA-33/66/100: a fast version of the old DMA channel. UltraDMA is also called UltraATA. Without a proper UltraDMA controller, your system cannot take advantage of higher data transfer rates of the new UltraDMA/UltraATA hard drives.

USB (Universal Serial Bus): a versatile port. This one port type can function as a serial, parallel, mouse, keyboard or joystick port. It is fast enough to support video transfer, and is capable of supporting up to 127 daisy-chained peripheral devices.

VGA (Video Graphics Array): the PC video display standard

V-SYNC: controls the vertical scanning properties of the monitor.

ZCR (Zero Channel RAID): PCI card that allows a RAID card to use the onboard SCSI chip, thus lowering cost of RAID solution

ZIF Socket (Zero Insertion Force socket): these sockets make it possible to insert CPUs without damaging the sensitive CPU pins. The CPU is lightly placed in an open ZIF socket, and a lever is pulled down. This shifts the processor over and down, guiding it into the board and locking it into place.

Technical Support

If a problem arises with your system, you should first turn to your dealer for direct support. Your system has most likely been configured or designed by them and they should have the best idea of what hardware and software your system contains. Hence, they should be of the most assistance for you. Furthermore, if you purchased your system from a dealer near you, take the system to them directly to have it serviced instead of attempting to do so yourself (which can have expensive consequences).

If these options are not available for you then TYAN® Computer Corporation can help. Besides designing innovative and quality products for over a decade, TYAN has continuously offered customers service beyond their expectations. TYAN®'s website (**www.tyan.com**) provides easy-to-access FAQ searches and online Trouble Ticket creation as well as Instant Chat capabilities with our Support Agents. TYAN® also provides easy-to-access resources such as in-depth Linux Online Support sections with downloadable Linux drivers and comprehensive compatibility reports for chassis, memory and much more. With all these convenient resources just a few keystrokes away, users can easily find the latest software and operating system components to keep their systems running as powerful and productive as possible. TYAN® also ranks high for its commitment to fast and friendly customer support through email. By offering plenty of options for users, TYAN® serves multiple market segments with the industry's most competitive services to support them.

"TYAN's tech support is some of the most impressive we've seen, with great response time and exceptional organization in general" - Anandtech.com

Help Resources:

1. See the beep codes section of this manual.
2. See the TYAN® website for FAQ's, bulletins, driver updates, and other information: **http://www.tyan.com**
3. Contact your dealer for help BEFORE calling TYAN®.
4. Check the TYAN® user group in Google Forum: **alt.comp.periphs.mainboard.TYAN**

Returning Merchandise for Service

During the warranty period, contact your distributor or system vendor FIRST for any product problems. This warranty only covers normal customer use and does not cover damages incurred during shipping or failure due to the alteration, misuse, abuse, or improper maintenance of products.

NOTE:
A receipt or copy of your invoice marked with the date of purchase is required before any warranty service can be rendered. You may obtain service by calling the manufacturer for a Return Merchandise Authorization (RMA) number. The RMA number Should be prominently displayed on the outside of the shipping carton and the package should be mailed prepaid. TYAN® will pay to have the board shipped back to you.

Notice for the USA
Compliance Information Statement (Declaration of Conformity Procedure) DoC
FCC Part 15: This device complies with part 15 of the FCC Rules

Operation is subject to the following conditions:

This device may not cause harmful interference, and this device must accept any interference received including interference that may cause undesired operation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try one or more of the following measures:
Reorient or relocate the receiving antenna.
Increase the separation between the equipment and the receiver.
Plug the equipment into an outlet on a circuit different from that of the receiver.
Consult the dealer on an experienced radio/television technician for help.

Notice for Canada
This apparatus complies with the Class B limits for radio interference as specified in the Canadian Department of Communications Radio Interference Regulations. (Cet appareil est conforme aux norms de Classe B d'interference radio tel que specifie par le Ministere Canadien des Communications dans les reglements d'ineteference radio.)

CAUTION: Lithium battery included with this board. Do not puncture, mutilate, or dispose of battery in fire. There is danger of an explosion if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by manufacturer. Dispose of used battery according to manufacturer instructions and in accordance with your local regulations.

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